

Assignment 2

2023-05-16

```
library (ggplot2)

## Warning: package 'ggplot2' was built under R version 4.2.3

library (tidyverse)

## Warning: package 'tidyverse' was built under R version 4.2.3
## Warning: package 'tibble' was built under R version 4.2.3
## Warning: package 'tidyr' was built under R version 4.2.3
## Warning: package 'readr' was built under R version 4.2.3
## Warning: package 'dplyr' was built under R version 4.2.3
## Warning: package 'forcats' was built under R version 4.2.3
## Warning: package 'lubridate' was built under R version 4.2.3

## — Attaching core tidyverse packages — tidyverse
2.0.0 —
## ✓ dplyr      1.1.2      ✓ readr      2.1.4
## ✓ forcats   1.0.0      ✓ stringr    1.5.0
## ✓ lubridate 1.9.2      ✓ tibble     3.2.1
## ✓ purrr     1.0.1      ✓ tidyr      1.3.0
## — Conflicts —
tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the [8];http://conflicted.r-lib.org/conflicted package[8]; to force
all conflicts to become errors

weight_df <- read_csv("C:/Users/Nischal/Downloads/datasetass1.csv")

## Rows: 6068 Columns: 8
## — Column specification
##
## Delimiter: ","
## chr (1): gender
## dbl (7): subjectid, height, height_selfreport, weight, weight_selfreport,
ag...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
weight_df
```

```
## # A tibble: 6,068 × 8
```

```
##   subjectid gender height height_selfreport weight weight_selfreport  
age
```

```
##      <dbl> <chr>   <dbl>          <dbl> <dbl>          <dbl>
```

```
<dbl>
```

```
## 1    10027 Male    178.          180.   81.5           81.7
```

```
41
```

```
## 2    10032 Male    170.          173.   72.6           72.6
```

```
35
```

```
## 3    10033 Male    174.          173.   92.9           93.0
```

```
42
```

```
## 4    10092 Male    166.          168.   79.4           79.4
```

```
31
```

```
## 5    10093 Male    191.          196.   94.6           96.6
```

```
21
```

```
## 6    10115 Male    172            175.   80.2           79.4
```

```
39
```

```
## 7    10117 Male    181            183.  116.           113.
```

```
32
```

```
## 8    10237 Male    185            188.   95.4           95.7
```

```
23
```

```
## 9    10242 Male    178.          178.   99.5           99.8
```

```
36
```

```
## 10   10244 Male    181.          183.   70.2           72.6
```

```
23
```

```
## # i 6,058 more rows
```

```
## # i 1 more variable: race <dbl>
```

```
glimpse(weight_df)
```

```
## Rows: 6,068
```

```
## Columns: 8
```

```
## $ subjectid      <dbl> 10027, 10032, 10033, 10092, 10093, 10115, 10117,  
102...
```

```
## $ gender         <chr> "Male", "Male", "Male", "Male", "Male", "Male",  
"Mal...
```

```
## $ height         <dbl> 177.6, 170.2, 173.5, 165.5, 191.4, 172.0, 181.0,  
185...
```

```
## $ height_selfreport <dbl> 180.34, 172.72, 172.72, 167.64, 195.58, 175.26,  
182....
```

```
## $ weight         <dbl> 81.5, 72.6, 92.9, 79.4, 94.6, 80.2, 116.2, 95.4,  
99....
```

```
## $ weight_selfreport <dbl> 81.66969, 72.59528, 93.01270, 79.40109,  
96.64247, 79...
```

```
## $ age            <dbl> 41, 35, 42, 31, 21, 39, 32, 23, 36, 23, 32, 28,  
36, ...
```

```
## $ race           <dbl> 1, 1, 2, 1, 2, 1, 2, 1, 1, 1, 1, 2, 1, 1, 2,  
1, 1...
```

#CHAPTER 3 HW #-----Selecting Variables-----

```
select(weight_df, subjectid, gender, age, race)
```

```
## # A tibble: 6,068 × 4
##   subjectid gender    age  race
##   <dbl> <chr>   <dbl> <dbl>
## 1    10027 Male     41     1
## 2    10032 Male     35     1
## 3    10033 Male     42     2
## 4    10092 Male     31     1
## 5    10093 Male     21     2
## 6    10115 Male     39     1
## 7    10117 Male     32     2
## 8    10237 Male     23     1
## 9    10242 Male     36     1
## 10   10244 Male     23     1
## # i 6,058 more rows
```

```
select(weight_df, gender:race)
```

```
## # A tibble: 6,068 × 7
##   gender height height_selfreport weight weight_selfreport    age  race
##   <chr>   <dbl>           <dbl>   <dbl>           <dbl> <dbl> <dbl>
## 1 Male    178.           180.    81.5           81.7    41     1
## 2 Male    170.           173.    72.6           72.6    35     1
## 3 Male    174.           173.    92.9           93.0    42     2
## 4 Male    166.           168.    79.4           79.4    31     1
## 5 Male    191.           196.    94.6           96.6    21     2
## 6 Male    172           175.    80.2           79.4    39     1
## 7 Male    181           183.   116.           113.    32     2
## 8 Male    185           188.    95.4           95.7    23     1
## 9 Male    178.           178.    99.5           99.8    36     1
## 10 Male   181.           183.    70.2           72.6    23     1
## # i 6,058 more rows
```

```
select(weight_df, 1:3) # this is to select from a range 1 to 3
```

```
## # A tibble: 6,068 × 3
##   subjectid gender height
##   <dbl> <chr>   <dbl>
## 1    10027 Male    178.
## 2    10032 Male    170.
## 3    10033 Male    174.
## 4    10092 Male    166.
## 5    10093 Male    191.
## 6    10115 Male    172
## 7    10117 Male    181
## 8    10237 Male    185
## 9    10242 Male    178.
```

```
## 10      10244 Male      181.
## # i 6,058 more rows

select(weight_df, starts_with('h'))

## # A tibble: 6,068 × 2
##   height height_selfreport
##   <dbl>         <dbl>
## 1   178.           180.
## 2   170.           173.
## 3   174.           173.
## 4   166.           168.
## 5   191.           196.
## 6   172.           175.
## 7   181.           183.
## 8   185.           188.
## 9   178.           178.
## 10  181.           183.
## # i 6,058 more rows

select(weight_df, ends_with('t'))

## # A tibble: 6,068 × 4
##   height height_selfreport weight weight_selfreport
##   <dbl>         <dbl> <dbl>         <dbl>
## 1   178.           180.   81.5           81.7
## 2   170.           173.   72.6           72.6
## 3   174.           173.   92.9           93.0
## 4   166.           168.   79.4           79.4
## 5   191.           196.   94.6           96.6
## 6   172.           175.   80.2           79.4
## 7   181.           183.  116.           113.
## 8   185.           188.   95.4           95.7
## 9   178.           178.   99.5           99.8
## 10  181.           183.   70.2           72.6
## # i 6,058 more rows

select(weight_df, contains('selfreport'))

## # A tibble: 6,068 × 2
##   height_selfreport weight_selfreport
##   <dbl>         <dbl>
## 1           180.           81.7
## 2           173.           72.6
## 3           173.           93.0
## 4           168.           79.4
## 5           196.           96.6
## 6           175.           79.4
## 7           183.          113.
## 8           188.           95.7
## 9           178.           99.8
```

```

## 10          183.          72.6
## # i 6,058 more rows

select(weight_df, matches('height|weight')) #matches ... or .... / represents
"OR"

## # A tibble: 6,068 × 4
##   height height_selfreport weight weight_selfreport
##   <dbl>      <dbl> <dbl>      <dbl>
## 1   178.        180.   81.5        81.7
## 2   170.        173.   72.6        72.6
## 3   174.        173.   92.9        93.0
## 4   166.        168.   79.4        79.4
## 5   191.        196.   94.6        96.6
## 6   172         175.   80.2        79.4
## 7   181         183.  116.        113.
## 8   185         188.   95.4        95.7
## 9   178.        178.   99.5        99.8
## 10  181.        183.   70.2        72.6
## # i 6,058 more rows

select(weight_df, -age)

## # A tibble: 6,068 × 7
##   subjectid gender height height_selfreport weight weight_selfreport
##   <dbl> <chr>   <dbl>      <dbl> <dbl>      <dbl>
## 1  10027 Male    178.        180.   81.5        81.7
## 2  10032 Male    170.        173.   72.6        72.6
## 3  10033 Male    174.        173.   92.9        93.0
## 4  10092 Male    166.        168.   79.4        79.4
## 5  10093 Male    191.        196.   94.6        96.6
## 6  10115 Male    172         175.   80.2        79.4
## 7  10117 Male    181         183.  116.        113.
## 8  10237 Male    185         188.   95.4        95.7
## 9  10242 Male    178.        178.   99.5        99.8
## 10 10244 Male    181.        183.   70.2        72.6
## # i 6,058 more rows

select(weight_df, -(4:7))

```

```
## # A tibble: 6,068 × 4
##   subjectid gender height  race
##   <dbl> <chr>   <dbl> <dbl>
## 1    10027 Male    178.    1
## 2    10032 Male    170.    1
## 3    10033 Male    174.    2
## 4    10092 Male    166.    1
## 5    10093 Male    191.    2
## 6    10115 Male    172     1
## 7    10117 Male    181     2
## 8    10237 Male    185     1
## 9    10242 Male    178.    1
## 10   10244 Male    181.    1
## # i 6,058 more rows
```

```
select(weight_df, -contains('selfreport'))
```

```
## # A tibble: 6,068 × 6
##   subjectid gender height weight  age  race
##   <dbl> <chr>   <dbl> <dbl> <dbl> <dbl>
## 1    10027 Male    178.   81.5   41    1
## 2    10032 Male    170.   72.6   35    1
## 3    10033 Male    174.   92.9   42    2
## 4    10092 Male    166.   79.4   31    1
## 5    10093 Male    191.   94.6   21    2
## 6    10115 Male    172   80.2   39    1
## 7    10117 Male    181  116.   32    2
## 8    10237 Male    185   95.4   23    1
## 9    10242 Male    178.   99.5   36    1
## 10   10244 Male    181.   70.2   23    1
## # i 6,058 more rows
```

```
select(weight_df, age, height, race)
```

```
## # A tibble: 6,068 × 3
##   age height  race
##   <dbl> <dbl> <dbl>
## 1    41   178.    1
## 2    35   170.    1
## 3    42   174.    2
## 4    31   166.    1
## 5    21   191.    2
## 6    39   172     1
## 7    32   181     2
## 8    23   185     1
## 9    36   178.    1
## 10   23   181.    1
## # i 6,058 more rows
```

```
select(weight_df, height, subjectid, age, everything())
```

```
## # A tibble: 6,068 × 8
##   height subjectid age gender height_selfreport weight
weight_selfreport
##   <dbl>      <dbl> <dbl> <chr>          <dbl> <dbl>
<dbl>
## 1 178.      10027  41 Male          180.  81.5
81.7
## 2 170.      10032  35 Male          173.  72.6
72.6
## 3 174.      10033  42 Male          173.  92.9
93.0
## 4 166.      10092  31 Male          168.  79.4
79.4
## 5 191.      10093  21 Male          196.  94.6
96.6
## 6 172       10115  39 Male          175.  80.2
79.4
## 7 181       10117  32 Male          183. 116.
## 8 185       10237  23 Male          188.  95.4
95.7
## 9 178.      10242  36 Male          178.  99.5
99.8
## 10 181.     10244  23 Male          183.  70.2
72.6
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

select(weight_df, race, everything(), -age, gender)

## # A tibble: 6,068 × 7
##   race subjectid gender height height_selfreport weight
weight_selfreport
##   <dbl>      <dbl> <chr>   <dbl>          <dbl> <dbl>
<dbl>
## 1 1 10027 Male 178.          180.  81.5
81.7
## 2 1 10032 Male 170.          173.  72.6
72.6
## 3 2 10033 Male 174.          173.  92.9
93.0
## 4 1 10092 Male 166.          168.  79.4
79.4
## 5 2 10093 Male 191.          196.  94.6
96.6
## 6 1 10115 Male 172           175.  80.2
79.4
## 7 2 10117 Male 181           183. 116.
## 8 1 10237 Male 185           188.  95.4
95.7
## 9 1 10242 Male 178.          178.  99.5
```

```

99.8
## 10      1      10244 Male      181.      183.      70.2
72.6
## # i 6,058 more rows

select_if(weight_df, is.character)

## # A tibble: 6,068 × 1
##   gender
##   <chr>
## 1 Male
## 2 Male
## 3 Male
## 4 Male
## 5 Male
## 6 Male
## 7 Male
## 8 Male
## 9 Male
## 10 Male
## # i 6,058 more rows

select_if(weight_df, is.numeric)

## # A tibble: 6,068 × 7
##   subjectid height height_selfreport weight weight_selfreport age race
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 10027 178. 180. 81.5 81.7 41 1
## 2 10032 170. 173. 72.6 72.6 35 1
## 3 10033 174. 173. 92.9 93.0 42 2
## 4 10092 166. 168. 79.4 79.4 31 1
## 5 10093 191. 196. 94.6 96.6 21 2
## 6 10115 172 175. 80.2 79.4 39 1
## 7 10117 181 183. 116. 113. 32 2
## 8 10237 185 188. 95.4 95.7 23 1
## 9 10242 178. 178. 99.5 99.8 36 1
## 10 10244 181. 183. 70.2 72.6 23 1
## # i 6,058 more rows

has_low_mean <- function(x){
  is.numeric(x) && (mean(x, na.rm = T) < 150)
}

select_if(weight_df, has_low_mean)

## # A tibble: 6,068 × 4
##   weight weight_selfreport age race
##   <dbl> <dbl> <dbl> <dbl>
## 1 81.5 81.7 41 1
## 2 72.6 72.6 35 1
## 3 92.9 93.0 42 2

```



```
## 4 79.4 79.4 31 1
## 5 94.6 96.6 21 2
## 6 80.2 79.4 39 1
## 7 116. 113. 32 2
## 8 95.4 95.7 23 1
## 9 99.5 99.8 36 1
## 10 70.2 72.6 23 1
## # i 6,058 more rows
```

```
select(weight_df, section=race, mass=weight)
```

```
## # A tibble: 6,068 × 2
##   section mass
##   <dbl> <dbl>
## 1 1 81.5
## 2 1 72.6
## 3 2 92.9
## 4 1 79.4
## 5 2 94.6
## 6 1 80.2
## 7 2 116.
## 8 1 95.4
## 9 1 99.5
## 10 1 70.2
## # i 6,058 more rows
```

```
rename_all(weight_df, ~str_replace_all(., '\\.', '_'))
```

```
## # A tibble: 6,068 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr> <dbl> <dbl> <dbl> <dbl>
<dbl>
## 1 10027 Male 178. 180. 81.5 81.7
41
## 2 10032 Male 170. 173. 72.6 72.6
35
## 3 10033 Male 174. 173. 92.9 93.0
42
## 4 10092 Male 166. 168. 79.4 79.4
31
## 5 10093 Male 191. 196. 94.6 96.6
21
## 6 10115 Male 172 175. 80.2 79.4
39
## 7 10117 Male 181 183. 116. 113.
32
## 8 10237 Male 185 188. 95.4 95.7
23
## 9 10242 Male 178. 178. 99.5 99.8
36
```

```
## 10      10244 Male      181.      183.    70.2      72.6
23
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

rename_at(weight_df,
  vars(matches('^self|report')),
  ~str_replace_all(., 'selfreport', 'sr'))

## # A tibble: 6,068 × 8
##   subjectid gender height height_sr weight weight_sr age race
##   <dbl> <chr>   <dbl>   <dbl>  <dbl>   <dbl> <dbl> <dbl>
## 1      10027 Male     178.     180.   81.5     81.7   41     1
## 2      10032 Male     170.     173.   72.6     72.6   35     1
## 3      10033 Male     174.     173.   92.9     93.0   42     2
## 4      10092 Male     166.     168.   79.4     79.4   31     1
## 5      10093 Male     191.     196.   94.6     96.6   21     2
## 6      10115 Male     172      175.   80.2     79.4   39     1
## 7      10117 Male     181     183.  116.     113.   32     2
## 8      10237 Male     185     188.   95.4     95.7   23     1
## 9      10242 Male     178.     178.   99.5     99.8   36     1
## 10     10244 Male     181.     183.   70.2     72.6   23     1
## # i 6,058 more rows

rename_if(weight_df, is.character, str_to_upper)

## # A tibble: 6,068 × 8
##   subjectid GENDER height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>           <dbl>  <dbl>           <dbl>
<dbl>
## 1      10027 Male     178.           180.   81.5           81.7
41
## 2      10032 Male     170.           173.   72.6           72.6
35
## 3      10033 Male     174.           173.   92.9           93.0
42
## 4      10092 Male     166.           168.   79.4           79.4
31
## 5      10093 Male     191.           196.   94.6           96.6
21
## 6      10115 Male     172           175.   80.2           79.4
39
## 7      10117 Male     181           183.  116.           113.
32
## 8      10237 Male     185           188.   95.4           95.7
23
## 9      10242 Male     178.           178.   99.5           99.8
36
## 10     10244 Male     181.           183.   70.2           72.6
23
```

```
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

slice(weight_df, c(1, 10, 150, 200, 250))

## # A tibble: 5 × 8
##   subjectid gender height height_selfreport weight weight_selfreport age
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl> <dbl>
<dbl>
## 1    10027 Male    178.         180.   81.5         81.7   41
1
## 2    10244 Male    181.         183.   70.2         72.6   23
1
## 3    10848 Male    193.         193.   80.6         80.8   21
1
## 4    11161 Male    175.         180.    84         84.8   33
2
## 5    11290 Male    161.         163.   56.8         59.0   30
6

slice(weight_df, 1:10)

## # A tibble: 10 × 8
##   subjectid gender height height_selfreport weight weight_selfreport age
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl>
<dbl>
## 1    10027 Male    178.         180.   81.5         81.7
41
## 2    10032 Male    170.         173.   72.6         72.6
35
## 3    10033 Male    174.         173.   92.9         93.0
42
## 4    10092 Male    166.         168.   79.4         79.4
31
## 5    10093 Male    191.         196.   94.6         96.6
21
## 6    10115 Male    172          175.   80.2         79.4
39
## 7    10117 Male    181          183.  116.         113.
32
## 8    10237 Male    185          188.   95.4         95.7
23
## 9    10242 Male    178.         178.   99.5         99.8
36
## 10   10244 Male    181.         183.   70.2         72.6
23
## # i 1 more variable: race <dbl>

slice(weight_df, -(1:10))
```

```
## # A tibble: 6,058 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>           <dbl> <dbl>           <dbl>
<dbl>
## 1    10246 Male    178             178.  88.2            88.5
32
## 2    10260 Male    173.            175.  70.1            70.3
28
## 3    10265 Male    181.            183.  104.            104.
36
## 4    10272 Male    186.            190.  112.            113.
26
## 5    10286 Male    171.            173.  89.6            90.7
22
## 6    10293 Male    174             175.  92.4            88.9
38
## 7    10303 Male    169.            170.  77.9            77.1
26
## 8    10304 Male    177.            178.  96.9            97.5
34
## 9    10305 Male    175.            178.  76              75.8
31
## 10   10310 Male    179.            178.  90.8            91.7
32
## # i 6,048 more rows
## # i 1 more variable: race <dbl>

slice(weight_df, 100:n()) # from the 100th row to nth row so in this case
when it is left empty it will go to final

## # A tibble: 5,969 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>           <dbl> <dbl>           <dbl>
<dbl>
## 1    10701 Male    176.            180.  96.8            97.5
27
## 2    10704 Male    179.            183.  86.6            86.2
44
## 3    10705 Male    176.            175.  74.2            72.6
19
## 4    10707 Male    179             180.  92              95.3
29
## 5    10710 Male    178.            183.  81.4            81.7
22
## 6    10712 Male    184.            185.  64.6            65.3
28
## 7    10713 Male    188             190.  95.4            97.5
31
```

```
## 8      10716 Male      174.      175.  85.3      86.2
28
## 9      10720 Male      173.      175.  90      89.4
31
## 10     10721 Male      182.      185.  83.5      83.0
34
## # i 5,959 more rows
## # i 1 more variable: race <dbl>
```

```
slice(weight_df, (n()-10):n()) #the last 10 rows
```

```
## # A tibble: 11 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>           <dbl> <dbl>           <dbl>
<dbl>
## 1      29493 Female   164.           165.  75.8           76.7
26
## 2      29494 Female   158.           157.  65.2           64.0
41
## 3      29495 Female   160.           163.  69.9           63.5
30
## 4      29496 Female   153.           155.  47.4           47.6
30
## 5      29497 Female   159.           160.  55.8           56.3
32
## 6      29498 Female   172.           173.  76.8           75.3
25
## 7      29501 Female   169.           170.  83.2           81.7
51
## 8      29502 Female   161.           160.  71.7           68.1
40
## 9      29503 Female   164.           168.  76.2           76.2
40
## 10     29511 Female   162.           160.  63.2           60.3
31
## 11     920103 Female   164.           168.  61            59.9
27
## # i 1 more variable: race <dbl>
```

```
#-----Filtering Variables-----
```

```
filter(weight_df, age == '21')
```

```
## # A tibble: 383 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>           <dbl> <dbl>           <dbl>
<dbl>
## 1      10093 Male      191.           196.  94.6           96.6
21
```

```

## 2      10367 Male      188.      190.      71.5      76.2
21
## 3      10471 Male      167.      168.      66.7      65.8
21
## 4      10638 Male      175      180.      85.2      86.2
21
## 5      10662 Male      158.      157.      62.7      62.6
21
## 6      10689 Male      184.      190.      69.6      74.9
21
## 7      10692 Male      179.      183.      77.5      77.1
21
## 8      10762 Male      166.      168.      78.9      77.1
21
## 9      10845 Male      178.      180.      88.7      88.5
21
## 10     10848 Male      193.      193.      80.6      80.8
21
## # i 373 more rows
## # i 1 more variable: race <dbl>

filter(weight_df, age == '21', gender=='Female')

## # A tibble: 129 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>    <dbl>          <dbl> <dbl>          <dbl>
<dbl>
## 1      10038 Female    166.          163.   53.4          54.4
21
## 2      10061 Female    164.          165.   73.2          74.9
21
## 3      11394 Female    162           163.   69.1          69.9
21
## 4      11814 Female    165.          165.   57.2          58.1
21
## 5      13441 Female    168.          168.    61           62.2
21
## 6      13540 Female    158.          160.   61.1          61.7
21
## 7      13542 Female    170.          170.   86.6          81.7
21
## 8      13555 Female    156.          157.   54.9          54.0
21
## 9      13603 Female    163.          163.   64.9          63.5
21
## 10     13605 Female    155.          157.   59.2          59.9
21
## # i 119 more rows
## # i 1 more variable: race <dbl>

```

```

filter(weight_df, age == '21', gender=='Female', height <= 150)

## # A tibble: 5 × 8
##   subjectid gender height height_selfreport weight weight_selfreport age
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl> <dbl>
## 1 25147 Female 148.         150.  57.5         59.0  21
## 2 25339 Female 149          150.  48.4         46.7  21
## 3 25764 Female 146.         147.  44.5         46.7  21
## 4 28159 Female 148.         150.  35.8         43.1  21
## 5 28259 Female 150.         152.  50.9         49.0  21

filter(weight_df, age == '21' & gender=='Female' & height <= 150)

## # A tibble: 5 × 8
##   subjectid gender height height_selfreport weight weight_selfreport age
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl> <dbl>
## 1 25147 Female 148.         150.  57.5         59.0  21
## 2 25339 Female 149          150.  48.4         46.7  21
## 3 25764 Female 146.         147.  44.5         46.7  21
## 4 28159 Female 148.         150.  35.8         43.1  21
## 5 28259 Female 150.         152.  50.9         49.0  21

filter(weight_df, age < 21 | height > 160)

## # A tibble: 5,457 × 8
##   subjectid gender height height_selfreport weight weight_selfreport age
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl>
## 1 10027 Male 178.         180.  81.5         81.7
## 2 10032 Male 170.         173.  72.6         72.6
## 3 10033 Male 174.         173.  92.9         93.0
## 4 10092 Male 166.         168.  79.4         79.4

```

```

## 5      10093 Male      191.      196.      94.6      96.6
21
## 6      10115 Male      172      175.      80.2      79.4
39
## 7      10117 Male      181      183.      116.      113.
32
## 8      10237 Male      185      188.      95.4      95.7
23
## 9      10242 Male      178.      178.      99.5      99.8
36
## 10     10244 Male      181.      183.      70.2      72.6
23
## # i 5,447 more rows
## # i 1 more variable: race <dbl>

filter(weight_df, age %in% 28:44)

## # A tibble: 2,617 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>           <dbl> <dbl>           <dbl>
<dbl>
## 1      10027 Male      178.           180.   81.5           81.7
41
## 2      10032 Male      170.           173.   72.6           72.6
35
## 3      10033 Male      174.           173.   92.9           93.0
42
## 4      10092 Male      166.           168.   79.4           79.4
31
## 5      10115 Male      172            175.   80.2           79.4
39
## 6      10117 Male      181            183.  116.           113.
32
## 7      10242 Male      178.           178.   99.5           99.8
36
## 8      10246 Male      178            178.   88.2           88.5
32
## 9      10260 Male      173.           175.   70.1           70.3
28
## 10     10265 Male      181.           183.  104.           104.
36
## # i 2,607 more rows
## # i 1 more variable: race <dbl>

#this allows us to filter the age from range 28 to 44 in variable age

filter_at(weight_df,
  vars(matches('height|weight_selfreport')),
  all_vars(. < median(., na.rm=T)))

```



```
## # A tibble: 3,015 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl>
<dbl>
## 1    10032 Male    170.         173.   72.6         72.6
35
## 2    10244 Male    181.         183.   70.2         72.6
23
## 3    10260 Male    173.         175.   70.1         70.3
28
## 4    10303 Male    169.         170.   77.9         77.1
26
## 5    10305 Male    175.         178.   76           75.8
31
## 6    10312 Male    178           180.   71.4         74.9
28
## 7    10319 Male    168           168.   74.6         74.0
24
## 8    10341 Male    163           163.   67.5         61.3
32
## 9    10367 Male    188.         190.   71.5         76.2
21
## 10   10371 Male    168.         165.   66.5         68.1
37
## # i 3,005 more rows
## # i 1 more variable: race <dbl>
```

#allows us to select variables according to their properties, rather than their names, and then apply filtering commands to the selected variables

```
filter_if(weight_df,
  is.numeric,
  all_vars(. < median(., na.rm=T)))
```

```
## # A tibble: 0 × 8
## # i 8 variables: subjectid <dbl>, gender <chr>, height <dbl>,
## #   height_selfreport <dbl>, weight <dbl>, weight_selfreport <dbl>, age
<dbl>,
## #   race <dbl>
```

#———RESHAPING DATA FRAMES———

```
mutate(weight_df, weight = weight_selfreport == height_selfreport)
```

```
## # A tibble: 6,068 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
##   <dbl> <chr>   <dbl>         <dbl> <lgl>         <dbl>
<dbl>
## 1    10027 Male    178.         180. FALSE         81.7
```

```

41
## 2      10032 Male      170.      173. FALSE      72.6
35
## 3      10033 Male      174.      173. FALSE      93.0
42
## 4      10092 Male      166.      168. FALSE      79.4
31
## 5      10093 Male      191.      196. FALSE      96.6
21
## 6      10115 Male      172      175. FALSE      79.4
39
## 7      10117 Male      181      183. FALSE      113.
32
## 8      10237 Male      185      188. FALSE      95.7
23
## 9      10242 Male      178.      178. FALSE      99.8
36
## 10     10244 Male      181.      183. FALSE      72.6
23
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

#create a new variable that takes the value true whenever the variables
mentioned above are the same, in this case it is not

##mutate(weight_df, fate = height(weight)) i dont understand this code

mutate_all(weight_df, as.character)

## # A tibble: 6,068 × 8
##   subjectid gender height height_selfreport weight weight_selfreport age
##   <chr>      <chr> <chr> <chr>          <chr> <chr>
<chr>
## 1 10027      Male  177.6  180.34          81.5  81.6696914700544  41
## 2 10032      Male  170.2  172.72          72.6  72.5952813067151  35
## 3 10033      Male  173.5  172.72          92.9  93.0127041742287  42
## 4 10092      Male  165.5  167.64          79.4  79.4010889292196  31
## 5 10093      Male  191.4  195.58          94.6  96.6424682395644  21
## 6 10115      Male  172     175.26          80.2  79.4010889292196  39
## 7 10117      Male  181     182.88          116.2 113.430127041742  32
## 8 10237      Male  185     187.96          95.4  95.7350272232305  23
## 9 10242      Male  177.7  177.8           99.5  99.8185117967332  36
## 10 10244      Male  181.1  182.88          70.2  72.5952813067151  23
## # i 6,058 more rows
## # i 1 more variable: race <chr>

#Logging the variables function
mutate_at(weight_df, vars(matches('height|height_selfreport')), log)

## # A tibble: 6,068 × 8
##   subjectid gender height height_selfreport weight weight_selfreport

```

```

age
##      <dbl> <chr>   <dbl>           <dbl> <dbl>           <dbl>
<dbl>
## 1    10027 Male    5.18             5.19  81.5            81.7
41
## 2    10032 Male    5.14             5.15  72.6            72.6
35
## 3    10033 Male    5.16             5.15  92.9            93.0
42
## 4    10092 Male    5.11             5.12  79.4            79.4
31
## 5    10093 Male    5.25             5.28  94.6            96.6
21
## 6    10115 Male    5.15             5.17  80.2            79.4
39
## 7    10117 Male    5.20             5.21  116.            113.
32
## 8    10237 Male    5.22             5.24  95.4            95.7
23
## 9    10242 Male    5.18             5.18  99.5            99.8
36
## 10   10244 Male    5.20             5.21  70.2            72.6
23
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

mutate_if(weight_df, is.character, as.factor)

## # A tibble: 6,068 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##      <dbl> <fct>   <dbl>           <dbl> <dbl>           <dbl>
<dbl>
## 1    10027 Male    178.            180.  81.5            81.7
41
## 2    10032 Male    170.            173.  72.6            72.6
35
## 3    10033 Male    174.            173.  92.9            93.0
42
## 4    10092 Male    166.            168.  79.4            79.4
31
## 5    10093 Male    191.            196.  94.6            96.6
21
## 6    10115 Male    172             175.  80.2            79.4
39
## 7    10117 Male    181             183.  116.            113.
32
## 8    10237 Male    185             188.  95.4            95.7
23
## 9    10242 Male    178.            178.  99.5            99.8

```

```

36
## 10      10244 Male      181.          183.    70.2          72.6
23
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

#factors variables into catagorical variable

#this isnt working but its a good fumction to have
##mutate(weight_df,
## size = if_else(height > 180,
## 'tall',
## 'short')

##most of transmutate codes didnt work, i will followup about this with the
TA

arrange(weight_df, age, race)

## # A tibble: 6,068 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>    <dbl>          <dbl> <dbl>          <dbl>
<dbl>
## 1      26138 Female   171.          170.    74.3          65.3
17
## 2      26714 Male     174.          178.    82.6          82.6
17
## 3      12247 Male     172.          173.    65.7          65.8
18
## 4      12428 Male     171.          168.    69.2          68.1
18
## 5      12467 Male     176.          178.    62.1          65.8
18
## 6      14371 Male     179           185.    81.9          81.7
18
## 7      14376 Male     181           185.    86.1          79.4
18
## 8      14389 Male     179.          183.    84           83.9
18
## 9      14393 Male     177.          183.    67           70.3
18
## 10     14488 Male     172.          173.    93.5          95.3
18
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

arrange(weight_df, age, desc(race)) #desc is descending

## # A tibble: 6,068 × 8
##   subjectid gender height height_selfreport weight weight_selfreport

```

```

age
##      <dbl> <chr>    <dbl>          <dbl> <dbl>          <dbl>
<dbl>
## 1    26714 Male     174.          178.  82.6          82.6
17
## 2    26138 Female   171.          170.  74.3          65.3
17
## 3    20272 Male     171.          173.  63            64.0
18
## 4    13084 Male     173.          180.  72.3          64.4
18
## 5    26621 Male     170.          173.  61.4          59.4
18
## 6    27122 Male     171.          175.  66.7          68.1
18
## 7    16084 Female   160           163.  69.9          64.4
18
## 8    18513 Female   159           160.  60.9          61.3
18
## 9    27076 Female   151.          150.  54.2          52.2
18
## 10   26169 Male     165.          165.  67.3          70.3
18
## # i 6,058 more rows
## # i 1 more variable: race <dbl>

sample_frac(weight_df, 0.1) #this allows us to sample a portion of the data
frame, 10% in this case

## # A tibble: 607 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##      <dbl> <chr>    <dbl>          <dbl> <dbl>          <dbl>
<dbl>
## 1    24163 Male     185.          190.  76.8          79.4
20
## 2    27196 Male     170.          173.  74.3          77.1
21
## 3    29390 Female   166.          163.  78            77.1
25
## 4    12703 Female   169.          170.  62.2          61.7
20
## 5    14522 Male     186.          185.  98.5          96.6
30
## 6    29492 Female   170.          173.  68.7          68.1
44
## 7    17974 Female   156.          155.  59.7          59.9
21
## 8    28304 Female   164.          165.  82.1          81.7
28

```

```

## 9      15150 Male      165.      168.  70.3      71.2
20
## 10     23735 Male      176.      178.  88      86.2
28
## # i 597 more rows
## # i 1 more variable: race <dbl>

sample_frac(weight_df, 0.1, replace=FALSE) #sampling will occur without
replacing

## # A tibble: 607 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl>
<dbl>
## 1      25524 Female   163.         163.  73.4         74.9
19
## 2      27115 Male     172.         173.  77.2         77.1
24
## 3      28687 Female   177.         180.  66.9         70.3
32
## 4      28991 Male     173.         175.  80.1         79.4
25
## 5      27840 Female   154          155.  58.9         59.0
27
## 6      25523 Female   168.         168.  68.1         65.8
21
## 7      13943 Male     169.         168.  96.6         95.3
45
## 8      16941 Female   168.         170.  77          77.1
27
## 9      14005 Male     166          168.  84.4         84.4
42
## 10     18460 Female   171.         173.  73.9         78.0
21
## # i 597 more rows
## # i 1 more variable: race <dbl>

sample_n(weight_df, 15)

## # A tibble: 15 × 8
##   subjectid gender height height_selfreport weight weight_selfreport
age
##   <dbl> <chr>   <dbl>         <dbl> <dbl>         <dbl>
<dbl>
## 1      14887 Male     185.         188. 103.         99.8
24
## 2      27421 Male     180.         185. 100.         97.5
32
## 3      29151 Male     181          183.  93          90.7
42

```

```
## 4      16115 Female  159.      160.  83.2      81.7
37
## 5      28170 Female  168.      168.  75.5      74.9
22
## 6      20702 Male   166.      168.  58.1      59.0
19
## 7      11051 Male   177.      180.  82.2      83.9
37
## 8      11537 Female  161.      160.  74.8      79.4
35
## 9      15970 Male   167.      168.  82.5      83.0
30
## 10     12974 Male   178.      180.  79.7      80.3
49
## 11     20156 Male   166.      168.  76.2      77.1
20
## 12     26996 Male   173.      175.  107.      107.
22
## 13     27807 Male   187.      188.  94.8      96.6
32
## 14     18316 Female  152.      152.  57.7      58.1
18
## 15     17860 Female  163.      165.  64.2      63.5
41
## # i 1 more variable: race <dbl>
```

#CHAPTER 4 homework

```
#-----
```

```
ship_df <- read_csv("C:/Users/Nischal/Downloads/TitanicSurvival.csv")

## New names:
## Rows: 1309 Columns: 5
## — Column specification
## _____ Delimiter: ","
chr
## (4): ...1, survived, sex, passengerClass dbl (1): age
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this
message.
## • `` -> `...1`

ship_df

## # A tibble: 1,309 × 5
##   ...1      survived sex      age passengerClass
##   <chr>      <chr>   <chr>   <dbl> <chr>
## 1 Allen, Miss. Elisabeth Walton yes     female 29     1st
## 2 Allison, Master. Hudson Trevor yes     male   0.917 1st
## 3 Allison, Miss. Helen Loraine no      female 2      1st
```

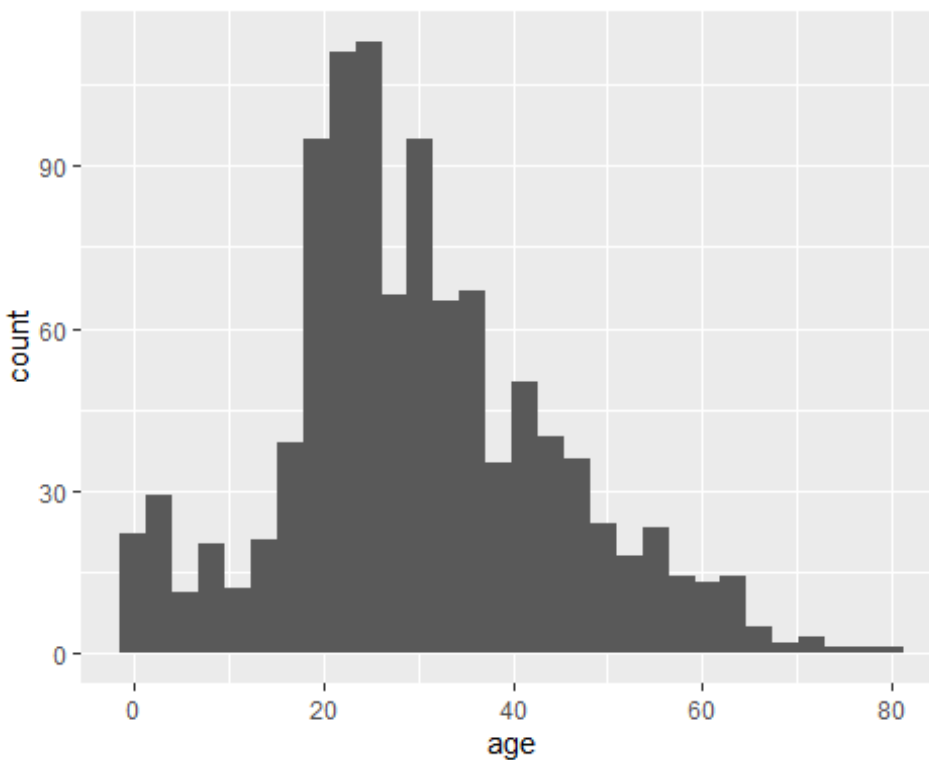
```
## 4 Allison, Mr. Hudson Joshua Crei no male 30 1st
## 5 Allison, Mrs. Hudson J C (Bessi no female 25 1st
## 6 Anderson, Mr. Harry yes male 48 1st
## 7 Andrews, Miss. Kornelia Theodos yes female 63 1st
## 8 Andrews, Mr. Thomas Jr no male 39 1st
## 9 Appleton, Mrs. Edward Dale (Cha yes female 53 1st
## 10 Artagaveytia, Mr. Ramon no male 71 1st
## # i 1,299 more rows
```

#using histogram to visualize the value because linechart would not show any for these type of variables.

```
ggplot(ship_df,
mapping = aes(x = age)
) + geom_histogram()
```

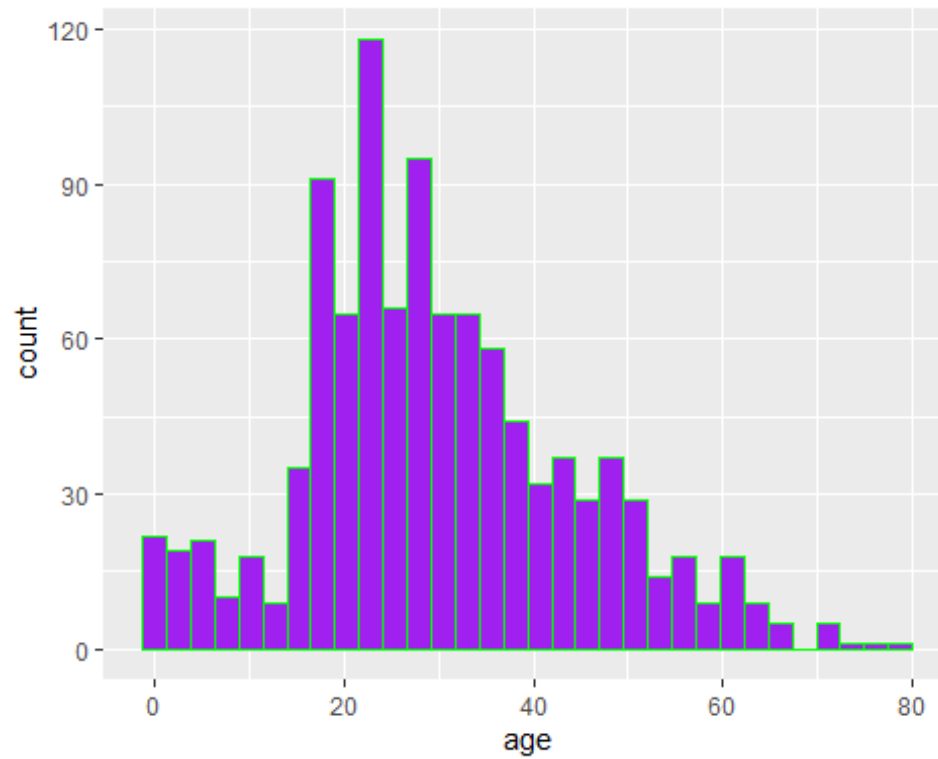
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

```
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```

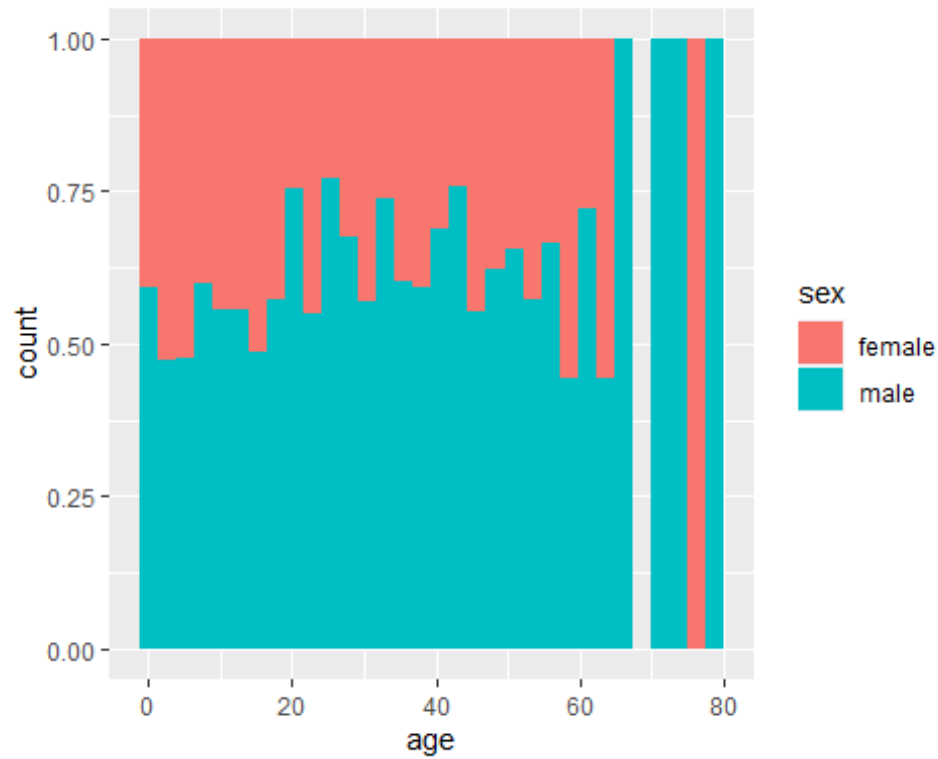


```
ggplot(ship_df,
mapping = aes(x = age, fill = sex)
) + geom_histogram(binwidth = 2.54, colour = 'green', fill = 'purple')
```

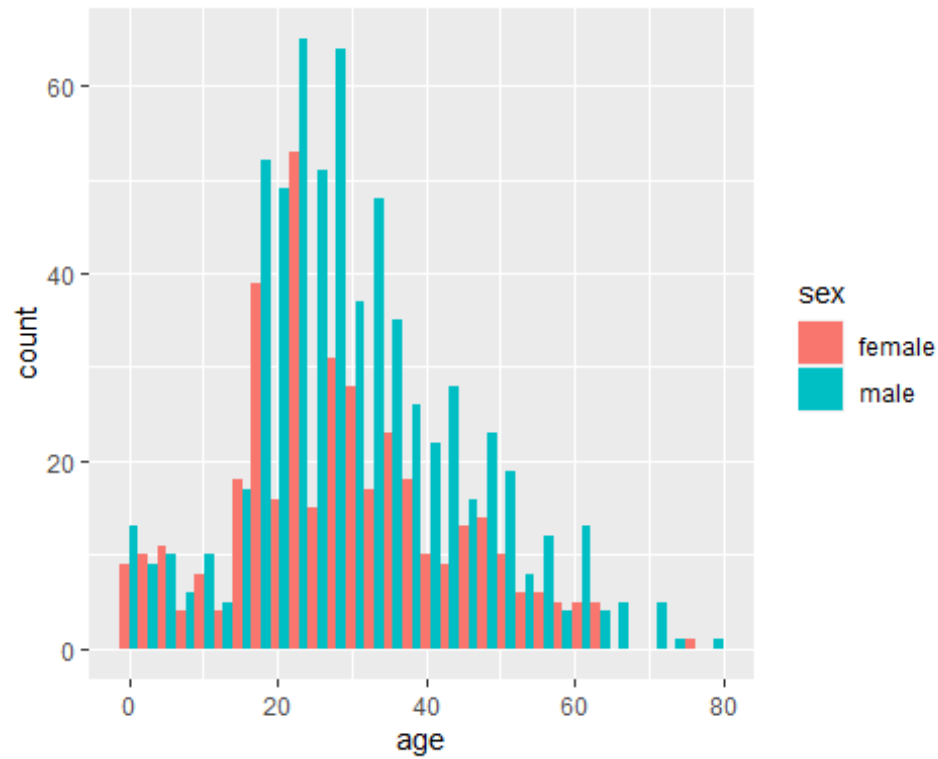
```
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```

```
ggplot(ship_df,  
  mapping = aes(x = age, fill = sex)  
) + geom_histogram(binwidth = 2.54, position = 'fill')  
  
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).  
## Warning: Removed 2 rows containing missing values (`geom_bar()`).
```

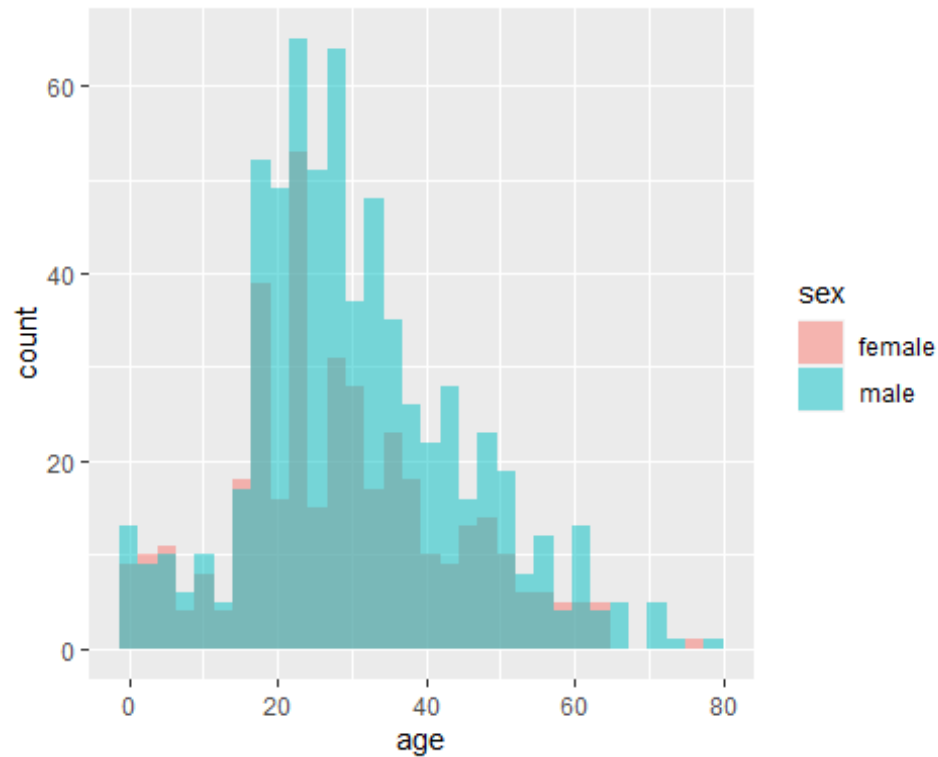


```
ggplot(ship_df,  
  mapping = aes(x = age, fill = sex)  
) + geom_histogram(binwidth = 2.54, position = 'dodge')  
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```



#if we want two seperate histogram, position = dodge

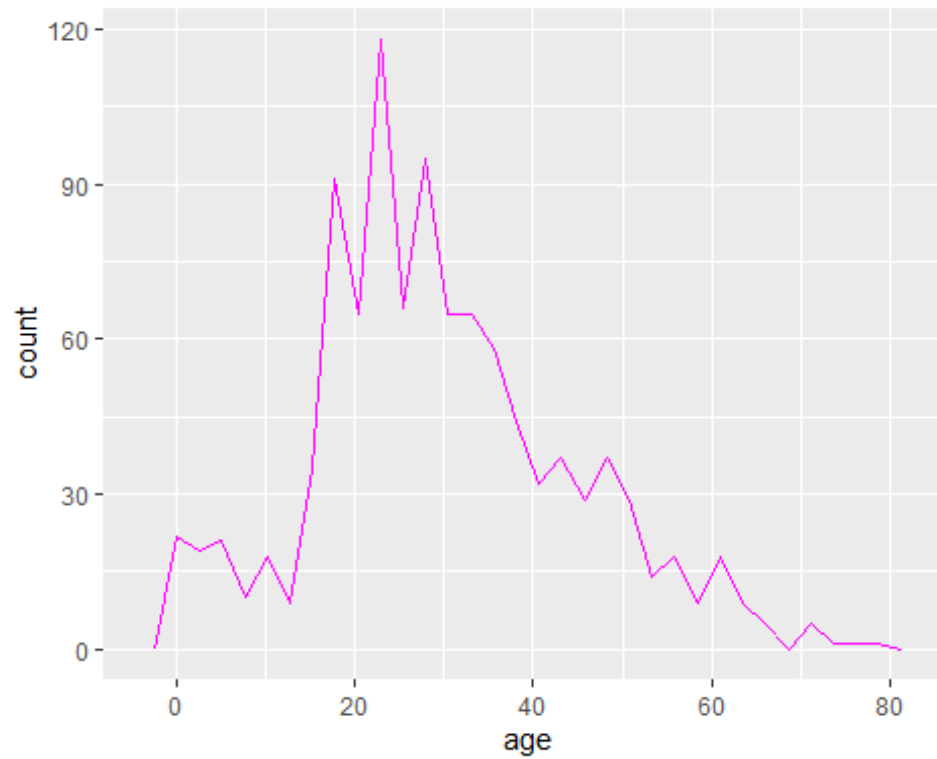
```
ggplot(ship_df,  
mapping = aes(x = age, fill = sex)  
) + geom_histogram(binwidth = 2.54, position = 'identity', alpha= 0.5)  
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```



```
#avoid complete occlusion = alpha
#position = identity will put both male and female side by side
```

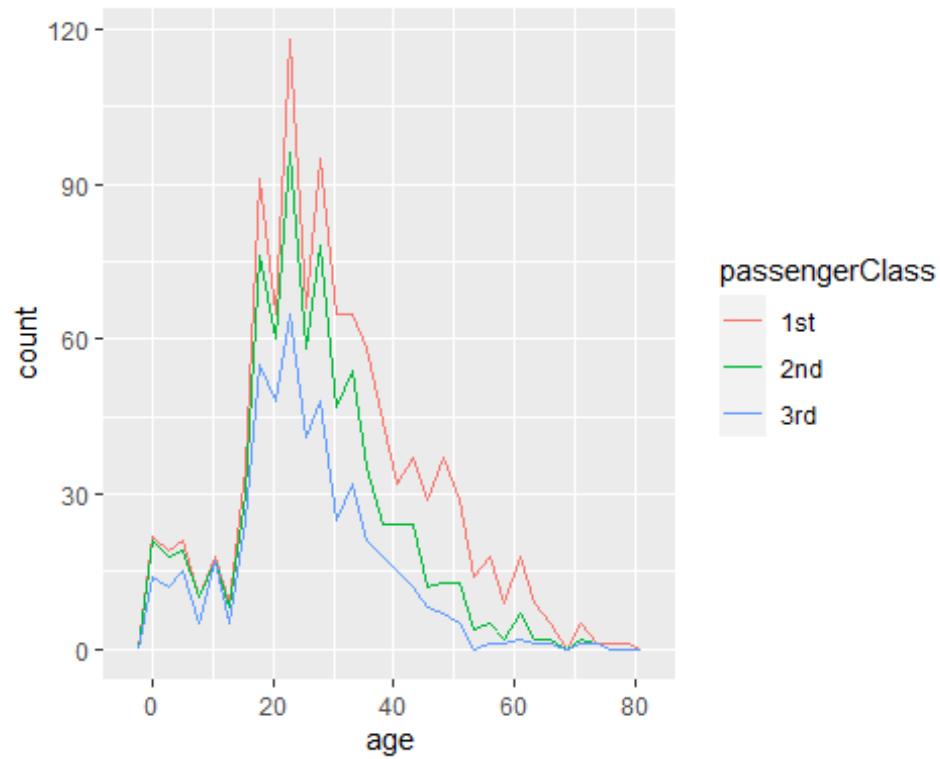
```
#frequency polygons
```

```
ggplot(ship_df,
  mapping = aes(x = age)
) + geom_freqpoly(binwidth = 2.54, colour = 'magenta')
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```



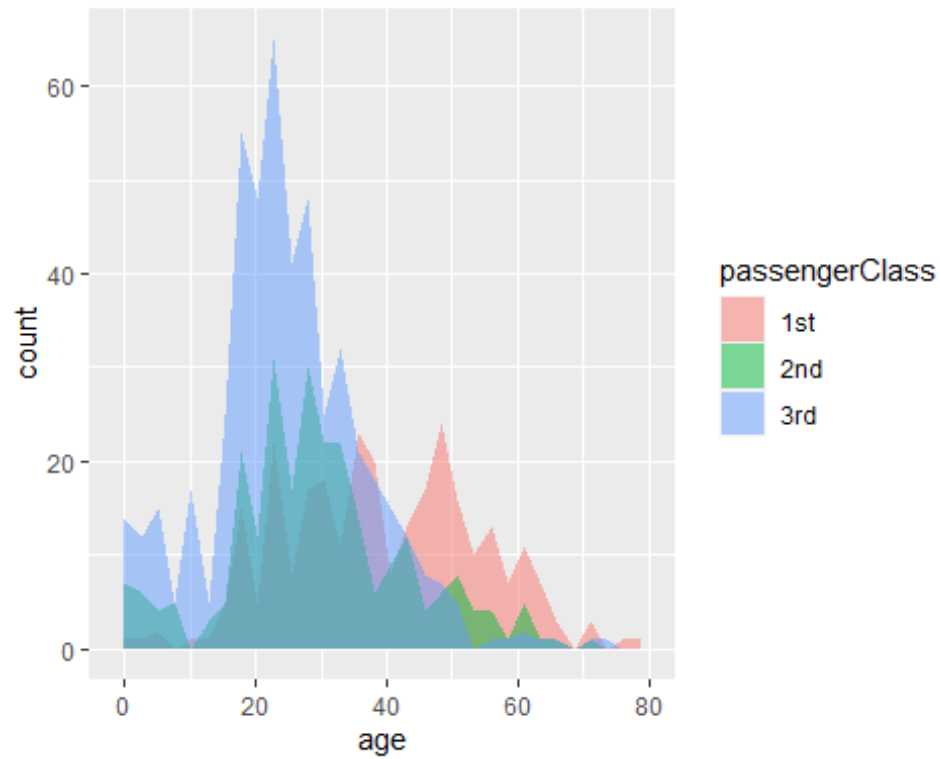
#stacked polygon chart

```
ggplot(ship_df,  
  mapping = aes(x = age, colour = passengerClass)  
) + geom_freqpoly(binwidth = 2.54, position = 'stack')  
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```



```
ggplot(ship_df,  
  mapping = aes(x = age, fill = passengerClass)  
) + geom_area(binwidth = 2.54, stat = "bin", position = 'identity', alpha =  
  0.5)
```

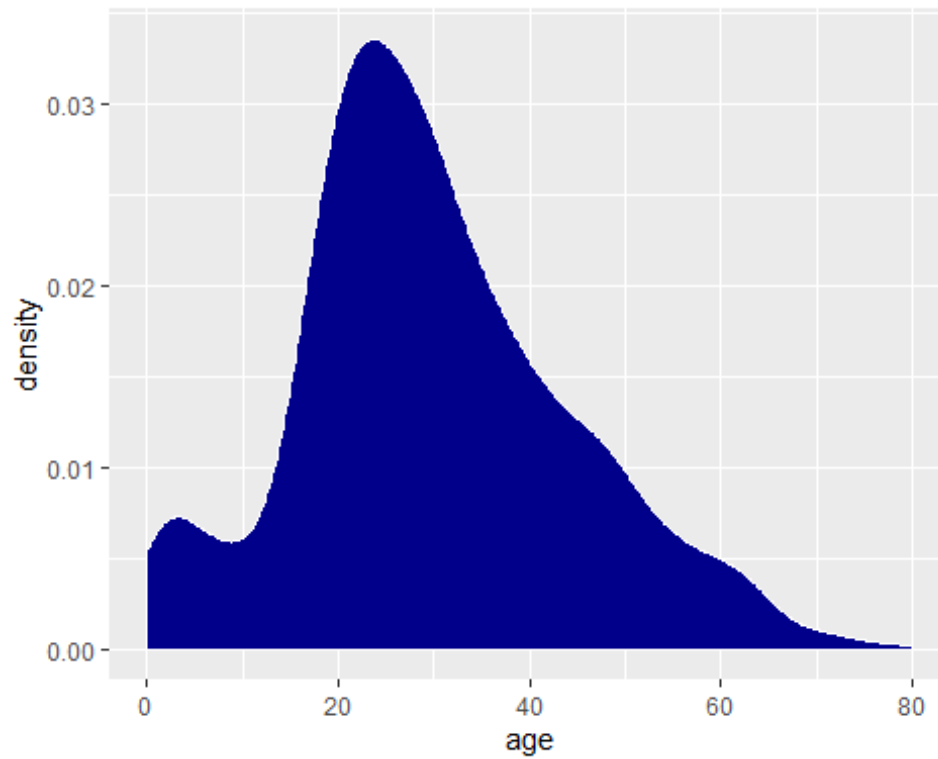
```
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```



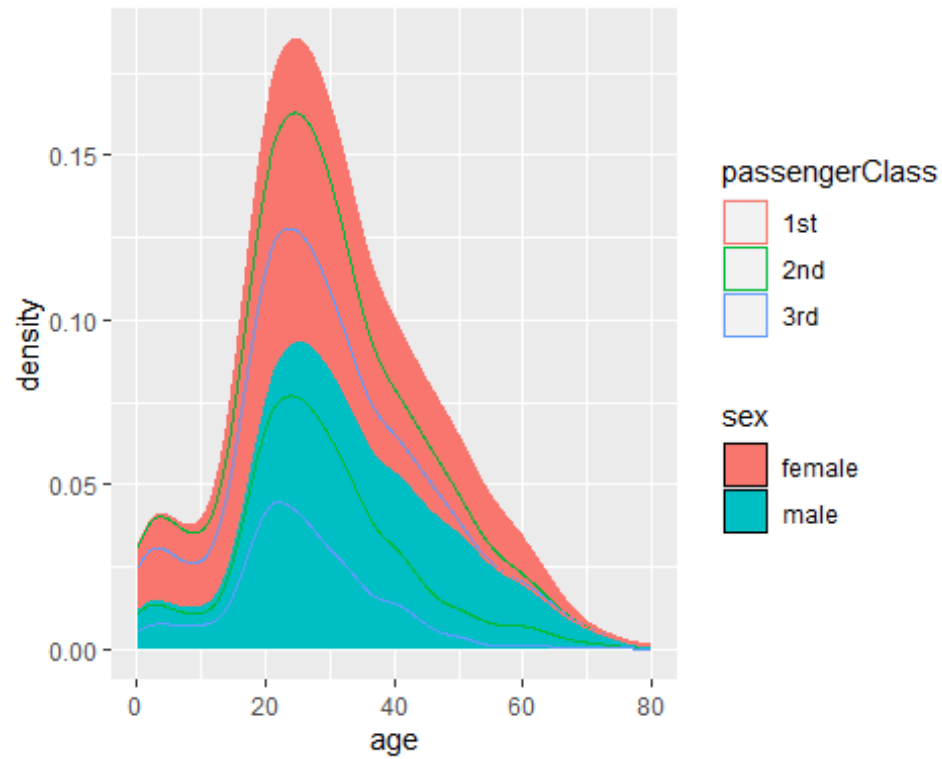
#area plot is made like this, this looks quite nice

#density plot

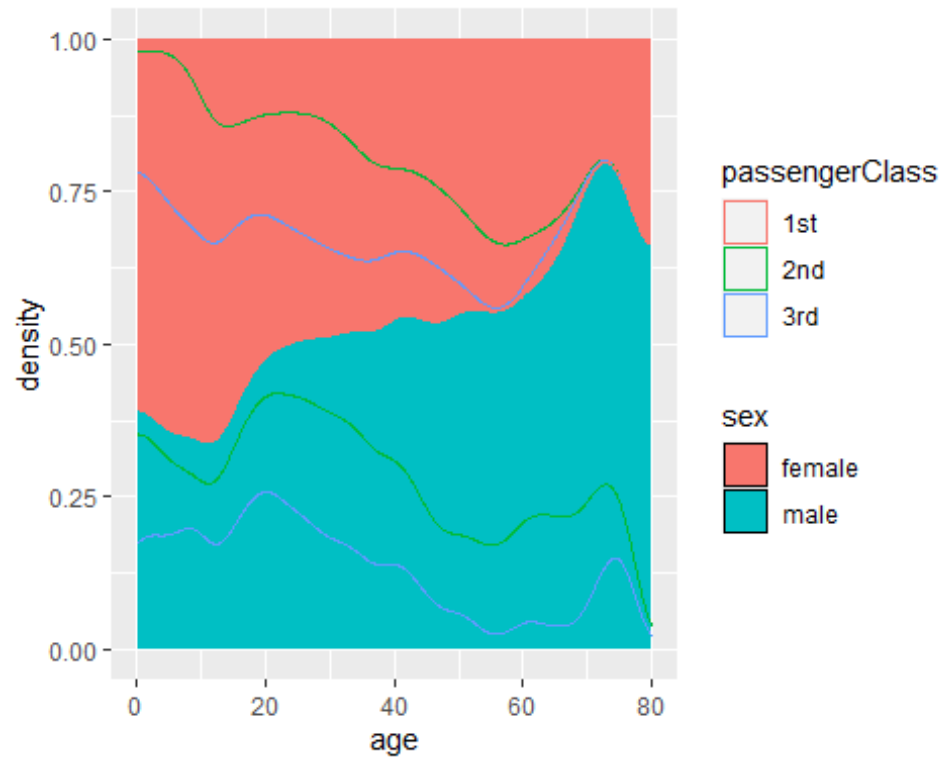
```
ggplot(ship_df,  
mapping = aes(x = age)  
) + geom_density(colour = 'white', fill = 'darkblue')  
## Warning: Removed 263 rows containing non-finite values (`stat_density()`).
```



```
ggplot(ship_df,  
mapping = aes(x = age, fill = sex, colour = passengerClass)  
) + geom_density(position = 'stack')  
## Warning: Removed 263 rows containing non-finite values (`stat_density()`).
```

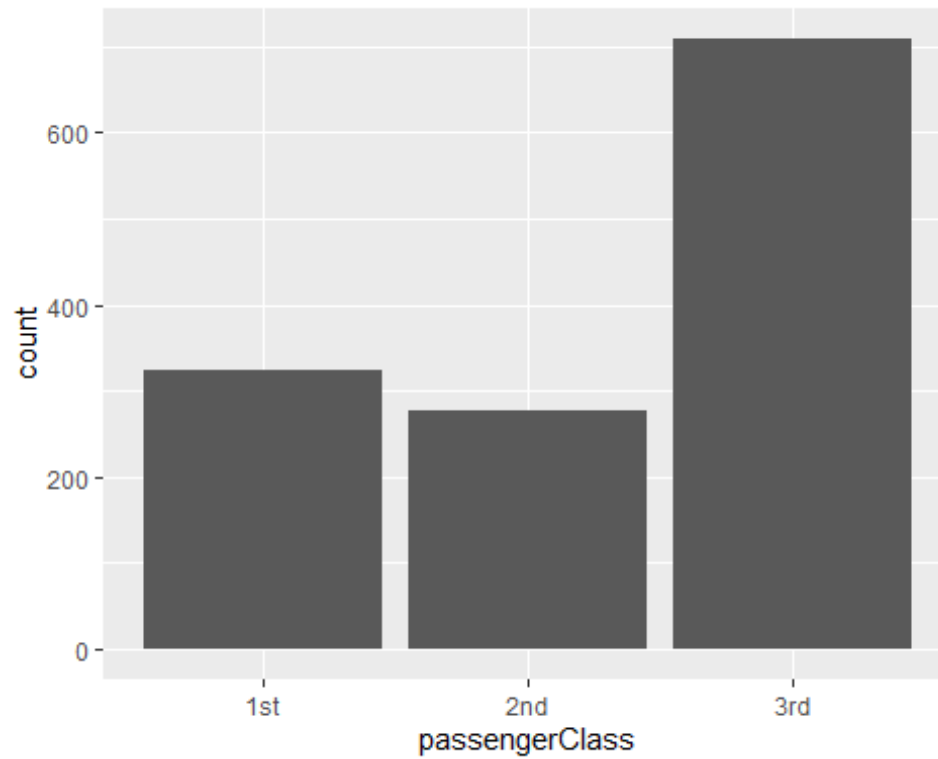



```
ggplot(ship_df,  
mapping = aes(x = age, fill = sex, colour = passengerClass)  
) + geom_density(position = 'fill')  
## Warning: Removed 263 rows containing non-finite values (`stat_density()`).
```

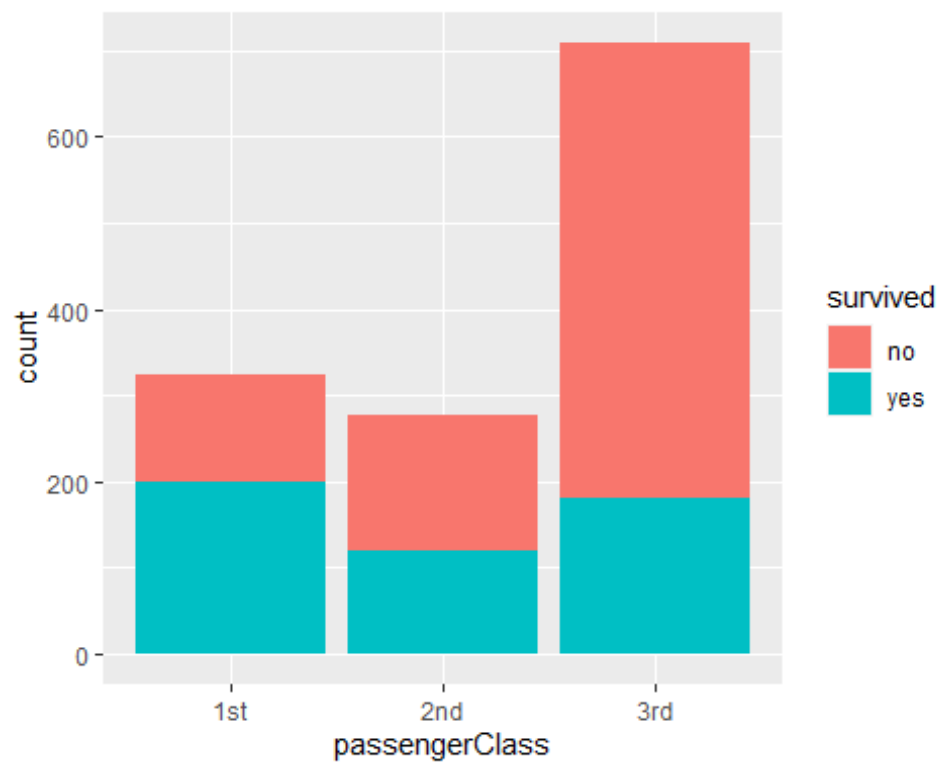


#Barplot

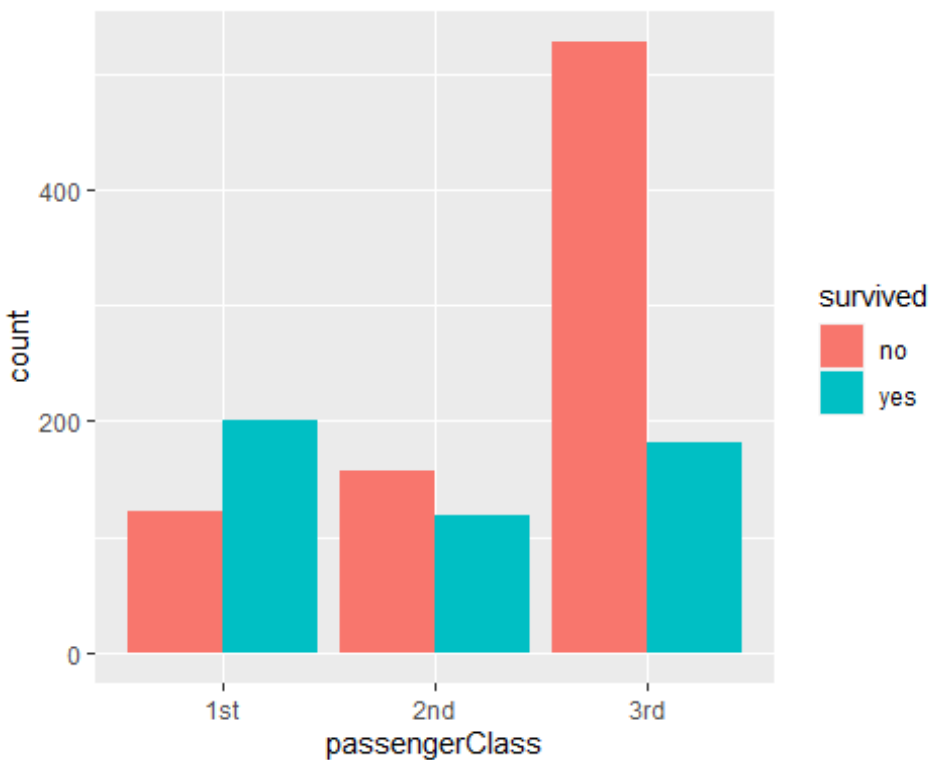
```
ggplot(ship_df,  
  mapping = aes(x = passengerClass)  
) + geom_bar()
```



```
ggplot(ship_df,  
mapping = aes(x = passengerClass, fill = survived)  
) + geom_bar()
```



```
ggplot(ship_df,
mapping = aes(x = passengerClass, fill = survived)
) + geom_bar(position = 'dodge')
```



```
carcar_df <-
read_csv("C:/Users/Nischal/Downloads/carprice.csv") %>% group_by(Type) %>%
summarise(price = mean(Price))

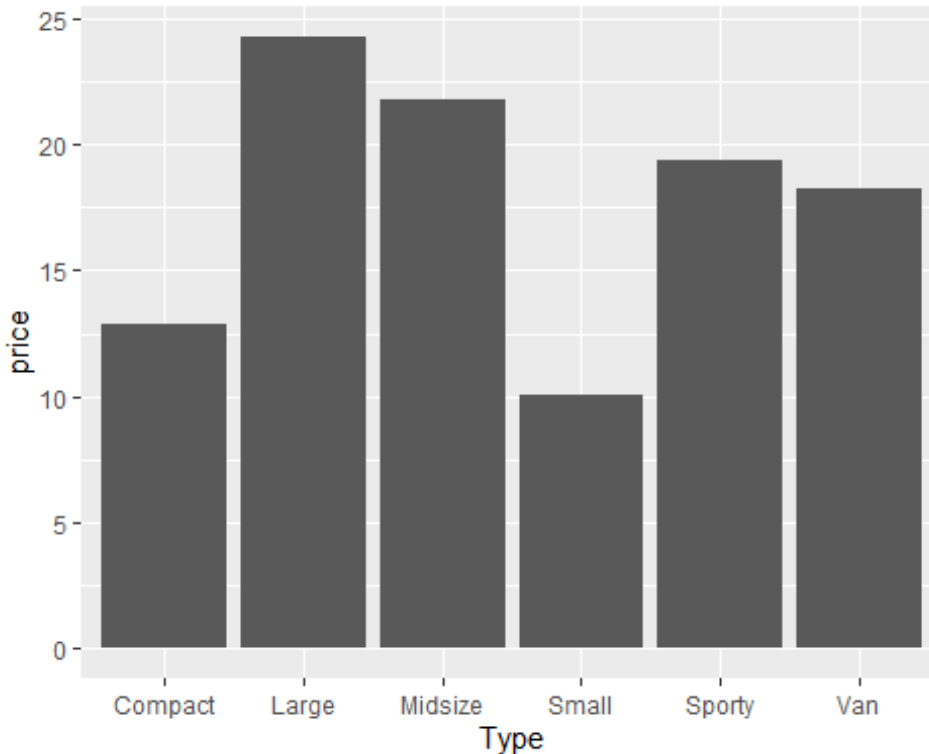
## New names:
## Rows: 48 Columns: 10
## — Column specification
## _____ Delimiter: ","
chr
## (1): Type dbl (9): ...1, Min.Price, Price, Max.Price, Range.Price,
RoughRange,
## gpm100,...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this
message.
## • `` -> `...1`

carcar_df

## # A tibble: 6 × 2
##   Type      price
##   <chr>    <dbl>
## 1 Compact  12.8
```

```
## 2 Large      24.3
## 3 Midsize    21.8
## 4 Small      10.0
## 5 Sporty     19.4
## 6 Van        18.3

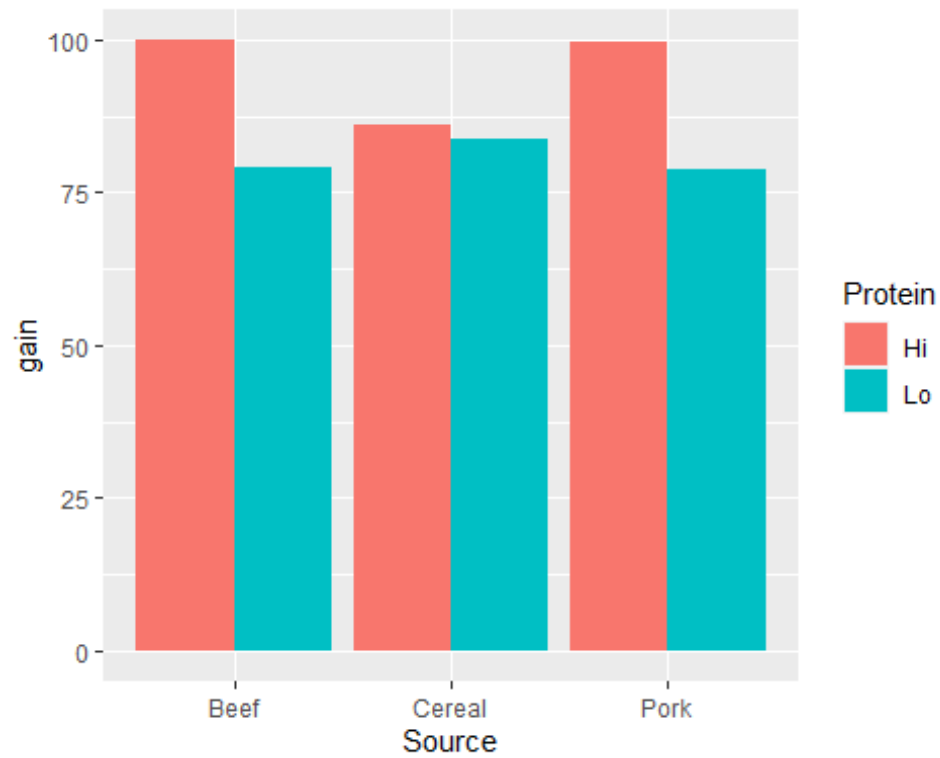
ggplot(carcar_df,
mapping = aes(x = Type, y = price)
) + geom_bar(stat = 'identity')
```



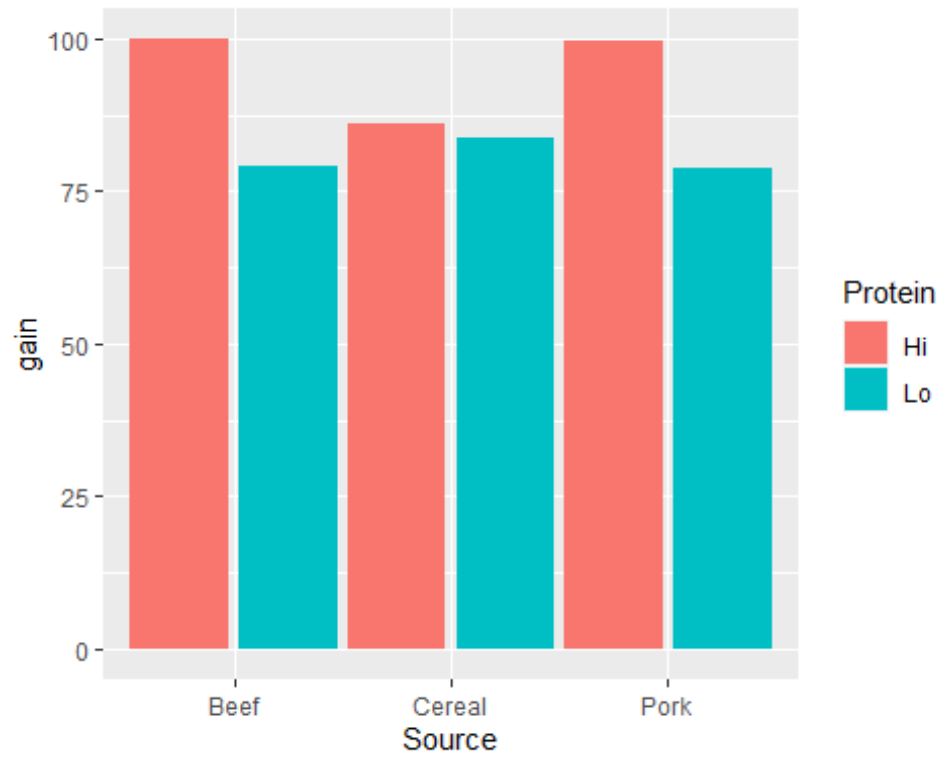
```
#this is the perfect way to group by low and high and join them together
fat_rats <- read_csv("C:/Users/Nischal/Downloads/FatRats.csv") %>%
  group_by(Protein, Source) %>%
  summarize(gain = mean(Gain), se = sd(Gain)/sqrt(n()))

## New names:
## Rows: 60 Columns: 4
## — Column specification
## _____ Delimiter: ","
chr
## (2): Protein, Source dbl (2): ...1, Gain
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this
message.
## `summarise()` has grouped output by 'Protein'. You can override using the
## `.groups` argument.
## • ` ` -> `...1`
```

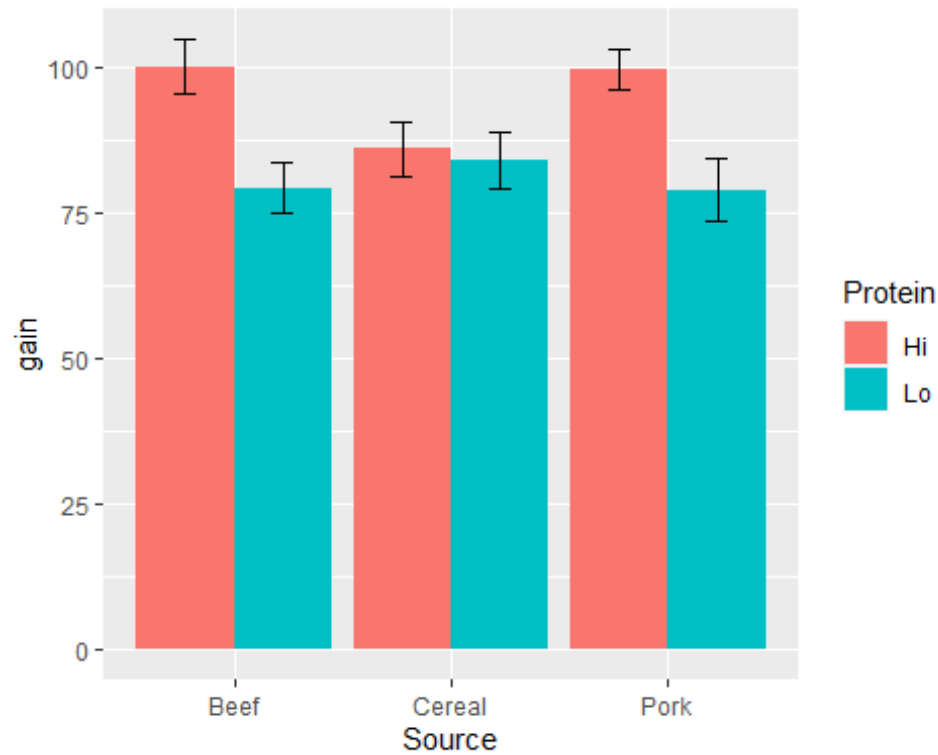
```
ggplot(fat_rats,
  mapping = aes(x = Source, y = gain, fill = Protein)
) + geom_bar(stat = 'identity', position = 'dodge')
```



```
ggplot(fat_rats,
  mapping = aes(x = Source, y = gain, fill = Protein)
) + geom_bar(stat = 'identity', position = position_dodge(width = 1.0))
```



```
ggplot(fat_rats,  
  mapping = aes(x = Source, y = gain, fill = Protein, ymin = gain - se, ymax =  
  gain + se)  
  ) + geom_bar(stat = 'identity', position = 'dodge') +  
  geom_errorbar(width = 0.2, position = position_dodge(width = 0.9))
```

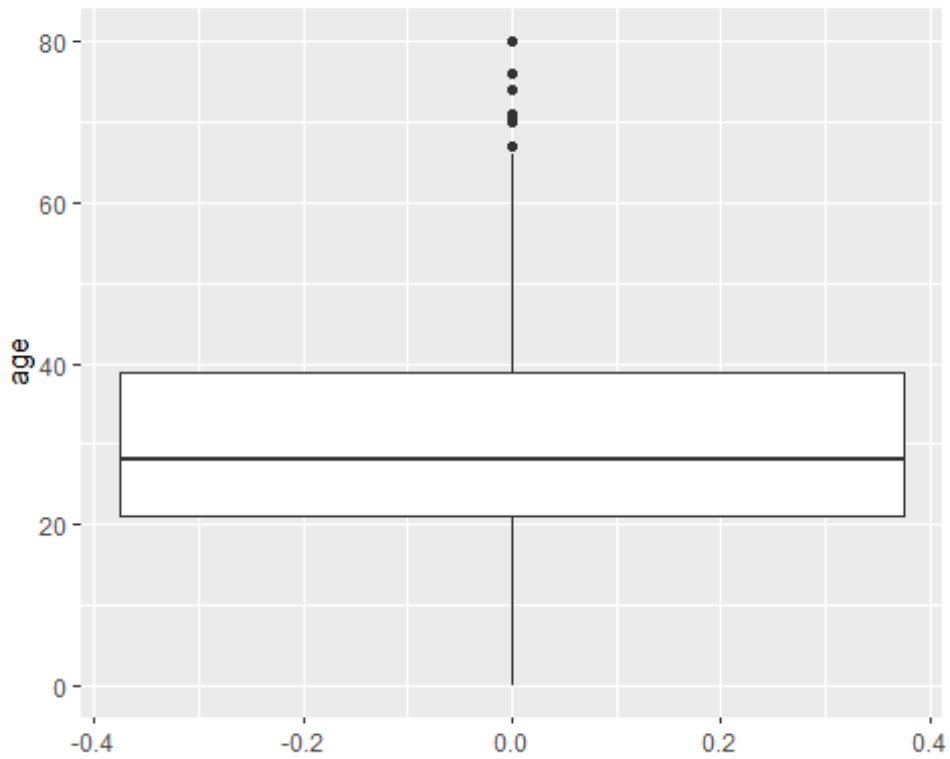


```
# we can add error bars to the barcharts  
#the function also can miniupluate the width of the cars
```

```
#Tukey barplots
```

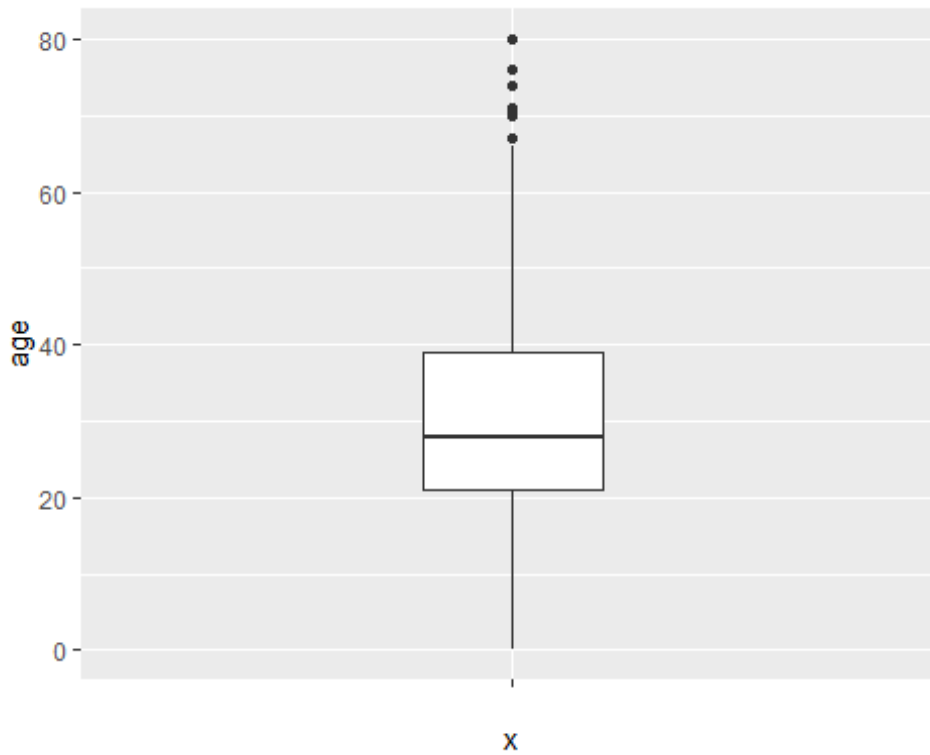
```
ggplot(ship_df,  
  mapping = aes(y = age)  
) + geom_boxplot()
```

```
## Warning: Removed 263 rows containing non-finite values (`stat_boxplot()`).
```

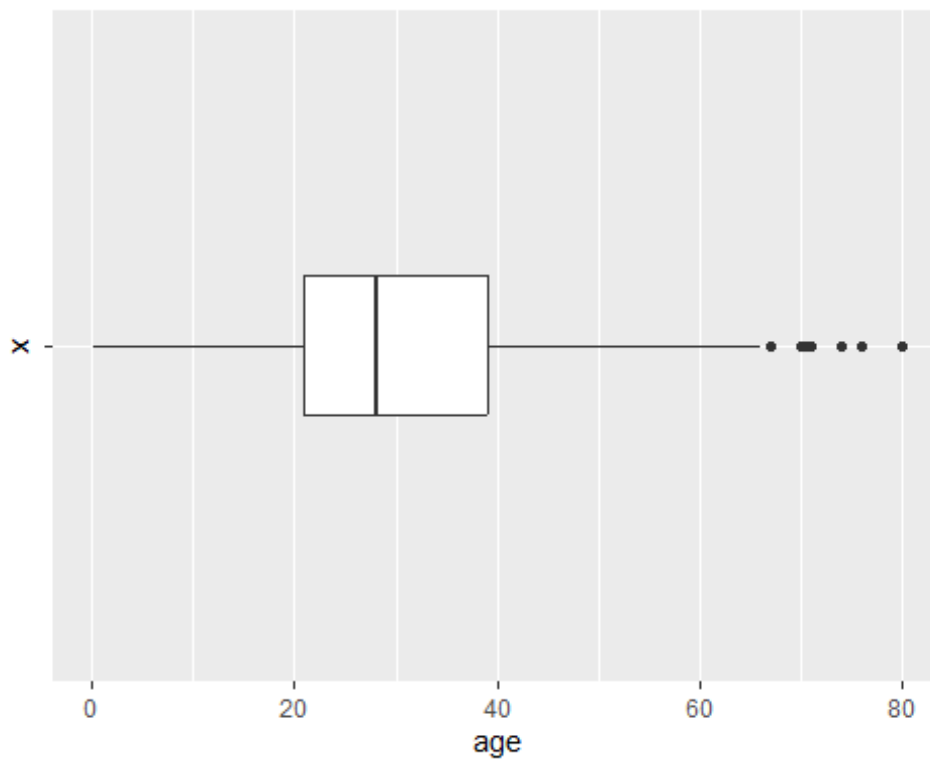
```
ggplot(ship_df,  
mapping = aes(x = '', y = age)  
) + geom_boxplot(width = 0.25)
```

```
## Warning: Removed 263 rows containing non-finite values (`stat_boxplot()`).
```



```
ggplot(ship_df,  
  mapping = aes(x = '', y = age)  
) + geom_boxplot(width = 0.25) +  
  coord_flip()
```

```
## Warning: Removed 263 rows containing non-finite values (`stat_boxplot()`).
```

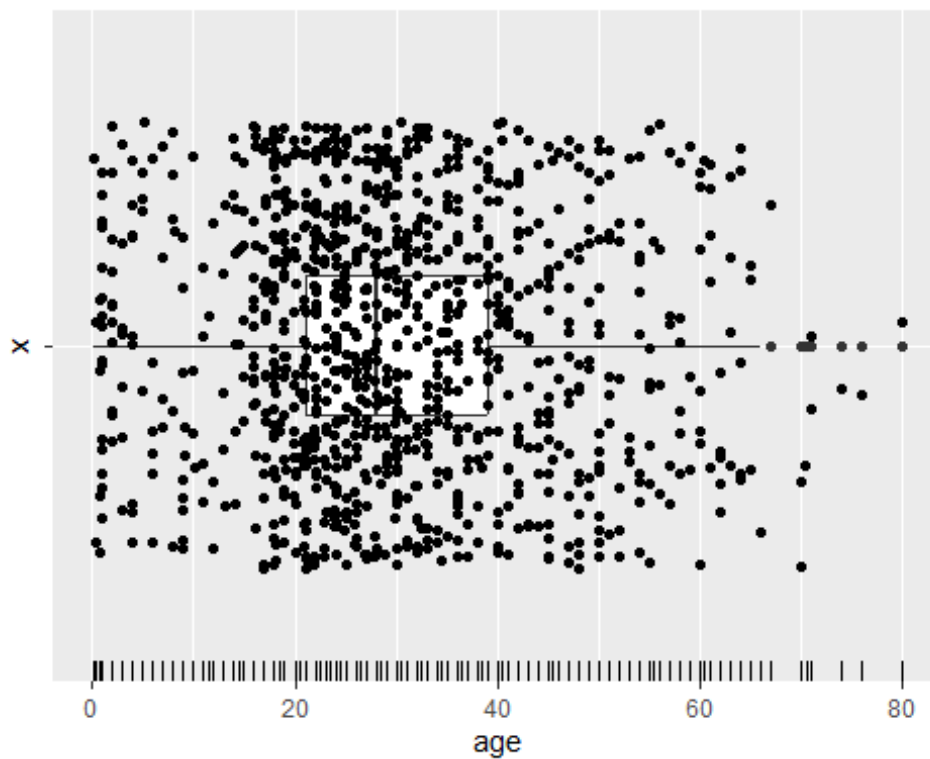


#this flips the plot

```
ggplot(ship_df,  
mapping = aes(x = '', y = age)  
) + geom_boxplot(width = 0.25) +  
coord_flip() +  
geom_jitter() +  
geom_rug(sides = 'l')
```

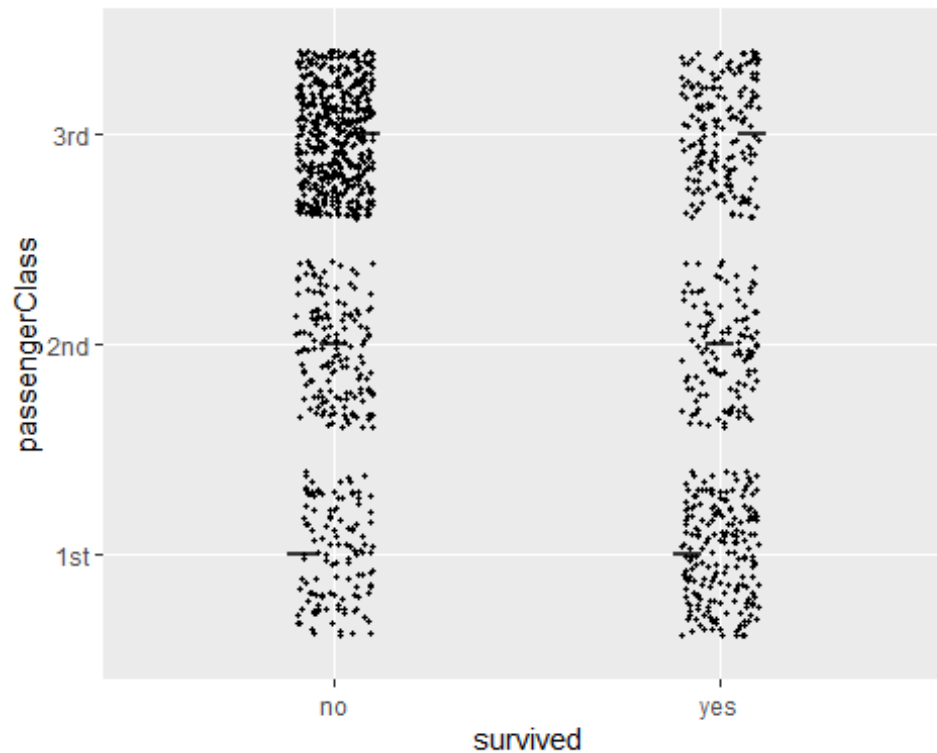
Warning: Removed 263 rows containing non-finite values (`stat\_boxplot()`).

Warning: Removed 263 rows containing missing values (`geom\_point()`).



#jitter shows all data, rug shows all the lines on the bottom plot, y axis now x because of the coord_flip

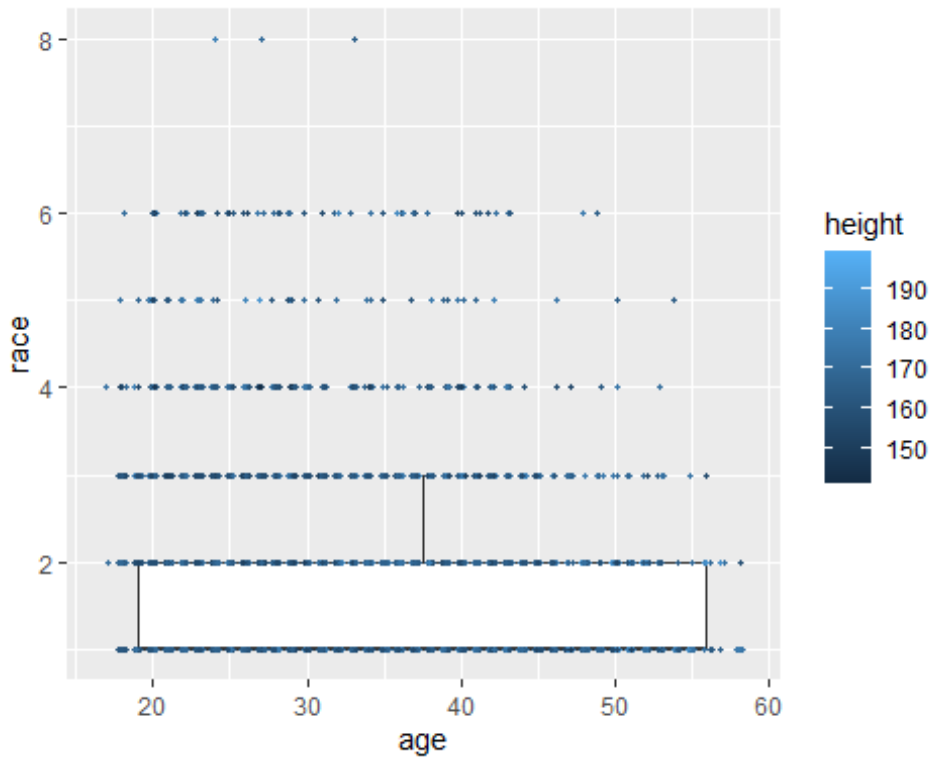
```
ggplot(ship_df,  
  mapping = aes(x = survived, y = passengerClass)  
) + geom_boxplot(width = 0.25, outlier.shape = NA) +  
  geom_jitter(width = 0.1, size = 0.75)
```



```
weight_df %>%
mutate(weighte = as.factor(age)) %>%
ggplot(mapping = aes(x = age, y = race, colour = height))
) + geom_boxplot(outlier.shape = NA, varwidth = T) +
geom_jitter(position = position_jitterdodge(0.5), size = 0.75)

## Warning: Continuous x aesthetic
## i did you forget `aes(group = ...)`?

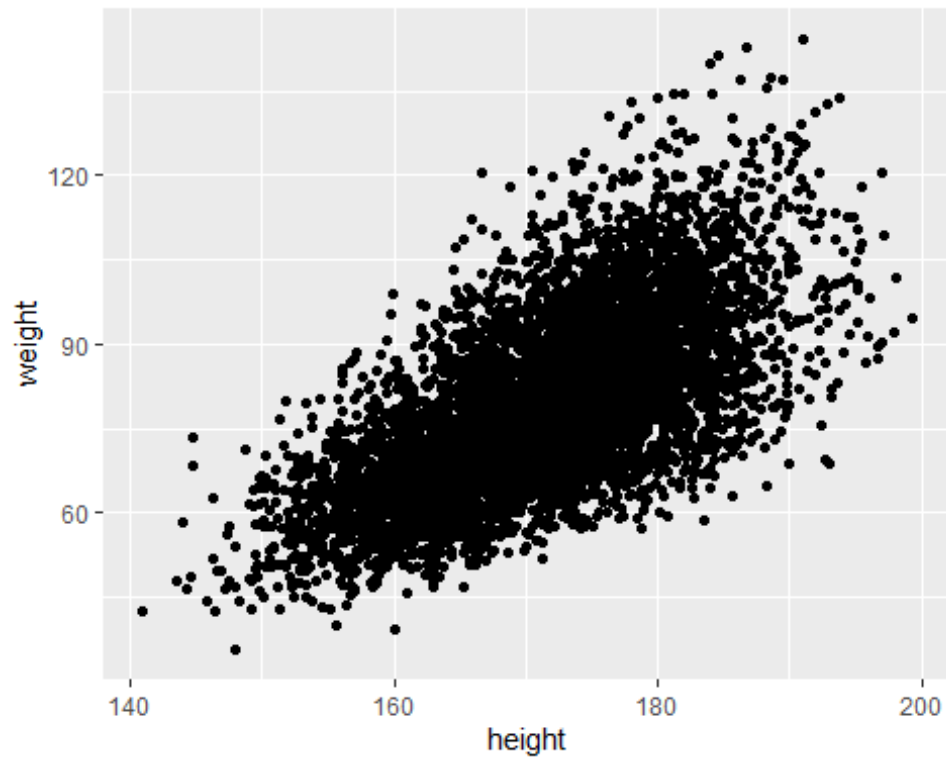
## Warning: The following aesthetics were dropped during statistical
transformation: colour
## i This can happen when ggplot fails to infer the correct grouping
structure in
## the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
## variable into a factor?
```



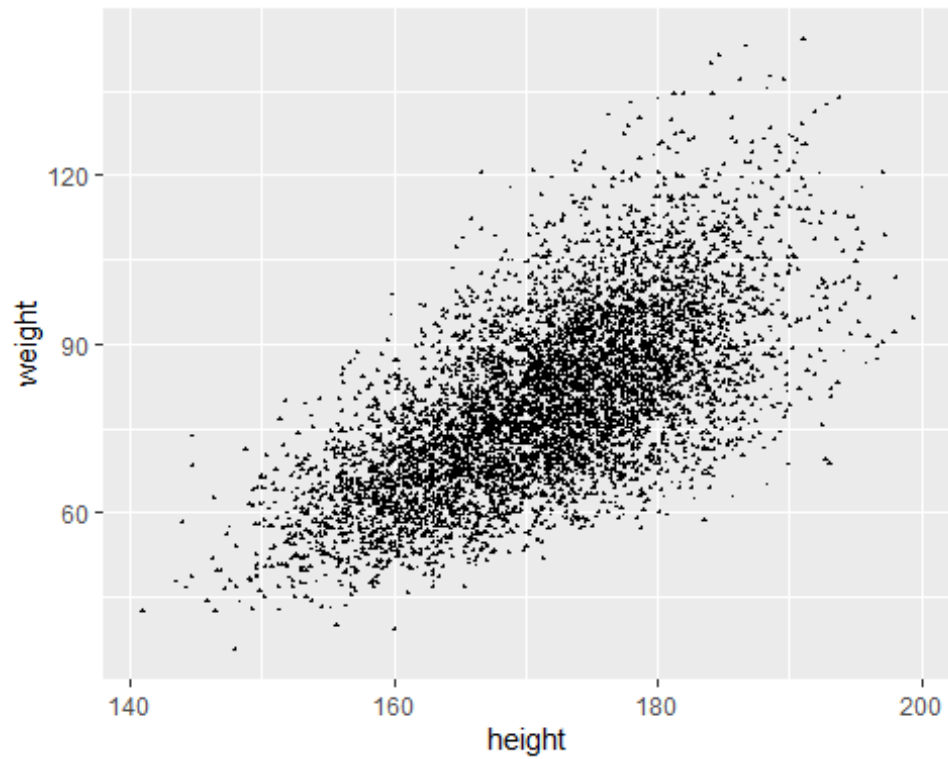
```
#
```

```
#scatterplots
```

```
ggplot(weight_df,  
mapping = aes(x=height, y=weight)  
) + geom_point()
```



```
ggplot(weight_df,  
mapping = aes(x=height, y=weight)  
) + geom_point(size =0.5, shape='triangle') #shape of the points are now  
triangle
```



```
ggplot(weight_df,  
mapping = aes(x=height, y=weight, colour = gender)  
) + geom_point(size = 0.5)
```




```
#adding marginal distributions
```

```
ggplot(weight_df,  
mapping = aes(x=height, y=weight, colour=gender)  
) +  
geom_point(size = 0.5) +  
geom_rug(alpha = 0.5, size = 1/10)
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



```
p <- ggplot(weight_df,  
mapping = aes(x=height, y=weight, colour=gender)  
) + geom_point(size = 0.5)
```

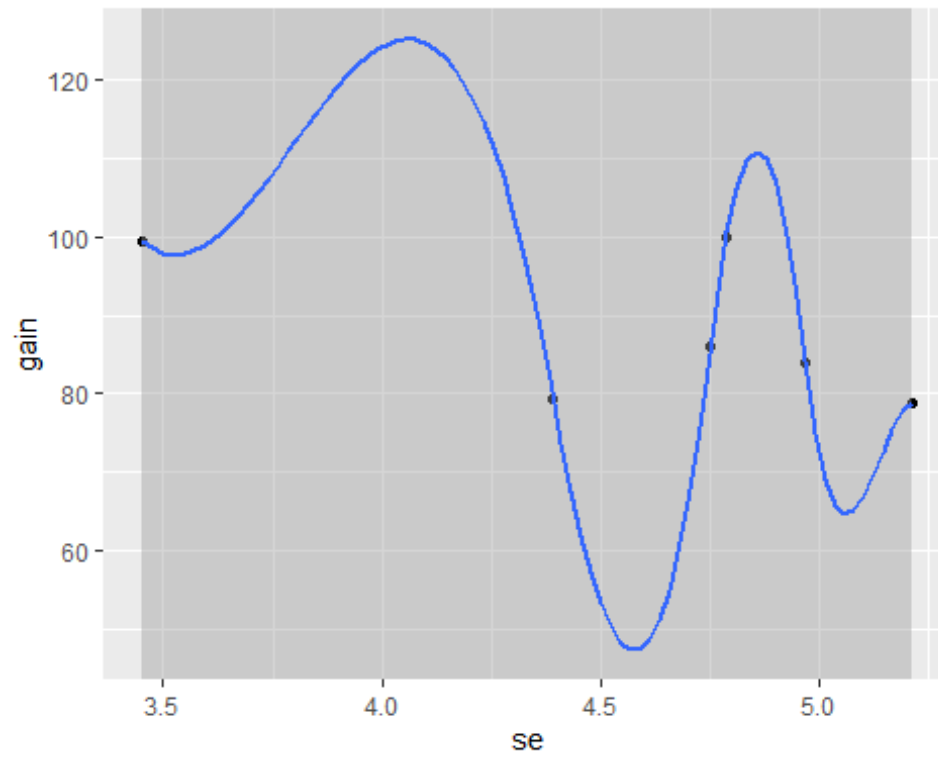
```
#ggMarginal(p)
```

```
#ggMarginal(p, type = 'histogram', groupColour = T, groupFill != T,  
#position = 'identity', alpha = 0.5)
```

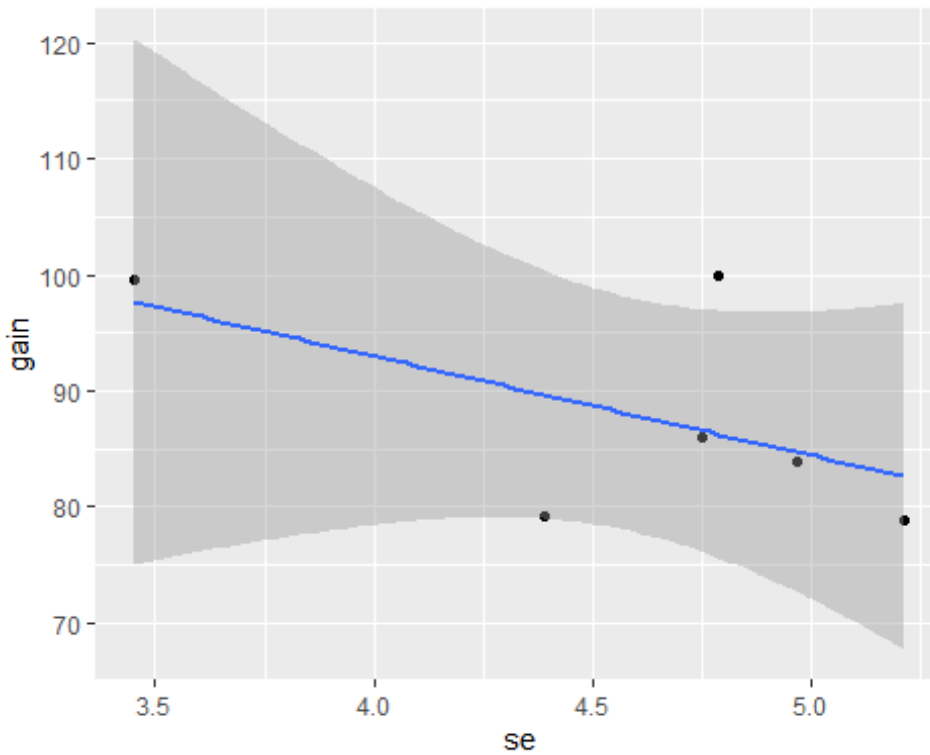
```
#this adds smoothing function
```

```
ggplot(fat_rats,  
mapping = aes(x = se, y = gain)  
) + geom_point() + geom_smooth()
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

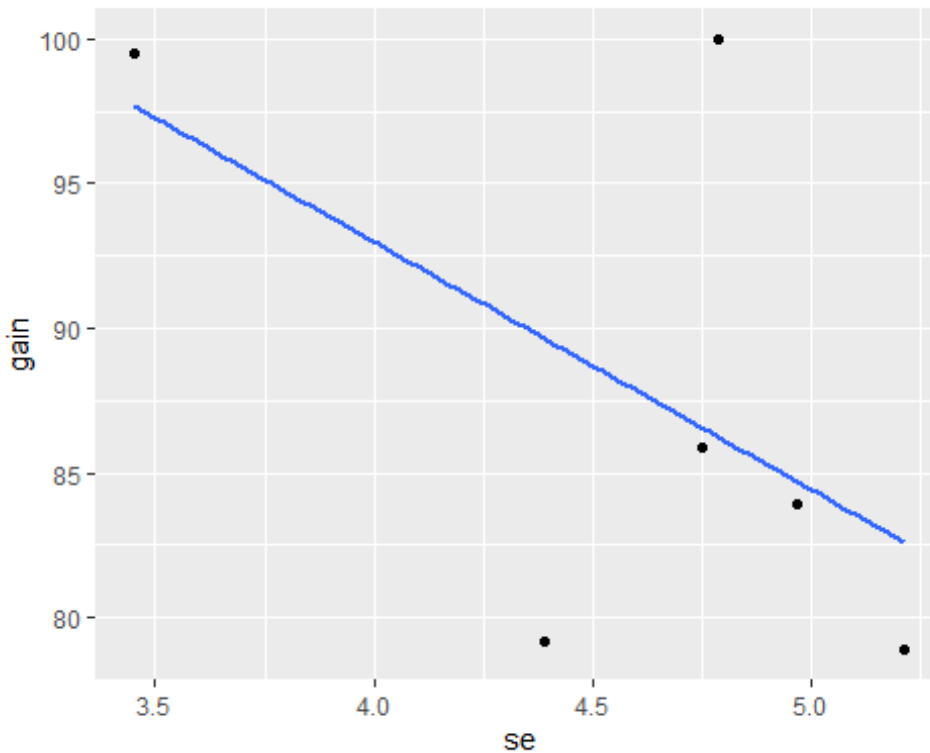


```
ggplot(fat_rats,  
  mapping = aes(x = se, y = gain)  
  ) + geom_point() + geom_smooth(method = 'lm')  
## `geom_smooth()` using formula = 'y ~ x'
```



#method lm adds a linear fit to the scatterplot

```
ggplot(fat_rats,  
  mapping = aes(x = se, y = gain)  
) + geom_point() + geom_smooth(method = 'lm', se = F)  
## `geom_smooth()` using formula = 'y ~ x'
```

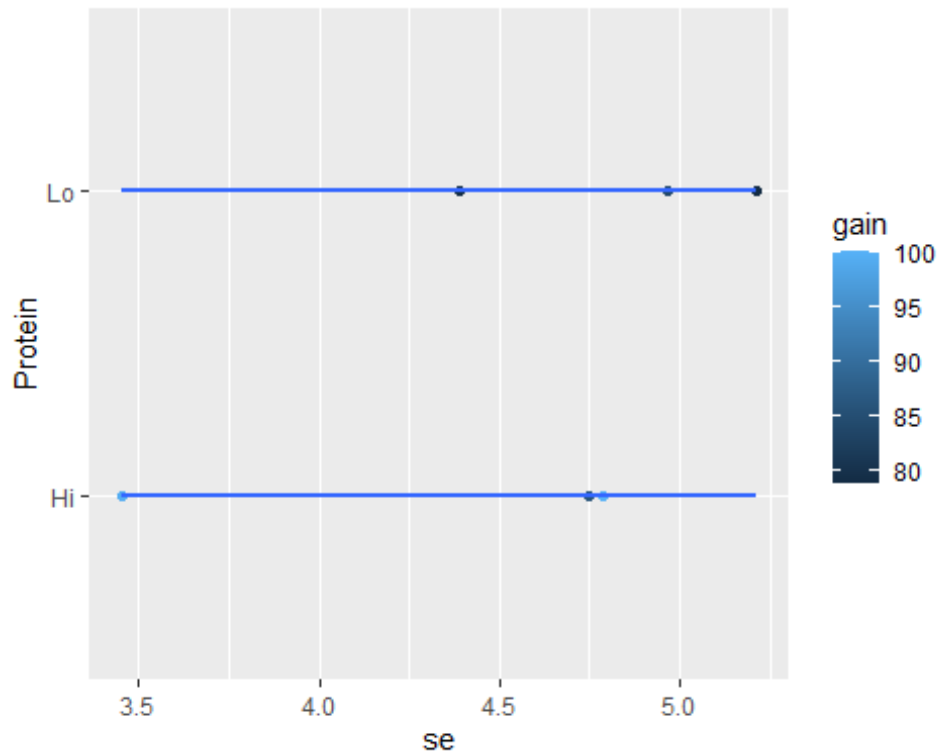


#se = f is turning off the standard error so no more clouded area cause our plot's x axis is standard error

```
ggplot(fat_rats,
mapping = aes(x = se, y = Protein, colour = gain)
) + geom_point() + geom_smooth(method = 'lm', se = F, fullrange = T)

## `geom_smooth()` using formula = 'y ~ x'

## Warning: The following aesthetics were dropped during statistical
transformation: colour
## i This can happen when ggplot fails to infer the correct grouping
structure in
## the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
## variable into a factor?
```



#this needs a different type of data set

#adding labels

```
car_df <- read_csv("C:/Users/Nischal/Downloads/carprice.csv")

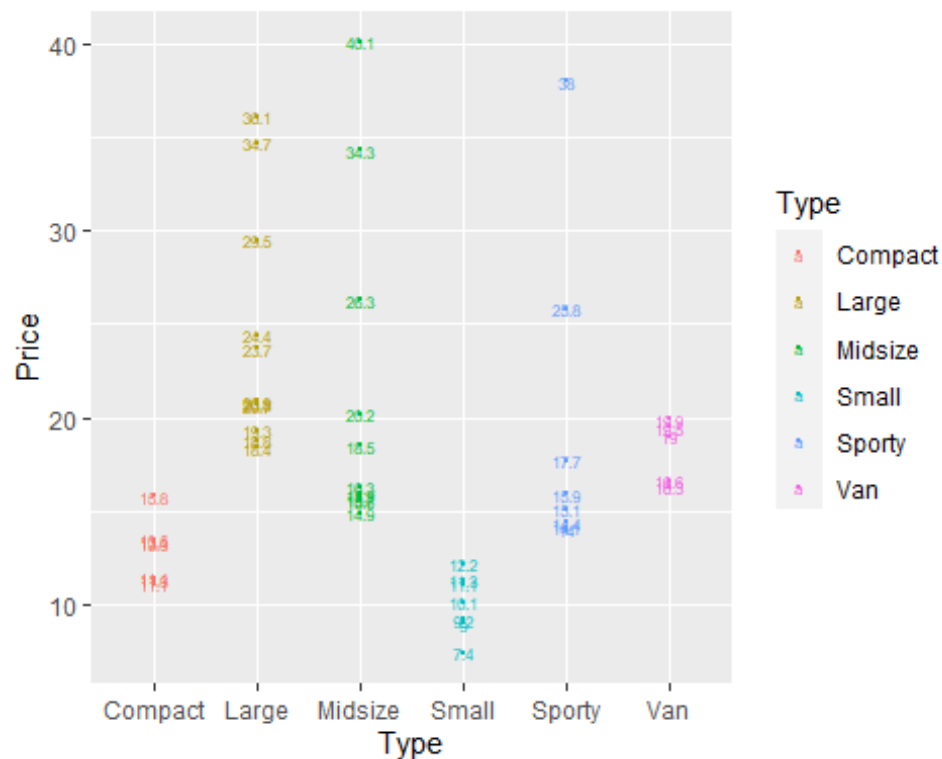
## New names:
## Rows: 48 Columns: 10
## — Column specification
## _____ Delimiter: ","
chr
## (1): Type dbl (9): ...1, Min.Price, Price, Max.Price, Range.Price,
RoughRange,
## gpm100,...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this
message.
## • `` -> `...1`

car_df

## # A tibble: 48 × 10
##   ...1 Type   Min.Price Price Max.Price Range.Price RoughRange gpm100
MPG.city
##   <dbl> <chr>   <dbl> <dbl>   <dbl>   <dbl>   <dbl> <dbl>
<dbl>
## 1     6 Midsi...   14.2  15.7    17.3     3.1     3.09   3.8
22
```

```
## 2      7 Large      19.9  20.8      21.7      1.8      1.79      4.2
19
## 3      8 Large      22.6  23.7      24.9      2.3      2.31      4.9
16
## 4      9 Midsi...    26.3  26.3      26.3      0      -0.01      4.3
19
## 5     10 Large      33      34.7      36.3      3.3      3.3      4.9
16
## 6     11 Midsi...    37.5  40.1      42.7      5.2      5.18      4.9
16
## 7     12 Compa...     8.5  13.4      18.3      9.8      9.8      3.3
25
## 8     13 Compa...    11.4  11.4      11.4      0      -0.01      3.4
25
## 9     14 Sporty     13.4  15.1      16.8      3.4      3.38      4.2
19
## 10    15 Midsi...    13.4  15.9      18.4      5      5.01      4
21
## # i 38 more rows
## # i 1 more variable: MPG.highway <dbl>
```

```
car_df %>%
  ggplot(
    mapping = aes(x = Type, y = Price, label = Price, colour = Type)
  ) + geom_point(size = 0.5) +
    geom_text(size = 2)
```

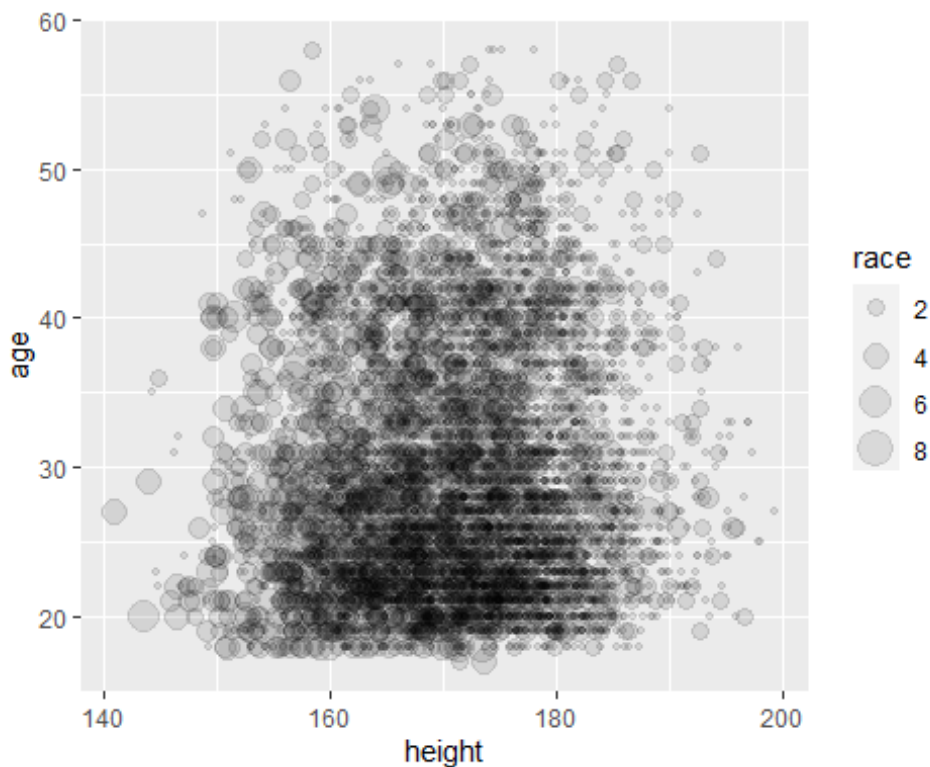


#ask TA about this code. I understand the variables im using wont make much sense but ifelse cannot be found

```
#car_df %>%  
#mutate(extreme = ifelse(Min.Price < 12 | Max.Price > 19, province, '')) %>%  
#ggplot(mapping = aes(x = Type, y = , Label = gpm100, colour = MPG.city)  
#) +  
#geom_point(size = 0.5) +  
#geom_label_repel(size = 2)
```

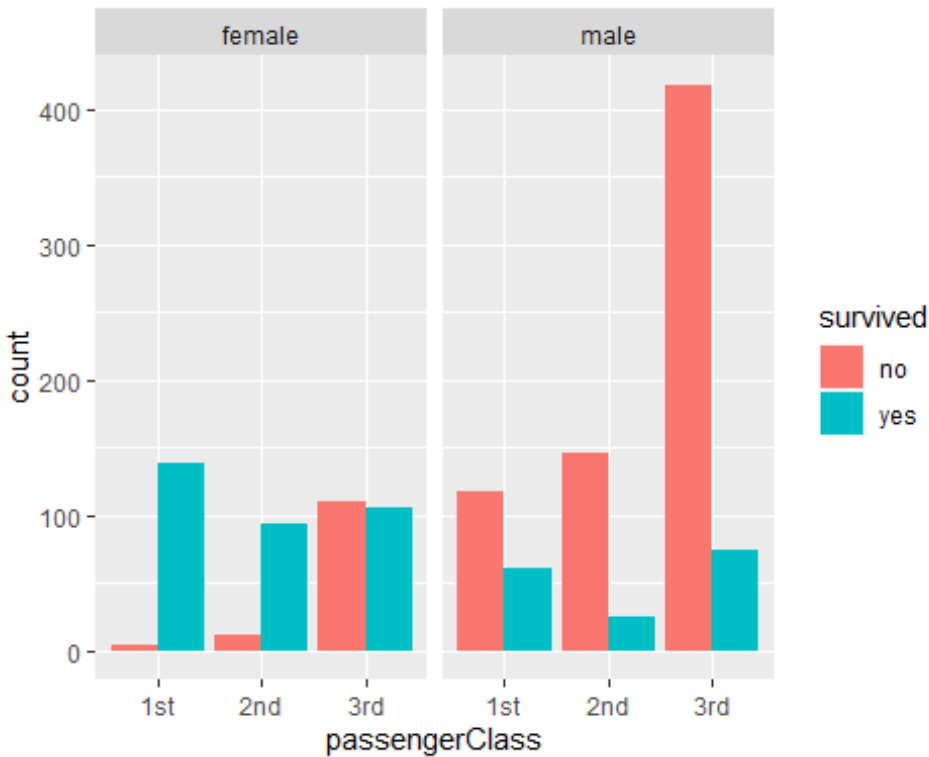
#Bubbleplot

```
weight_df %>%  
ggplot(mapping = aes(x = height, y = age, size = race))  
) +  
geom_point(alpha = 0.1)
```



#Facet plots

```
ggplot(ship_df,  
mapping = aes(x = passengerClass, fill = survived)  
) + geom_bar(position = 'dodge') +  
facet_wrap(~sex) #now grouped by sex
```

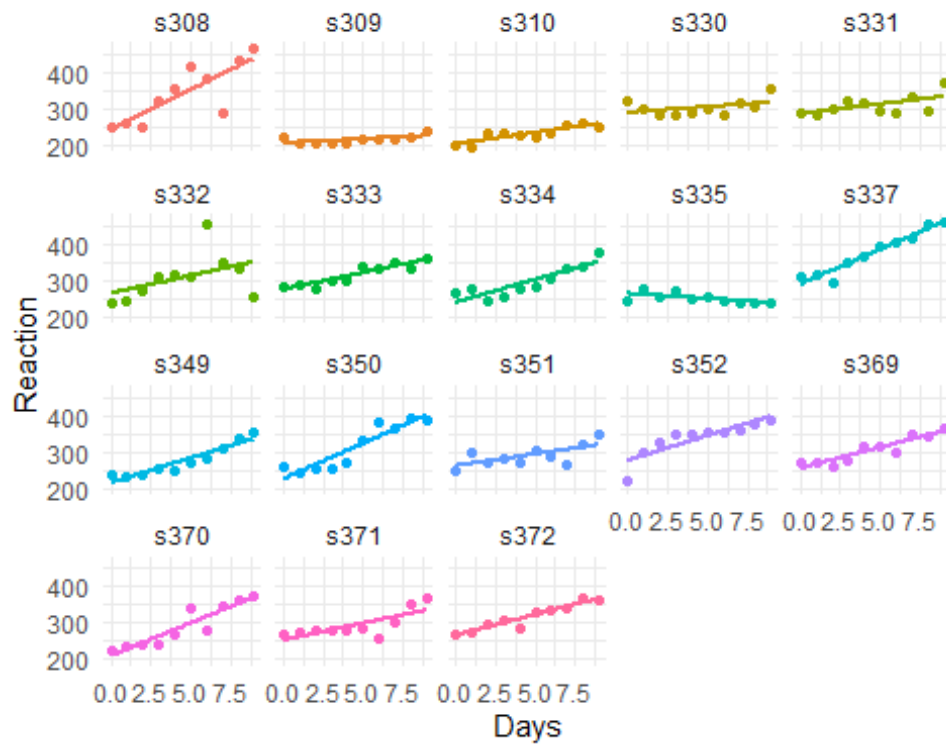


```
sleepstudy_df <- read_csv("C:/Users/Nischal/Downloads/sleepstudy.csv")

## Rows: 180 Columns: 3
## — Column specification
## Delimiter: ","
## chr (1): Subject
## dbl (2): Reaction, Days
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
message.

ggplot(sleepstudy_df,
  mapping = aes(x = Days, y = Reaction, colour = Subject)
) + geom_point() +
  geom_smooth(method = 'lm', se = F) +
  facet_wrap(~Subject) +
  theme_minimal() +
  theme(legend.position = 'none')

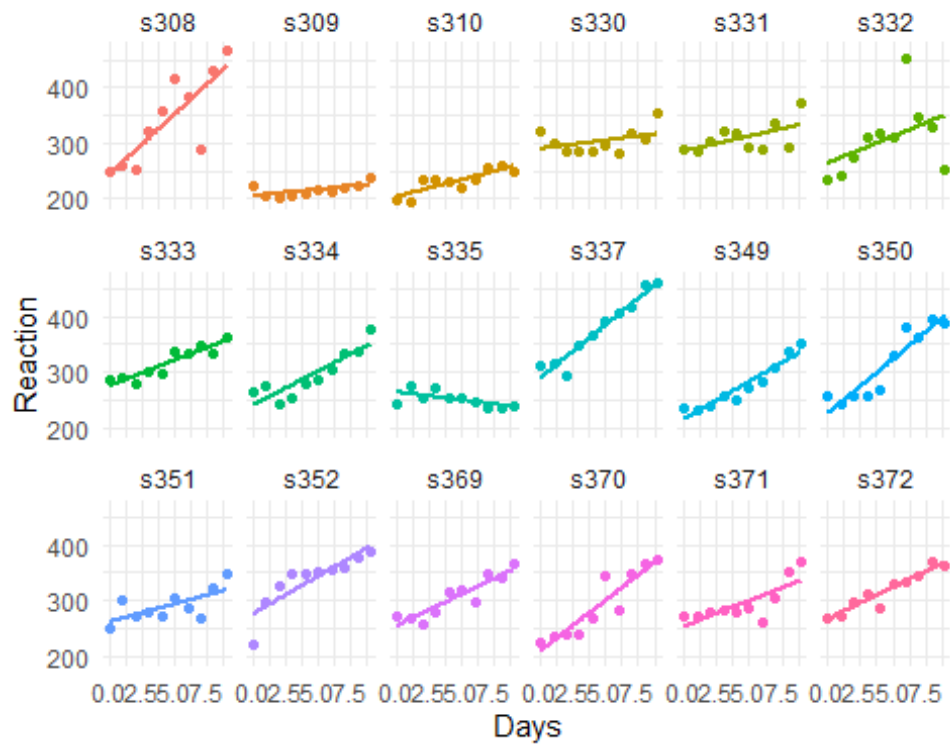
## `geom_smooth()` using formula = 'y ~ x'
```

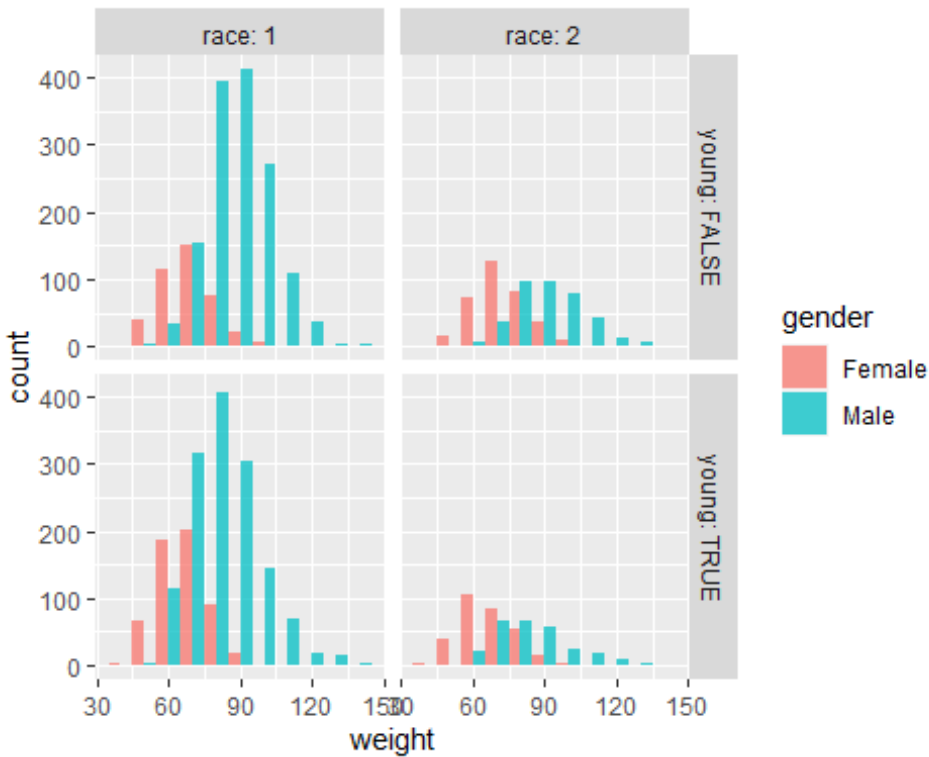
#this code chunk is very cool, basically allows best fit line for all 18 subjects in the experiment

```
ggplot(sleepstudy_df,
mapping = aes(x = Days, y = Reaction, colour = Subject)) + geom_point() +
geom_smooth(method = 'lm', se = F) +
facet_wrap(~Subject, nrow = 3) +
theme_minimal() +
theme(legend.position = 'none')

## `geom_smooth()` using formula = 'y ~ x'
```



```
weight_df %>%
  filter(race %in% c(1, 2)) %>%
  mutate(young = age < median(age),
         race = as.factor(race)) %>%
  ggplot(mapping = aes(x = weight, fill = gender)) +
  geom_histogram(binwidth = 10, position = 'dodge', alpha = 0.75) +
  facet_grid(young ~ race, labeller = label_both)
```



#we show histograms of weight for males and females who are either young (defined as below the median age) or not, and in one of two different racial categories

#——Chapter 5 HW —————

```
ship_df <- read.csv('C:/Users/Nischal/Downloads/titanicsurvival.csv')
ship_df
```

##	passengerClass	X	survived	sex	age
## 1	Allen, Miss. Elisabeth Walton	yes	female	29.0000	
1st					
## 2	Allison, Master. Hudson Trevor	yes	male	0.9167	
1st					
## 3	Allison, Miss. Helen Loraine	no	female	2.0000	
1st					
## 4	Allison, Mr. Hudson Joshua Crei	no	male	30.0000	
1st					
## 5	Allison, Mrs. Hudson J C (Bessi	no	female	25.0000	
1st					
## 6	Anderson, Mr. Harry	yes	male	48.0000	
1st					
## 7	Andrews, Miss. Kornelia Theodos	yes	female	63.0000	
1st					
## 8	Andrews, Mr. Thomas Jr	no	male	39.0000	
1st					

## 9	Appleton, Mrs. Edward Dale (Cha	yes female 53.0000
1st		
## 10	Artagaveytia, Mr. Ramon	no male 71.0000
1st		
## 11	Astor, Col. John Jacob	no male 47.0000
1st		
## 12	Astor, Mrs. John Jacob (Madelei	yes female 18.0000
1st		
## 13	Aubart, Mme. Leontine Pauline	yes female 24.0000
1st		
## 14	Barber, Miss. Ellen Nellie	yes female 26.0000
1st		
## 15	Barkworth, Mr. Algernon Henry W	yes male 80.0000
1st		
## 16	Baumann, Mr. John D	no male NA
1st		
## 17	Baxter, Mr. Quigg Edmond	no male 24.0000
1st		
## 18	Baxter, Mrs. James (Helene DeLa	yes female 50.0000
1st		
## 19	Bazzani, Miss. Albina	yes female 32.0000
1st		
## 20	Beattie, Mr. Thomson	no male 36.0000
1st		
## 21	Beckwith, Mr. Richard Leonard	yes male 37.0000
1st		
## 22	Beckwith, Mrs. Richard Leonard	yes female 47.0000
1st		
## 23	Behr, Mr. Karl Howell	yes male 26.0000
1st		
## 24	Bidois, Miss. Rosalie	yes female 42.0000
1st		
## 25	Bird, Miss. Ellen	yes female 29.0000
1st		
## 26	Birnbaum, Mr. Jakob	no male 25.0000
1st		
## 27	Bishop, Mr. Dickinson H	yes male 25.0000
1st		
## 28	Bishop, Mrs. Dickinson H (Helen	yes female 19.0000
1st		
## 29	Bissette, Miss. Amelia	yes female 35.0000
1st		
## 30	Bjornstrom-Steffansson, Mr. Mau	yes male 28.0000
1st		
## 31	Blackwell, Mr. Stephen Weart	no male 45.0000
1st		
## 32	Blank, Mr. Henry	yes male 40.0000
1st		
## 33	Bonnell, Miss. Caroline	yes female 30.0000
1st		

## 34	Bonnell, Miss. Elizabeth	yes female 58.0000
1st		
## 35	Borebank, Mr. John James	no male 42.0000
1st		
## 36	Bowen, Miss. Grace Scott	yes female 45.0000
1st		
## 37	Bowerman, Miss. Elsie Edith	yes female 22.0000
1st		
## 38	Bradley, Mr. George (George Ar	yes male NA
1st		
## 39	Brady, Mr. John Bertram	no male 41.0000
1st		
## 40	Brandeis, Mr. Emil	no male 48.0000
1st		
## 41	Brewe, Dr. Arthur Jackson	no male NA
1st		
## 42	Brown, Mrs. James Joseph (Marga	yes female 44.0000
1st		
## 43	Brown, Mrs. John Murray (Caroli	yes female 59.0000
1st		
## 44	Bucknell, Mrs. William Robert (	yes female 60.0000
1st		
## 45	Burns, Miss. Elizabeth Margaret	yes female 41.0000
1st		
## 46	Butt, Major. Archibald Willingh	no male 45.0000
1st		
## 47	Cairns, Mr. Alexander	no male NA
1st		
## 48	Calderhead, Mr. Edward Penningt	yes male 42.0000
1st		
## 49	Candee, Mrs. Edward (Helen Chur	yes female 53.0000
1st		
## 50	Cardeza, Mr. Thomas Drake Marti	yes male 36.0000
1st		
## 51	Cardeza, Mrs. James Warburton M	yes female 58.0000
1st		
## 52	Carlsson, Mr. Frans Olof	no male 33.0000
1st		
## 53	Carrau, Mr. Francisco M	no male 28.0000
1st		
## 54	Carrau, Mr. Jose Pedro	no male 17.0000
1st		
## 55	Carter, Master. William Thornto	yes male 11.0000
1st		
## 56	Carter, Miss. Lucile Polk	yes female 14.0000
1st		
## 57	Carter, Mr. William Ernest	yes male 36.0000
1st		
## 58	Carter, Mrs. William Ernest (Lu	yes female 36.0000
1st		

## 59	Case, Mr. Howard Brown	no	male	49.0000
1st				
## 60	Cassebeer, Mrs. Henry Arthur Jr	yes	female	NA
1st				
## 61	Cavendish, Mr. Tyrell William	no	male	36.0000
1st				
## 62	Cavendish, Mrs. Tyrell William	yes	female	76.0000
1st				
## 63	Chaffee, Mr. Herbert Fuller	no	male	46.0000
1st				
## 64	Chaffee, Mrs. Herbert Fuller (C	yes	female	47.0000
1st				
## 65	Chambers, Mr. Norman Campbell	yes	male	27.0000
1st				
## 66	Chambers, Mrs. Norman Campbell	yes	female	33.0000
1st				
## 67	Chaudanson, Miss. Victorine	yes	female	36.0000
1st				
## 68	Cherry, Miss. Gladys	yes	female	30.0000
1st				
## 69	Chevre, Mr. Paul Romaine	yes	male	45.0000
1st				
## 70	Chibnall, Mrs. (Edith Martha Bo	yes	female	NA
1st				
## 71	Chisholm, Mr. Roderick Robert C	no	male	NA
1st				
## 72	Clark, Mr. Walter Miller	no	male	27.0000
1st				
## 73	Clark, Mrs. Walter Miller (Virg	yes	female	26.0000
1st				
## 74	Cleaver, Miss. Alice	yes	female	22.0000
1st				
## 75	Clifford, Mr. George Quincy	no	male	NA
1st				
## 76	Colley, Mr. Edward Pomeroy	no	male	47.0000
1st				
## 77	Compton, Miss. Sara Rebecca	yes	female	39.0000
1st				
## 78	Compton, Mr. Alexander Taylor J	no	male	37.0000
1st				
## 79	Compton, Mrs. Alexander Taylor	yes	female	64.0000
1st				
## 80	Cornell, Mrs. Robert Clifford (	yes	female	55.0000
1st				
## 81	Crafton, Mr. John Bertram	no	male	NA
1st				
## 82	Crosby, Capt. Edward Gifford	no	male	70.0000
1st				
## 83	Crosby, Miss. Harriet R	yes	female	36.0000
1st				

## 84	Crosby, Mrs. Edward Gifford (Ca	yes female 64.0000
1st		
## 85	Cumings, Mr. John Bradley	no male 39.0000
1st		
## 86	Cumings, Mrs. John Bradley (Flo	yes female 38.0000
1st		
## 87	Daly, Mr. Peter Denis	yes male 51.0000
1st		
## 88	Daniel, Mr. Robert Williams	yes male 27.0000
1st		
## 89	Daniels, Miss. Sarah	yes female 33.0000
1st		
## 90	Davidson, Mr. Thornton	no male 31.0000
1st		
## 91	Davidson, Mrs. Thornton (Orian	yes female 27.0000
1st		
## 92	Dick, Mr. Albert Adrian	yes male 31.0000
1st		
## 93	Dick, Mrs. Albert Adrian (Vera	yes female 17.0000
1st		
## 94	Dodge, Dr. Washington	yes male 53.0000
1st		
## 95	Dodge, Master. Washington	yes male 4.0000
1st		
## 96	Dodge, Mrs. Washington (Ruth Vi	yes female 54.0000
1st		
## 97	Douglas, Mr. Walter Donald	no male 50.0000
1st		
## 98	Douglas, Mrs. Frederick Charles	yes female 27.0000
1st		
## 99	Douglas, Mrs. Walter Donald (Ma	yes female 48.0000
1st		
## 100	Duff Gordon, Lady. (Lucille Chr	yes female 48.0000
1st		
## 101	Duff Gordon, Sir. Cosmo Edmund	yes male 49.0000
1st		
## 102	Dulles, Mr. William Crothers	no male 39.0000
1st		
## 103	Earnshaw, Mrs. Boulton (Olive P	yes female 23.0000
1st		
## 104	Endres, Miss. Caroline Louise	yes female 38.0000
1st		
## 105	Eustis, Miss. Elizabeth Mussey	yes female 54.0000
1st		
## 106	Evans, Miss. Edith Corse	no female 36.0000
1st		
## 107	Farthing, Mr. John	no male NA
1st		
## 108	Flegenheim, Mrs. Alfred (Antoin	yes female NA
1st		

## 109	Fleming, Miss. Margaret	yes female	NA
1st			
## 110	Flynn, Mr. John Irwin (Irving	yes male	36.0000
1st			
## 111	Foreman, Mr. Benjamin Laventall	no male	30.0000
1st			
## 112	Fortune, Miss. Alice Elizabeth	yes female	24.0000
1st			
## 113	Fortune, Miss. Ethel Flora	yes female	28.0000
1st			
## 114	Fortune, Miss. Mabel Helen	yes female	23.0000
1st			
## 115	Fortune, Mr. Charles Alexander	no male	19.0000
1st			
## 116	Fortune, Mr. Mark	no male	64.0000
1st			
## 117	Fortune, Mrs. Mark (Mary McDoug	yes female	60.0000
1st			
## 118	Francatelli, Miss. Laura Mabel	yes female	30.0000
1st			
## 119	Franklin, Mr. Thomas Parham	no male	NA
1st			
## 120	Frauenthal, Dr. Henry William	yes male	50.0000
1st			
## 121	Frauenthal, Mr. Isaac Gerald	yes male	43.0000
1st			
## 122	Frauenthal, Mrs. Henry William	yes female	NA
1st			
## 123	Frolicher, Miss. Hedwig Margari	yes female	22.0000
1st			
## 124	Frolicher-Stehli, Mr. Maxmillia	yes male	60.0000
1st			
## 125	Frolicher-Stehli, Mrs. Maxmilli	yes female	48.0000
1st			
## 126	Fry, Mr. Richard	no male	NA
1st			
## 127	Futrelle, Mr. Jacques Heath	no male	37.0000
1st			
## 128	Futrelle, Mrs. Jacques Heath (L	yes female	35.0000
1st			
## 129	Gee, Mr. Arthur H	no male	47.0000
1st			
## 130	Geiger, Miss. Amalie	yes female	35.0000
1st			
## 131	Gibson, Miss. Dorothy Winifred	yes female	22.0000
1st			
## 132	Gibson, Mrs. Leonard (Pauline C	yes female	45.0000
1st			
## 133	Giglio, Mr. Victor	no male	24.0000
1st			

## 134	Goldenberg, Mr. Samuel L	yes	male	49.0000
1st				
## 135	Goldenberg, Mrs. Samuel L (Edwi	yes	female	NA
1st				
## 136	Goldschmidt, Mr. George B	no	male	71.0000
1st				
## 137	Gracie, Col. Archibald IV	yes	male	53.0000
1st				
## 138	Graham, Miss. Margaret Edith	yes	female	19.0000
1st				
## 139	Graham, Mr. George Edward	no	male	38.0000
1st				
## 140	Graham, Mrs. William Thompson (	yes	female	58.0000
1st				
## 141	Greenfield, Mr. William Bertram	yes	male	23.0000
1st				
## 142	Greenfield, Mrs. Leo David (Bla	yes	female	45.0000
1st				
## 143	Guggenheim, Mr. Benjamin	no	male	46.0000
1st				
## 144	Harder, Mr. George Achilles	yes	male	25.0000
1st				
## 145	Harder, Mrs. George Achilles (D	yes	female	25.0000
1st				
## 146	Harper, Mr. Henry Sleeper	yes	male	48.0000
1st				
## 147	Harper, Mrs. Henry Sleeper (Myn	yes	female	49.0000
1st				
## 148	Harrington, Mr. Charles H	no	male	NA
1st				
## 149	Harris, Mr. Henry Birkhardt	no	male	45.0000
1st				
## 150	Harris, Mrs. Henry Birkhardt (I	yes	female	35.0000
1st				
## 151	Harrison, Mr. William	no	male	40.0000
1st				
## 152	Hassab, Mr. Hammad	yes	male	27.0000
1st				
## 153	Hawksford, Mr. Walter James	yes	male	NA
1st				
## 154	Hays, Miss. Margaret Bechstein	yes	female	24.0000
1st				
## 155	Hays, Mr. Charles Melville	no	male	55.0000
1st				
## 156	Hays, Mrs. Charles Melville (Cl	yes	female	52.0000
1st				
## 157	Head, Mr. Christopher	no	male	42.0000
1st				
## 158	Hilliard, Mr. Herbert Henry	no	male	NA
1st				

## 159	Hipkins, Mr. William Edward	no	male	55.0000
1st				
## 160	Hippach, Miss. Jean Gertrude	yes	female	16.0000
1st				
## 161	Hippach, Mrs. Louis Albert (Ida	yes	female	44.0000
1st				
## 162	Hogeboom, Mrs. John C (Anna And	yes	female	51.0000
1st				
## 163	Holverson, Mr. Alexander Oskar	no	male	42.0000
1st				
## 164	Holverson, Mrs. Alexander Oskar	yes	female	35.0000
1st				
## 165	Homer, Mr. Harry (Mr E Haven)	yes	male	35.0000
1st				
## 166	Hoyt, Mr. Frederick Maxfield	yes	male	38.0000
1st				
## 167	Hoyt, Mr. William Fisher	no	male	NA
1st				
## 168	Hoyt, Mrs. Frederick Maxfield (	yes	female	35.0000
1st				
## 169	Icard, Miss. Amelie	yes	female	38.0000
1st				
## 170	Isham, Miss. Ann Elizabeth	no	female	50.0000
1st				
## 171	Ismay, Mr. Joseph Bruce	yes	male	49.0000
1st				
## 172	Jones, Mr. Charles Cresson	no	male	46.0000
1st				
## 173	Julian, Mr. Henry Forbes	no	male	50.0000
1st				
## 174	Keeping, Mr. Edwin	no	male	32.5000
1st				
## 175	Kent, Mr. Edward Austin	no	male	58.0000
1st				
## 176	Kenyon, Mr. Frederick R	no	male	41.0000
1st				
## 177	Kenyon, Mrs. Frederick R (Mario	yes	female	NA
1st				
## 178	Kimball, Mr. Edwin Nelson Jr	yes	male	42.0000
1st				
## 179	Kimball, Mrs. Edwin Nelson Jr (	yes	female	45.0000
1st				
## 180	Klaber, Mr. Herman	no	male	NA
1st				
## 181	Kreuchen, Miss. Emilie	yes	female	39.0000
1st				
## 182	Leader, Dr. Alice (Farnham)	yes	female	49.0000
1st				
## 183	LeRoy, Miss. Bertha	yes	female	30.0000
1st				

## 184	Lesurer, Mr. Gustave J	yes	male	35.0000
1st				
## 185	Lewy, Mr. Ervin G	no	male	NA
1st				
## 186	Lindeberg-Lind, Mr. Erik Gustaf	no	male	42.0000
1st				
## 187	Lindstrom, Mrs. Carl Johan (Sig	yes	female	55.0000
1st				
## 188	Lines, Miss. Mary Conover	yes	female	16.0000
1st				
## 189	Lines, Mrs. Ernest H (Elizabeth	yes	female	51.0000
1st				
## 190	Long, Mr. Milton Clyde	no	male	29.0000
1st				
## 191	Longley, Miss. Gretchen Fiske	yes	female	21.0000
1st				
## 192	Loring, Mr. Joseph Holland	no	male	30.0000
1st				
## 193	Lurette, Miss. Elise	yes	female	58.0000
1st				
## 194	Madill, Miss. Georgette Alexand	yes	female	15.0000
1st				
## 195	Maguire, Mr. John Edward	no	male	30.0000
1st				
## 196	Maioni, Miss. Roberta	yes	female	16.0000
1st				
## 197	Marechal, Mr. Pierre	yes	male	NA
1st				
## 198	Marvin, Mr. Daniel Warner	no	male	19.0000
1st				
## 199	Marvin, Mrs. Daniel Warner (Mar	yes	female	18.0000
1st				
## 200	Mayne, Mlle. Berthe Antonine (	yes	female	24.0000
1st				
## 201	McCaffry, Mr. Thomas Francis	no	male	46.0000
1st				
## 202	McCarthy, Mr. Timothy J	no	male	54.0000
1st				
## 203	McGough, Mr. James Robert	yes	male	36.0000
1st				
## 204	Meyer, Mr. Edgar Joseph	no	male	28.0000
1st				
## 205	Meyer, Mrs. Edgar Joseph (Leila	yes	female	NA
1st				
## 206	Millet, Mr. Francis Davis	no	male	65.0000
1st				
## 207	Minahan, Dr. William Edward	no	male	44.0000
1st				
## 208	Minahan, Miss. Daisy E	yes	female	33.0000
1st				

## 209	Minahan, Mrs. William Edward (L	yes female 37.0000
1st		
## 210	Mock, Mr. Philipp Edmund	yes male 30.0000
1st		
## 211	Molson, Mr. Harry Markland	no male 55.0000
1st		
## 212	Moore, Mr. Clarence Bloomfield	no male 47.0000
1st		
## 213	Natsch, Mr. Charles H	no male 37.0000
1st		
## 214	Newell, Miss. Madeleine	yes female 31.0000
1st		
## 215	Newell, Miss. Marjorie	yes female 23.0000
1st		
## 216	Newell, Mr. Arthur Webster	no male 58.0000
1st		
## 217	Newsom, Miss. Helen Monypeny	yes female 19.0000
1st		
## 218	Nicholson, Mr. Arthur Ernest	no male 64.0000
1st		
## 219	Oliva y Ocana, Dona. Fermina	yes female 39.0000
1st		
## 220	Omont, Mr. Alfred Fernand	yes male NA
1st		
## 221	Ostby, Miss. Helene Ragnhild	yes female 22.0000
1st		
## 222	Ostby, Mr. Engelhart Cornelius	no male 65.0000
1st		
## 223	Ovies y Rodriguez, Mr. Servando	no male 28.5000
1st		
## 224	Parr, Mr. William Henry Marsh	no male NA
1st		
## 225	Partner, Mr. Austen	no male 45.5000
1st		
## 226	Payne, Mr. Vivian Ponsonby	no male 23.0000
1st		
## 227	Pears, Mr. Thomas Clinton	no male 29.0000
1st		
## 228	Pears, Mrs. Thomas (Edith Wearn	yes female 22.0000
1st		
## 229	Penasco y Castellana, Mr. Victo	no male 18.0000
1st		
## 230	Penasco y Castellana, Mrs. Vict	yes female 17.0000
1st		
## 231	Perreault, Miss. Anne	yes female 30.0000
1st		
## 232	Peuchen, Major. Arthur Godfrey	yes male 52.0000
1st		
## 233	Porter, Mr. Walter Chamberlain	no male 47.0000
1st		

## 234	Potter, Mrs. Thomas Jr (Lily Al	yes	female	56.0000
1st				
## 235	Reuchlin, Jonkheer. John George	no	male	38.0000
1st				
## 236	Rheims, Mr. George Alexander Lu	yes	male	NA
1st				
## 237	Ringhini, Mr. Sante	no	male	22.0000
1st				
## 238	Robbins, Mr. Victor	no	male	NA
1st				
## 239	Robert, Mrs. Edward Scott (Elis	yes	female	43.0000
1st				
## 240	Roebbling, Mr. Washington August	no	male	31.0000
1st				
## 241	Romaine, Mr. Charles Hallace (	yes	male	45.0000
1st				
## 242	Rood, Mr. Hugh Roscoe	no	male	NA
1st				
## 243	Rosenbaum, Miss. Edith Louise	yes	female	33.0000
1st				
## 244	Rosenshine, Mr. George (Mr Geo	no	male	46.0000
1st				
## 245	Ross, Mr. John Hugo	no	male	36.0000
1st				
## 246	Roths, the Countess. of (Lucy	yes	female	33.0000
1st				
## 247	Rothschild, Mr. Martin	no	male	55.0000
1st				
## 248	Rothschild, Mrs. Martin (Elizab	yes	female	54.0000
1st				
## 249	Rowe, Mr. Alfred G	no	male	33.0000
1st				
## 250	Ryerson, Master. John Borie	yes	male	13.0000
1st				
## 251	Ryerson, Miss. Emily Borie	yes	female	18.0000
1st				
## 252	Ryerson, Miss. Susan Parker Su	yes	female	21.0000
1st				
## 253	Ryerson, Mr. Arthur Larned	no	male	61.0000
1st				
## 254	Ryerson, Mrs. Arthur Larned (Em	yes	female	48.0000
1st				
## 255	Saalfeld, Mr. Adolphe	yes	male	NA
1st				
## 256	Sagesser, Mlle. Emma	yes	female	24.0000
1st				
## 257	Salomon, Mr. Abraham L	yes	male	NA
1st				
## 258	Schabert, Mrs. Paul (Emma Mock)	yes	female	35.0000
1st				

## 259	Serepeca, Miss. Augusta	yes female 30.0000
1st		
## 260	Seward, Mr. Frederic Kimber	yes male 34.0000
1st		
## 261	Shutes, Miss. Elizabeth W	yes female 40.0000
1st		
## 262	Silverthorne, Mr. Spencer Victo	yes male 35.0000
1st		
## 263	Silvey, Mr. William Baird	no male 50.0000
1st		
## 264	Silvey, Mrs. William Baird (Ali	yes female 39.0000
1st		
## 265	Simonius-Blumer, Col. Oberst Al	yes male 56.0000
1st		
## 266	Sloper, Mr. William Thompson	yes male 28.0000
1st		
## 267	Smart, Mr. John Montgomery	no male 56.0000
1st		
## 268	Smith, Mr. James Clinch	no male 56.0000
1st		
## 269	Smith, Mr. Lucien Philip	no male 24.0000
1st		
## 270	Smith, Mr. Richard William	no male NA
1st		
## 271	Smith, Mrs. Lucien Philip (Mary	yes female 18.0000
1st		
## 272	Snyder, Mr. John Pillsbury	yes male 24.0000
1st		
## 273	Snyder, Mrs. John Pillsbury (Ne	yes female 23.0000
1st		
## 274	Spedden, Master. Robert Douglas	yes male 6.0000
1st		
## 275	Spedden, Mr. Frederic Oakley	yes male 45.0000
1st		
## 276	Spedden, Mrs. Frederic Oakley (	yes female 40.0000
1st		
## 277	Spencer, Mr. William Augustus	no male 57.0000
1st		
## 278	Spencer, Mrs. William Augustus	yes female NA
1st		
## 279	Stahelin-Maeglin, Dr. Max	yes male 32.0000
1st		
## 280	Stead, Mr. William Thomas	no male 62.0000
1st		
## 281	Stengel, Mr. Charles Emil Henry	yes male 54.0000
1st		
## 282	Stengel, Mrs. Charles Emil Henr	yes female 43.0000
1st		
## 283	Stephenson, Mrs. Walter Bertram	yes female 52.0000
1st		

## 284	Stewart, Mr. Albert A	no	male	NA
1st				
## 285	Stone, Mrs. George Nelson (Mart	yes	female	62.0000
1st				
## 286	Straus, Mr. Isidor	no	male	67.0000
1st				
## 287	Straus, Mrs. Isidor (Rosalie Id	no	female	63.0000
1st				
## 288	Sutton, Mr. Frederick	no	male	61.0000
1st				
## 289	Swift, Mrs. Frederick Joel (Mar	yes	female	48.0000
1st				
## 290	Taussig, Miss. Ruth	yes	female	18.0000
1st				
## 291	Taussig, Mr. Emil	no	male	52.0000
1st				
## 292	Taussig, Mrs. Emil (Tillie Mand	yes	female	39.0000
1st				
## 293	Taylor, Mr. Elmer Zebley	yes	male	48.0000
1st				
## 294	Taylor, Mrs. Elmer Zebley (Juli	yes	female	NA
1st				
## 295	Thayer, Mr. John Borland	no	male	49.0000
1st				
## 296	Thayer, Mr. John Borland Jr	yes	male	17.0000
1st				
## 297	Thayer, Mrs. John Borland (Mari	yes	female	39.0000
1st				
## 298	Thorne, Mrs. Gertrude Maybelle	yes	female	NA
1st				
## 299	Tucker, Mr. Gilbert Milligan Jr	yes	male	31.0000
1st				
## 300	Uruchurtu, Don. Manuel E	no	male	40.0000
1st				
## 301	Van der hoef, Mr. Wyckoff	no	male	61.0000
1st				
## 302	Walker, Mr. William Anderson	no	male	47.0000
1st				
## 303	Ward, Miss. Anna	yes	female	35.0000
1st				
## 304	Warren, Mr. Frank Manley	no	male	64.0000
1st				
## 305	Warren, Mrs. Frank Manley (Anna	yes	female	60.0000
1st				
## 306	Weir, Col. John	no	male	60.0000
1st				
## 307	White, Mr. Percival Wayland	no	male	54.0000
1st				
## 308	White, Mr. Richard Frasar	no	male	21.0000
1st				

## 309	White, Mrs. John Stuart (Ella H	yes female 55.0000
1st		
## 310	Wick, Miss. Mary Natalie	yes female 31.0000
1st		
## 311	Wick, Mr. George Dennick	no male 57.0000
1st		
## 312	Wick, Mrs. George Dennick (Mary	yes female 45.0000
1st		
## 313	Widener, Mr. George Dunton	no male 50.0000
1st		
## 314	Widener, Mr. Harry Elkins	no male 27.0000
1st		
## 315	Widener, Mrs. George Dunton (El	yes female 50.0000
1st		
## 316	Willard, Miss. Constance	yes female 21.0000
1st		
## 317	Williams, Mr. Charles Duane	no male 51.0000
1st		
## 318	Williams, Mr. Richard Norris II	yes male 21.0000
1st		
## 319	Williams-Lambert, Mr. Fletcher	no male NA
1st		
## 320	Wilson, Miss. Helen Alice	yes female 31.0000
1st		
## 321	Woolner, Mr. Hugh	yes male NA
1st		
## 322	Wright, Mr. George	no male 62.0000
1st		
## 323	Young, Miss. Marie Grice	yes female 36.0000
1st		
## 324	Abelson, Mr. Samuel	no male 30.0000
2nd		
## 325	Abelson, Mrs. Samuel (Hannah Wi	yes female 28.0000
2nd		
## 326	Aldworth, Mr. Charles Augustus	no male 30.0000
2nd		
## 327	Andrew, Mr. Edgardo Samuel	no male 18.0000
2nd		
## 328	Andrew, Mr. Frank Thomas	no male 25.0000
2nd		
## 329	Angle, Mr. William A	no male 34.0000
2nd		
## 330	Angle, Mrs. William A (Florence	yes female 36.0000
2nd		
## 331	Ashby, Mr. John	no male 57.0000
2nd		
## 332	Bailey, Mr. Percy Andrew	no male 18.0000
2nd		
## 333	Baimbrigge, Mr. Charles Robert	no male 23.0000
2nd		

## 334	Ball, Mrs. (Ada E Hall)	yes female 36.0000
2nd		
## 335	Banfield, Mr. Frederick James	no male 28.0000
2nd		
## 336	Bateman, Rev. Robert James	no male 51.0000
2nd		
## 337	Beane, Mr. Edward	yes male 32.0000
2nd		
## 338	Beane, Mrs. Edward (Ethel Clark	yes female 19.0000
2nd		
## 339	Beauchamp, Mr. Henry James	no male 28.0000
2nd		
## 340	Becker, Master. Richard F	yes male 1.0000
2nd		
## 341	Becker, Miss. Marion Louise	yes female 4.0000
2nd		
## 342	Becker, Miss. Ruth Elizabeth	yes female 12.0000
2nd		
## 343	Becker, Mrs. Allen Oliver (Nell	yes female 36.0000
2nd		
## 344	Beesley, Mr. Lawrence	yes male 34.0000
2nd		
## 345	Bentham, Miss. Lilian W	yes female 19.0000
2nd		
## 346	Berriman, Mr. William John	no male 23.0000
2nd		
## 347	Botsford, Mr. William Hull	no male 26.0000
2nd		
## 348	Bowenur, Mr. Solomon	no male 42.0000
2nd		
## 349	Bracken, Mr. James H	no male 27.0000
2nd		
## 350	Brown, Miss. Amelia Mildred	yes female 24.0000
2nd		
## 351	Brown, Miss. Edith Eileen	yes female 15.0000
2nd		
## 352	Brown, Mr. Thomas William Solom	no male 60.0000
2nd		
## 353	Brown, Mrs. Thomas William Solo	yes female 40.0000
2nd		
## 354	Bryhl, Miss. Dagmar Jenny Ingeb	yes female 20.0000
2nd		
## 355	Bryhl, Mr. Kurt Arnold Gottfrid	no male 25.0000
2nd		
## 356	Buss, Miss. Kate	yes female 36.0000
2nd		
## 357	Butler, Mr. Reginald Fenton	no male 25.0000
2nd		
## 358	Byles, Rev. Thomas Roussel Davi	no male 42.0000
2nd		

## 359	Bystrom, Mrs. (Karolina)	yes female 42.0000
2nd		
## 360	Caldwell, Master. Alden Gates	yes male 0.8333
2nd		
## 361	Caldwell, Mr. Albert Francis	yes male 26.0000
2nd		
## 362	Caldwell, Mrs. Albert Francis (	yes female 22.0000
2nd		
## 363	Cameron, Miss. Clear Annie	yes female 35.0000
2nd		
## 364	Campbell, Mr. William	no male NA
2nd		
## 365	Carbines, Mr. William	no male 19.0000
2nd		
## 366	Carter, Mrs. Ernest Courtenay (	no female 44.0000
2nd		
## 367	Carter, Rev. Ernest Courtenay	no male 54.0000
2nd		
## 368	Chapman, Mr. Charles Henry	no male 52.0000
2nd		
## 369	Chapman, Mr. John Henry	no male 37.0000
2nd		
## 370	Chapman, Mrs. John Henry (Sara	no female 29.0000
2nd		
## 371	Christy, Miss. Julie Rachel	yes female 25.0000
2nd		
## 372	Christy, Mrs. (Alice Frances)	yes female 45.0000
2nd		
## 373	Clarke, Mr. Charles Valentine	no male 29.0000
2nd		
## 374	Clarke, Mrs. Charles V (Ada Mar	yes female 28.0000
2nd		
## 375	Coleridge, Mr. Reginald Charles	no male 29.0000
2nd		
## 376	Collander, Mr. Erik Gustaf	no male 28.0000
2nd		
## 377	Collett, Mr. Sidney C Stuart	yes male 24.0000
2nd		
## 378	Collyer, Miss. Marjorie Lottie	yes female 8.0000
2nd		
## 379	Collyer, Mr. Harvey	no male 31.0000
2nd		
## 380	Collyer, Mrs. Harvey (Charlotte	yes female 31.0000
2nd		
## 381	Cook, Mrs. (Selena Rogers)	yes female 22.0000
2nd		
## 382	Corbett, Mrs. Walter H (Irene C	no female 30.0000
2nd		
## 383	Corey, Mrs. Percy C (Mary Phyll	no female NA
2nd		

## 384	Cotterill, Mr. Henry Harry	no	male	21.0000
2nd				
## 385	Cunningham, Mr. Alfred Fleming	no	male	NA
2nd				
## 386	Davies, Master. John Morgan Jr	yes	male	8.0000
2nd				
## 387	Davies, Mr. Charles Henry	no	male	18.0000
2nd				
## 388	Davies, Mrs. John Morgan (Eliza	yes	female	48.0000
2nd				
## 389	Davis, Miss. Mary	yes	female	28.0000
2nd				
## 390	de Brito, Mr. Jose Joaquim	no	male	32.0000
2nd				
## 391	Deacon, Mr. Percy William	no	male	17.0000
2nd				
## 392	del Carlo, Mr. Sebastiano	no	male	29.0000
2nd				
## 393	del Carlo, Mrs. Sebastiano (Arg	yes	female	24.0000
2nd				
## 394	Denbury, Mr. Herbert	no	male	25.0000
2nd				
## 395	Dibden, Mr. William	no	male	18.0000
2nd				
## 396	Doling, Miss. Elsie	yes	female	18.0000
2nd				
## 397	Doling, Mrs. John T (Ada Julia	yes	female	34.0000
2nd				
## 398	Downton, Mr. William James	no	male	54.0000
2nd				
## 399	Drew, Master. Marshall Brines	yes	male	8.0000
2nd				
## 400	Drew, Mr. James Vivian	no	male	42.0000
2nd				
## 401	Drew, Mrs. James Vivian (Lulu T	yes	female	34.0000
2nd				
## 402	Duran y More, Miss. Asuncion	yes	female	27.0000
2nd				
## 403	Duran y More, Miss. Florentina	yes	female	30.0000
2nd				
## 404	Eitemiller, Mr. George Floyd	no	male	23.0000
2nd				
## 405	Enander, Mr. Ingvar	no	male	21.0000
2nd				
## 406	Fahlstrom, Mr. Arne Jonas	no	male	18.0000
2nd				
## 407	Faunthorpe, Mr. Harry	no	male	40.0000
2nd				
## 408	Faunthorpe, Mrs. Lizzie (Elizab	yes	female	29.0000
2nd				

## 409	Fillbrook, Mr. Joseph Charles	no	male	18.0000
2nd				
## 410	Fox, Mr. Stanley Hubert	no	male	36.0000
2nd				
## 411	Frost, Mr. Anthony Wood Archie	no	male	NA
2nd				
## 412	Funk, Miss. Annie Clemmer	no	female	38.0000
2nd				
## 413	Fynney, Mr. Joseph J	no	male	35.0000
2nd				
## 414	Gale, Mr. Harry	no	male	38.0000
2nd				
## 415	Gale, Mr. Shadrach	no	male	34.0000
2nd				
## 416	Garside, Miss. Ethel	yes	female	34.0000
2nd				
## 417	Gaskell, Mr. Alfred	no	male	16.0000
2nd				
## 418	Gavey, Mr. Lawrence	no	male	26.0000
2nd				
## 419	Gilbert, Mr. William	no	male	47.0000
2nd				
## 420	Giles, Mr. Edgar	no	male	21.0000
2nd				
## 421	Giles, Mr. Frederick Edward	no	male	21.0000
2nd				
## 422	Giles, Mr. Ralph	no	male	24.0000
2nd				
## 423	Gill, Mr. John William	no	male	24.0000
2nd				
## 424	Gillespie, Mr. William Henry	no	male	34.0000
2nd				
## 425	Givard, Mr. Hans Kristensen	no	male	30.0000
2nd				
## 426	Greenberg, Mr. Samuel	no	male	52.0000
2nd				
## 427	Hale, Mr. Reginald	no	male	30.0000
2nd				
## 428	Hamalainen, Master. Viljo	yes	male	0.6667
2nd				
## 429	Hamalainen, Mrs. William (Anna)	yes	female	24.0000
2nd				
## 430	Harbeck, Mr. William H	no	male	44.0000
2nd				
## 431	Harper, Miss. Annie Jessie Nin	yes	female	6.0000
2nd				
## 432	Harper, Rev. John	no	male	28.0000
2nd				
## 433	Harris, Mr. George	yes	male	62.0000
2nd				

## 434	Harris, Mr. Walter	no	male	30.0000
2nd				
## 435	Hart, Miss. Eva Miriam	yes	female	7.0000
2nd				
## 436	Hart, Mr. Benjamin	no	male	43.0000
2nd				
## 437	Hart, Mrs. Benjamin (Esther Ada	yes	female	45.0000
2nd				
## 438	Herman, Miss. Alice	yes	female	24.0000
2nd				
## 439	Herman, Miss. Kate	yes	female	24.0000
2nd				
## 440	Herman, Mr. Samuel	no	male	49.0000
2nd				
## 441	Herman, Mrs. Samuel (Jane Laver	yes	female	48.0000
2nd				
## 442	Hewlett, Mrs. (Mary D Kingcome)	yes	female	55.0000
2nd				
## 443	Hickman, Mr. Leonard Mark	no	male	24.0000
2nd				
## 444	Hickman, Mr. Lewis	no	male	32.0000
2nd				
## 445	Hickman, Mr. Stanley George	no	male	21.0000
2nd				
## 446	Hiltunen, Miss. Marta	no	female	18.0000
2nd				
## 447	Hocking, Miss. Ellen Nellie	yes	female	20.0000
2nd				
## 448	Hocking, Mr. Richard George	no	male	23.0000
2nd				
## 449	Hocking, Mr. Samuel James Metca	no	male	36.0000
2nd				
## 450	Hocking, Mrs. Elizabeth (Eliza	yes	female	54.0000
2nd				
## 451	Hodges, Mr. Henry Price	no	male	50.0000
2nd				
## 452	Hold, Mr. Stephen	no	male	44.0000
2nd				
## 453	Hold, Mrs. Stephen (Annie Marga	yes	female	29.0000
2nd				
## 454	Hood, Mr. Ambrose Jr	no	male	21.0000
2nd				
## 455	Hosono, Mr. Masabumi	yes	male	42.0000
2nd				
## 456	Howard, Mr. Benjamin	no	male	63.0000
2nd				
## 457	Howard, Mrs. Benjamin (Ellen Tr	no	female	60.0000
2nd				
## 458	Hunt, Mr. George Henry	no	male	33.0000
2nd				

## 459	Ilett, Miss. Bertha	yes female 17.0000
2nd		
## 460	Jacobsohn, Mr. Sidney Samuel	no male 42.0000
2nd		
## 461	Jacobsohn, Mrs. Sidney Samuel (	yes female 24.0000
2nd		
## 462	Jarvis, Mr. John Denzil	no male 47.0000
2nd		
## 463	Jefferys, Mr. Clifford Thomas	no male 24.0000
2nd		
## 464	Jefferys, Mr. Ernest Wilfred	no male 22.0000
2nd		
## 465	Jenkin, Mr. Stephen Curnow	no male 32.0000
2nd		
## 466	Jerwan, Mrs. Amin S (Marie Mart	yes female 23.0000
2nd		
## 467	Kantor, Mr. Sinai	no male 34.0000
2nd		
## 468	Kantor, Mrs. Sinai (Miriam Ster	yes female 24.0000
2nd		
## 469	Karnes, Mrs. J Frank (Claire Be	no female 22.0000
2nd		
## 470	Keane, Miss. Nora A	yes female NA
2nd		
## 471	Keane, Mr. Daniel	no male 35.0000
2nd		
## 472	Kelly, Mrs. Florence Fannie	yes female 45.0000
2nd		
## 473	Kirkland, Rev. Charles Leonard	no male 57.0000
2nd		
## 474	Knight, Mr. Robert J	no male NA
2nd		
## 475	Kvillner, Mr. Johan Henrik Joha	no male 31.0000
2nd		
## 476	Lahtinen, Mrs. William (Anna Sy	no female 26.0000
2nd		
## 477	Lahtinen, Rev. William	no male 30.0000
2nd		
## 478	Lamb, Mr. John Joseph	no male NA
2nd		
## 479	Laroche, Miss. Louise	yes female 1.0000
2nd		
## 480	Laroche, Miss. Simonne Marie An	yes female 3.0000
2nd		
## 481	Laroche, Mr. Joseph Philippe Le	no male 25.0000
2nd		
## 482	Laroche, Mrs. Joseph (Juliette	yes female 22.0000
2nd		
## 483	Lehmann, Miss. Bertha	yes female 17.0000
2nd		

## 484	Leitch, Miss. Jessie Wills	yes female	NA
2nd			
## 485	Lemore, Mrs. (Amelia Milley)	yes female	34.0000
2nd			
## 486	Levy, Mr. Rene Jacques	no male	36.0000
2nd			
## 487	Leyson, Mr. Robert William Norm	no male	24.0000
2nd			
## 488	Lingane, Mr. John	no male	61.0000
2nd			
## 489	Louch, Mr. Charles Alexander	no male	50.0000
2nd			
## 490	Louch, Mrs. Charles Alexander (	yes female	42.0000
2nd			
## 491	Mack, Mrs. (Mary)	no female	57.0000
2nd			
## 492	Malachard, Mr. Noel	no male	NA
2nd			
## 493	Mallet, Master. Andre	yes male	1.0000
2nd			
## 494	Mallet, Mr. Albert	no male	31.0000
2nd			
## 495	Mallet, Mrs. Albert (Antoinette	yes female	24.0000
2nd			
## 496	Mangiavacchi, Mr. Serafino Emil	no male	NA
2nd			
## 497	Matthews, Mr. William John	no male	30.0000
2nd			
## 498	Maybery, Mr. Frank Hubert	no male	40.0000
2nd			
## 499	McCrae, Mr. Arthur Gordon	no male	32.0000
2nd			
## 500	McCrie, Mr. James Matthew	no male	30.0000
2nd			
## 501	McKane, Mr. Peter David	no male	46.0000
2nd			
## 502	Mellinger, Miss. Madeleine Viol	yes female	13.0000
2nd			
## 503	Mellinger, Mrs. (Elizabeth Anne	yes female	41.0000
2nd			
## 504	Mellors, Mr. William John	yes male	19.0000
2nd			
## 505	Meyer, Mr. August	no male	39.0000
2nd			
## 506	Milling, Mr. Jacob Christian	no male	48.0000
2nd			
## 507	Mitchell, Mr. Henry Michael	no male	70.0000
2nd			
## 508	Montvila, Rev. Juozas	no male	27.0000
2nd			

## 509	Moraweck, Dr. Ernest	no	male	54.0000
2nd				
## 510	Morley, Mr. Henry Samuel (Mr H	no	male	39.0000
2nd				
## 511	Mudd, Mr. Thomas Charles	no	male	16.0000
2nd				
## 512	Myles, Mr. Thomas Francis	no	male	62.0000
2nd				
## 513	Nasser, Mr. Nicholas	no	male	32.5000
2nd				
## 514	Nasser, Mrs. Nicholas (Adele Ac	yes	female	14.0000
2nd				
## 515	Navratil, Master. Edmond Roger	yes	male	2.0000
2nd				
## 516	Navratil, Master. Michel M	yes	male	3.0000
2nd				
## 517	Navratil, Mr. Michel (Louis M	no	male	36.5000
2nd				
## 518	Nesson, Mr. Israel	no	male	26.0000
2nd				
## 519	Nicholls, Mr. Joseph Charles	no	male	19.0000
2nd				
## 520	Norman, Mr. Robert Douglas	no	male	28.0000
2nd				
## 521	Nourney, Mr. Alfred (Baron von	yes	male	20.0000
2nd				
## 522	Nye, Mrs. (Elizabeth Ramell)	yes	female	29.0000
2nd				
## 523	Otter, Mr. Richard	no	male	39.0000
2nd				
## 524	Oxenham, Mr. Percy Thomas	yes	male	22.0000
2nd				
## 525	Padro y Manent, Mr. Julian	yes	male	NA
2nd				
## 526	Pain, Dr. Alfred	no	male	23.0000
2nd				
## 527	Pallas y Castello, Mr. Emilio	yes	male	29.0000
2nd				
## 528	Parker, Mr. Clifford Richard	no	male	28.0000
2nd				
## 529	Parkes, Mr. Francis Frank	no	male	NA
2nd				
## 530	Parrish, Mrs. (Lutie Davis)	yes	female	50.0000
2nd				
## 531	Pengelly, Mr. Frederick William	no	male	19.0000
2nd				
## 532	Pernot, Mr. Rene	no	male	NA
2nd				
## 533	Peruschitz, Rev. Joseph Maria	no	male	41.0000
2nd				

## 534	Phillips, Miss. Alice Frances L	yes female 21.0000
2nd		
## 535	Phillips, Miss. Kate Florence (	yes female 19.0000
2nd		
## 536	Phillips, Mr. Escott Robert	no male 43.0000
2nd		
## 537	Pinsky, Mrs. (Rosa)	yes female 32.0000
2nd		
## 538	Ponesell, Mr. Martin	no male 34.0000
2nd		
## 539	Portaluppi, Mr. Emilio Ilario G	yes male 30.0000
2nd		
## 540	Pulbaum, Mr. Franz	no male 27.0000
2nd		
## 541	Quick, Miss. Phyllis May	yes female 2.0000
2nd		
## 542	Quick, Miss. Winifred Vera	yes female 8.0000
2nd		
## 543	Quick, Mrs. Frederick Charles (	yes female 33.0000
2nd		
## 544	Reeves, Mr. David	no male 36.0000
2nd		
## 545	Renouf, Mr. Peter Henry	no male 34.0000
2nd		
## 546	Renouf, Mrs. Peter Henry (Lilli	yes female 30.0000
2nd		
## 547	Reynaldo, Ms. Encarnacion	yes female 28.0000
2nd		
## 548	Richard, Mr. Emile	no male 23.0000
2nd		
## 549	Richards, Master. George Sibley	yes male 0.8333
2nd		
## 550	Richards, Master. William Rowe	yes male 3.0000
2nd		
## 551	Richards, Mrs. Sidney (Emily Ho	yes female 24.0000
2nd		
## 552	Ridsdale, Miss. Lucy	yes female 50.0000
2nd		
## 553	Rogers, Mr. Reginald Harry	no male 19.0000
2nd		
## 554	Rugg, Miss. Emily	yes female 21.0000
2nd		
## 555	Schmidt, Mr. August	no male 26.0000
2nd		
## 556	Sedgwick, Mr. Charles Frederick	no male 25.0000
2nd		
## 557	Sharp, Mr. Percival James R	no male 27.0000
2nd		
## 558	Shelley, Mrs. William (Imanita	yes female 25.0000
2nd		

## 559	Silven, Miss. Lyyli Karoliina	yes female 18.0000
2nd		
## 560	Sincock, Miss. Maude	yes female 20.0000
2nd		
## 561	Sinkkonen, Miss. Anna	yes female 30.0000
2nd		
## 562	Sjostedt, Mr. Ernst Adolf	no male 59.0000
2nd		
## 563	Slayter, Miss. Hilda Mary	yes female 30.0000
2nd		
## 564	Slemen, Mr. Richard James	no male 35.0000
2nd		
## 565	Smith, Miss. Marion Elsie	yes female 40.0000
2nd		
## 566	Sobey, Mr. Samuel James Hayden	no male 25.0000
2nd		
## 567	Stanton, Mr. Samuel Ward	no male 41.0000
2nd		
## 568	Stokes, Mr. Philip Joseph	no male 25.0000
2nd		
## 569	Swane, Mr. George	no male 18.5000
2nd		
## 570	Sweet, Mr. George Frederick	no male 14.0000
2nd		
## 571	Toomey, Miss. Ellen	yes female 50.0000
2nd		
## 572	Troupiansky, Mr. Moses Aaron	no male 23.0000
2nd		
## 573	Trout, Mrs. William H (Jessie L	yes female 28.0000
2nd		
## 574	Troutt, Miss. Edwina Celia Win	yes female 27.0000
2nd		
## 575	Turpin, Mr. William John Robert	no male 29.0000
2nd		
## 576	Turpin, Mrs. William John Rober	no female 27.0000
2nd		
## 577	Veal, Mr. James	no male 40.0000
2nd		
## 578	Walcroft, Miss. Nellie	yes female 31.0000
2nd		
## 579	Ware, Mr. John James	no male 30.0000
2nd		
## 580	Ware, Mr. William Jeffery	no male 23.0000
2nd		
## 581	Ware, Mrs. John James (Florence	yes female 31.0000
2nd		
## 582	Watson, Mr. Ennis Hastings	no male NA
2nd		
## 583	Watt, Miss. Bertha J	yes female 12.0000
2nd		

## 584	Watt, Mrs. James (Elizabeth Be	yes female 40.0000
2nd		
## 585	Webber, Miss. Susan	yes female 32.5000
2nd		
## 586	Weisz, Mr. Leopold	no male 27.0000
2nd		
## 587	Weisz, Mrs. Leopold (Mathilde F	yes female 29.0000
2nd		
## 588	Wells, Master. Ralph Lester	yes male 2.0000
2nd		
## 589	Wells, Miss. Joan	yes female 4.0000
2nd		
## 590	Wells, Mrs. Arthur Henry (Addi	yes female 29.0000
2nd		
## 591	West, Miss. Barbara J	yes female 0.9167
2nd		
## 592	West, Miss. Constance Mirium	yes female 5.0000
2nd		
## 593	West, Mr. Edwy Arthur	no male 36.0000
2nd		
## 594	West, Mrs. Edwy Arthur (Ada Mar	yes female 33.0000
2nd		
## 595	Wheadon, Mr. Edward H	no male 66.0000
2nd		
## 596	Wheeler, Mr. Edwin Frederick	no male NA
2nd		
## 597	Wilhelms, Mr. Charles	yes male 31.0000
2nd		
## 598	Williams, Mr. Charles Eugene	yes male NA
2nd		
## 599	Wright, Miss. Marion	yes female 26.0000
2nd		
## 600	Yrois, Miss. Henriette (Mrs Ha	no female 24.0000
2nd		
## 601	Abbing, Mr. Anthony	no male 42.0000
3rd		
## 602	Abbott, Master. Eugene Joseph	no male 13.0000
3rd		
## 603	Abbott, Mr. Rossmore Edward	no male 16.0000
3rd		
## 604	Abbott, Mrs. Stanton (Rosa Hunt	yes female 35.0000
3rd		
## 605	Abelseth, Miss. Karen Marie	yes female 16.0000
3rd		
## 606	Abelseth, Mr. Olaus Jorgensen	yes male 25.0000
3rd		
## 607	Abrahamsson, Mr. Abraham August	yes male 20.0000
3rd		
## 608	Abraham, Mrs. Joseph (Sophie Ha	yes female 18.0000
3rd		

## 609	Adahl, Mr. Mauritz Nils Martin	no	male	30.0000
3rd				
## 610	Adams, Mr. John	no	male	26.0000
3rd				
## 611	Ahlin, Mrs. Johan (Johanna Pers	no	female	40.0000
3rd				
## 612	Aks, Master. Philip Frank	yes	male	0.8333
3rd				
## 613	Aks, Mrs. Sam (Leah Rosen)	yes	female	18.0000
3rd				
## 614	Albimona, Mr. Nassef Cassem	yes	male	26.0000
3rd				
## 615	Alexander, Mr. William	no	male	26.0000
3rd				
## 616	Alhomaki, Mr. Ilmari Rudolf	no	male	20.0000
3rd				
## 617	Ali, Mr. Ahmed	no	male	24.0000
3rd				
## 618	Ali, Mr. William	no	male	25.0000
3rd				
## 619	Allen, Mr. William Henry	no	male	35.0000
3rd				
## 620	Allum, Mr. Owen George	no	male	18.0000
3rd				
## 621	Andersen, Mr. Albert Karvin	no	male	32.0000
3rd				
## 622	Andersen-Jensen, Miss. Carla Ch	yes	female	19.0000
3rd				
## 623	Andersson, Master. Sigvard Hara	no	male	4.0000
3rd				
## 624	Andersson, Miss. Ebba Iris Alfr	no	female	6.0000
3rd				
## 625	Andersson, Miss. Ellis Anna Mar	no	female	2.0000
3rd				
## 626	Andersson, Miss. Erna Alexandra	yes	female	17.0000
3rd				
## 627	Andersson, Miss. Ida Augusta Ma	no	female	38.0000
3rd				
## 628	Andersson, Miss. Ingeborg Const	no	female	9.0000
3rd				
## 629	Andersson, Miss. Sigrid Elisabe	no	female	11.0000
3rd				
## 630	Andersson, Mr. Anders Johan	no	male	39.0000
3rd				
## 631	Andersson, Mr. August Edvard (	yes	male	27.0000
3rd				
## 632	Andersson, Mr. Johan Samuel	no	male	26.0000
3rd				
## 633	Andersson, Mrs. Anders Johan (A	no	female	39.0000
3rd				

## 634	Andreasson, Mr. Paul Edvin	no	male	20.0000
3rd				
## 635	Angheloff, Mr. Minko	no	male	26.0000
3rd				
## 636	Arnold-Franchi, Mr. Josef	no	male	25.0000
3rd				
## 637	Arnold-Franchi, Mrs. Josef (Jos	no	female	18.0000
3rd				
## 638	Aronsson, Mr. Ernst Axel Algot	no	male	24.0000
3rd				
## 639	Asim, Mr. Adola	no	male	35.0000
3rd				
## 640	Asplund, Master. Carl Edgar	no	male	5.0000
3rd				
## 641	Asplund, Master. Clarence Gusta	no	male	9.0000
3rd				
## 642	Asplund, Master. Edvin Rojj Fel	yes	male	3.0000
3rd				
## 643	Asplund, Master. Filip Oscar	no	male	13.0000
3rd				
## 644	Asplund, Miss. Lillian Gertrud	yes	female	5.0000
3rd				
## 645	Asplund, Mr. Carl Oscar Vilhelm	no	male	40.0000
3rd				
## 646	Asplund, Mr. Johan Charles	yes	male	23.0000
3rd				
## 647	Asplund, Mrs. Carl Oscar (Selma	yes	female	38.0000
3rd				
## 648	Assaf Khalil, Mrs. Mariana (Mi	yes	female	45.0000
3rd				
## 649	Assaf, Mr. Gerios	no	male	21.0000
3rd				
## 650	Assam, Mr. Ali	no	male	23.0000
3rd				
## 651	Attalah, Miss. Malake	no	female	17.0000
3rd				
## 652	Attalah, Mr. Sleiman	no	male	30.0000
3rd				
## 653	Augustsson, Mr. Albert	no	male	23.0000
3rd				
## 654	Ayoub, Miss. Banoura	yes	female	13.0000
3rd				
## 655	Baccos, Mr. Raffull	no	male	20.0000
3rd				
## 656	Backstrom, Mr. Karl Alfred	no	male	32.0000
3rd				
## 657	Backstrom, Mrs. Karl Alfred (Ma	yes	female	33.0000
3rd				
## 658	Baclini, Miss. Eugenie	yes	female	0.7500
3rd				

## 659	Baclini, Miss. Helene Barbara	yes female	0.7500
3rd			
## 660	Baclini, Miss. Marie Catherine	yes female	5.0000
3rd			
## 661	Baclini, Mrs. Solomon (Latifa Q	yes female	24.0000
3rd			
## 662	Badman, Miss. Emily Louisa	yes female	18.0000
3rd			
## 663	Badt, Mr. Mohamed	no male	40.0000
3rd			
## 664	Balkic, Mr. Cerin	no male	26.0000
3rd			
## 665	Barah, Mr. Hanna Assi	yes male	20.0000
3rd			
## 666	Barbara, Miss. Saiide	no female	18.0000
3rd			
## 667	Barbara, Mrs. (Catherine David)	no female	45.0000
3rd			
## 668	Barry, Miss. Julia	no female	27.0000
3rd			
## 669	Barton, Mr. David John	no male	22.0000
3rd			
## 670	Beavan, Mr. William Thomas	no male	19.0000
3rd			
## 671	Bengtsson, Mr. John Viktor	no male	26.0000
3rd			
## 672	Berglund, Mr. Karl Ivar Sven	no male	22.0000
3rd			
## 673	Betros, Master. Seman	no male	NA
3rd			
## 674	Betros, Mr. Tannous	no male	20.0000
3rd			
## 675	Bing, Mr. Lee	yes male	32.0000
3rd			
## 676	Birkeland, Mr. Hans Martin Mons	no male	21.0000
3rd			
## 677	Bjorklund, Mr. Ernst Herbert	no male	18.0000
3rd			
## 678	Bostandyeff, Mr. Guentcho	no male	26.0000
3rd			
## 679	Boulos, Master. Akar	no male	6.0000
3rd			
## 680	Boulos, Miss. Nourelain	no female	9.0000
3rd			
## 681	Boulos, Mr. Hanna	no male	NA
3rd			
## 682	Boulos, Mrs. Joseph (Sultana)	no female	NA
3rd			
## 683	Bourke, Miss. Mary	no female	NA
3rd			

## 684	Bourke, Mr. John	no	male	40.0000
3rd				
## 685	Bourke, Mrs. John (Catherine)	no	female	32.0000
3rd				
## 686	Bowen, Mr. David John Dai	no	male	21.0000
3rd				
## 687	Bradley, Miss. Bridget Delia	yes	female	22.0000
3rd				
## 688	Braf, Miss. Elin Ester Maria	no	female	20.0000
3rd				
## 689	Braund, Mr. Lewis Richard	no	male	29.0000
3rd				
## 690	Braund, Mr. Owen Harris	no	male	22.0000
3rd				
## 691	Brobeck, Mr. Karl Rudolf	no	male	22.0000
3rd				
## 692	Brocklebank, Mr. William Alfred	no	male	35.0000
3rd				
## 693	Buckley, Miss. Katherine	no	female	18.5000
3rd				
## 694	Buckley, Mr. Daniel	yes	male	21.0000
3rd				
## 695	Burke, Mr. Jeremiah	no	male	19.0000
3rd				
## 696	Burns, Miss. Mary Delia	no	female	18.0000
3rd				
## 697	Cacic, Miss. Manda	no	female	21.0000
3rd				
## 698	Cacic, Miss. Marija	no	female	30.0000
3rd				
## 699	Cacic, Mr. Jego Grga	no	male	18.0000
3rd				
## 700	Cacic, Mr. Luka	no	male	38.0000
3rd				
## 701	Calic, Mr. Jovo	no	male	17.0000
3rd				
## 702	Calic, Mr. Petar	no	male	17.0000
3rd				
## 703	Canavan, Miss. Mary	no	female	21.0000
3rd				
## 704	Canavan, Mr. Patrick	no	male	21.0000
3rd				
## 705	Cann, Mr. Ernest Charles	no	male	21.0000
3rd				
## 706	Caram, Mr. Joseph	no	male	NA
3rd				
## 707	Caram, Mrs. Joseph (Maria Elias	no	female	NA
3rd				
## 708	Carlsson, Mr. August Sigfrid	no	male	28.0000
3rd				

## 709 3rd	Carlsson, Mr. Carl Robert	no male 24.0000
## 710 3rd	Carr, Miss. Helen Ellen	yes female 16.0000
## 711 3rd	Carr, Miss. Jeannie	no female 37.0000
## 712 3rd	Carver, Mr. Alfred John	no male 28.0000
## 713 3rd	Celotti, Mr. Francesco	no male 24.0000
## 714 3rd	Charters, Mr. David	no male 21.0000
## 715 3rd	Chip, Mr. Chang	yes male 32.0000
## 716 3rd	Christmann, Mr. Emil	no male 29.0000
## 717 3rd	Chronopoulos, Mr. Apostolos	no male 26.0000
## 718 3rd	Chronopoulos, Mr. Demetrios	no male 18.0000
## 719 3rd	Coelho, Mr. Domingos Fernandeo	no male 20.0000
## 720 3rd	Cohen, Mr. Gurshon Gus	yes male 18.0000
## 721 3rd	Colbert, Mr. Patrick	no male 24.0000
## 722 3rd	Coleff, Mr. Peju	no male 36.0000
## 723 3rd	Coleff, Mr. Satio	no male 24.0000
## 724 3rd	Conlon, Mr. Thomas Henry	no male 31.0000
## 725 3rd	Connaghton, Mr. Michael	no male 31.0000
## 726 3rd	Connolly, Miss. Kate (A)	yes female 22.0000
## 727 3rd	Connolly, Miss. Kate (B)	no female 30.0000
## 728 3rd	Connors, Mr. Patrick	no male 70.5000
## 729 3rd	Cook, Mr. Jacob	no male 43.0000
## 730 3rd	Cor, Mr. Bartol	no male 35.0000
## 731 3rd	Cor, Mr. Ivan	no male 27.0000
## 732 3rd	Cor, Mr. Liudevrit	no male 19.0000
## 733 3rd	Corn, Mr. Harry	no male 30.0000

## 734	Coutts, Master. Eden Leslie Ne	yes	male	9.0000
3rd				
## 735	Coutts, Master. William Loch W	yes	male	3.0000
3rd				
## 736	Coutts, Mrs. William (Winnie M	yes	female	36.0000
3rd				
## 737	Coxon, Mr. Daniel	no	male	59.0000
3rd				
## 738	Crease, Mr. Ernest James	no	male	19.0000
3rd				
## 739	Cribb, Miss. Laura Alice	yes	female	17.0000
3rd				
## 740	Cribb, Mr. John Hatfield	no	male	44.0000
3rd				
## 741	Culumovic, Mr. Jeso	no	male	17.0000
3rd				
## 742	Daher, Mr. Shedid	no	male	22.5000
3rd				
## 743	Dahl, Mr. Karl Edwart	yes	male	45.0000
3rd				
## 744	Dahlberg, Miss. Gerda Ulrika	no	female	22.0000
3rd				
## 745	Dakic, Mr. Branko	no	male	19.0000
3rd				
## 746	Daly, Miss. Margaret Marcella	yes	female	30.0000
3rd				
## 747	Daly, Mr. Eugene Patrick	yes	male	29.0000
3rd				
## 748	Danbom, Master. Gilbert Sigvard	no	male	0.3333
3rd				
## 749	Danbom, Mr. Ernst Gilbert	no	male	34.0000
3rd				
## 750	Danbom, Mrs. Ernst Gilbert (Ann	no	female	28.0000
3rd				
## 751	Danoff, Mr. Yoto	no	male	27.0000
3rd				
## 752	Dantcheff, Mr. Ristiu	no	male	25.0000
3rd				
## 753	Davies, Mr. Alfred J	no	male	24.0000
3rd				
## 754	Davies, Mr. Evan	no	male	22.0000
3rd				
## 755	Davies, Mr. John Samuel	no	male	21.0000
3rd				
## 756	Davies, Mr. Joseph	no	male	17.0000
3rd				
## 757	Davison, Mr. Thomas Henry	no	male	NA
3rd				
## 758	Davison, Mrs. Thomas Henry (Mar	yes	female	NA
3rd				

## 759	de Messemaeker, Mr. Guillaume J	yes	male	36.5000
3rd				
## 760	de Messemaeker, Mrs. Guillaume	yes	female	36.0000
3rd				
## 761	de Mulder, Mr. Theodore	yes	male	30.0000
3rd				
## 762	de Pelsmaeker, Mr. Alfons	no	male	16.0000
3rd				
## 763	Dean, Master. Bertram Vere	yes	male	1.0000
3rd				
## 764	Dean, Miss. Elizabeth Gladys M	yes	female	0.1667
3rd				
## 765	Dean, Mr. Bertram Frank	no	male	26.0000
3rd				
## 766	Dean, Mrs. Bertram (Eva Georget	yes	female	33.0000
3rd				
## 767	Delalic, Mr. Redjo	no	male	25.0000
3rd				
## 768	Demetri, Mr. Marinko	no	male	NA
3rd				
## 769	Denkoff, Mr. Mitto	no	male	NA
3rd				
## 770	Dennis, Mr. Samuel	no	male	22.0000
3rd				
## 771	Dennis, Mr. William	no	male	36.0000
3rd				
## 772	Devaney, Miss. Margaret Delia	yes	female	19.0000
3rd				
## 773	Dika, Mr. Mirko	no	male	17.0000
3rd				
## 774	Dimic, Mr. Jovan	no	male	42.0000
3rd				
## 775	Dintcheff, Mr. Valtcho	no	male	43.0000
3rd				
## 776	Doharr, Mr. Tannous	no	male	NA
3rd				
## 777	Dooley, Mr. Patrick	no	male	32.0000
3rd				
## 778	Dorking, Mr. Edward Arthur	yes	male	19.0000
3rd				
## 779	Dowdell, Miss. Elizabeth	yes	female	30.0000
3rd				
## 780	Doyle, Miss. Elizabeth	no	female	24.0000
3rd				
## 781	Drapkin, Miss. Jennie	yes	female	23.0000
3rd				
## 782	Drazenoic, Mr. Jozef	no	male	33.0000
3rd				
## 783	Duane, Mr. Frank	no	male	65.0000
3rd				

## 784	Duquemin, Mr. Joseph	yes	male	24.0000
3rd				
## 785	Dyker, Mr. Adolf Fredrik	no	male	23.0000
3rd				
## 786	Dyker, Mrs. Adolf Fredrik (Anna	yes	female	22.0000
3rd				
## 787	Edvardsson, Mr. Gustaf Hjalmar	no	male	18.0000
3rd				
## 788	Eklund, Mr. Hans Linus	no	male	16.0000
3rd				
## 789	Ekstrom, Mr. Johan	no	male	45.0000
3rd				
## 790	Elias, Mr. Dibo	no	male	NA
3rd				
## 791	Elias, Mr. Joseph	no	male	39.0000
3rd				
## 792	Elias, Mr. Joseph Jr	no	male	17.0000
3rd				
## 793	Elias, Mr. Tannous	no	male	15.0000
3rd				
## 794	Elsbury, Mr. William James	no	male	47.0000
3rd				
## 795	Emanuel, Miss. Virginia Ethel	yes	female	5.0000
3rd				
## 796	Emir, Mr. Farred Chehab	no	male	NA
3rd				
## 797	Everett, Mr. Thomas James	no	male	40.5000
3rd				
## 798	Farrell, Mr. James	no	male	40.5000
3rd				
## 799	Finoli, Mr. Luigi	yes	male	NA
3rd				
## 800	Fischer, Mr. Eberhard Thelander	no	male	18.0000
3rd				
## 801	Fleming, Miss. Honora	no	female	NA
3rd				
## 802	Flynn, Mr. James	no	male	NA
3rd				
## 803	Flynn, Mr. John	no	male	NA
3rd				
## 804	Foley, Mr. Joseph	no	male	26.0000
3rd				
## 805	Foley, Mr. William	no	male	NA
3rd				
## 806	Foo, Mr. Choong	yes	male	NA
3rd				
## 807	Ford, Miss. Doolina Margaret D	no	female	21.0000
3rd				
## 808	Ford, Miss. Robina Maggie Ruby	no	female	9.0000
3rd				

## 809	Ford, Mr. Arthur	no	male	NA
3rd				
## 810	Ford, Mr. Edward Watson	no	male	18.0000
3rd				
## 811	Ford, Mr. William Neal	no	male	16.0000
3rd				
## 812	Ford, Mrs. Edward (Margaret Ann	no	female	48.0000
3rd				
## 813	Fox, Mr. Patrick	no	male	NA
3rd				
## 814	Franklin, Mr. Charles (Charles	no	male	NA
3rd				
## 815	Gallagher, Mr. Martin	no	male	25.0000
3rd				
## 816	Garfirth, Mr. John	no	male	NA
3rd				
## 817	Gheorgheff, Mr. Stanio	no	male	NA
3rd				
## 818	Gilinski, Mr. Eliezer	no	male	22.0000
3rd				
## 819	Gilnagh, Miss. Katherine Katie	yes	female	16.0000
3rd				
## 820	Glynn, Miss. Mary Agatha	yes	female	NA
3rd				
## 821	Goldsmith, Master. Frank John W	yes	male	9.0000
3rd				
## 822	Goldsmith, Mr. Frank John	no	male	33.0000
3rd				
## 823	Goldsmith, Mr. Nathan	no	male	41.0000
3rd				
## 824	Goldsmith, Mrs. Frank John (Emi	yes	female	31.0000
3rd				
## 825	Goncalves, Mr. Manuel Estanslas	no	male	38.0000
3rd				
## 826	Goodwin, Master. Harold Victor	no	male	9.0000
3rd				
## 827	Goodwin, Master. Sidney Leonard	no	male	1.0000
3rd				
## 828	Goodwin, Master. William Freder	no	male	11.0000
3rd				
## 829	Goodwin, Miss. Jessie Allis	no	female	10.0000
3rd				
## 830	Goodwin, Miss. Lillian Amy	no	female	16.0000
3rd				
## 831	Goodwin, Mr. Charles Edward	no	male	14.0000
3rd				
## 832	Goodwin, Mr. Charles Frederick	no	male	40.0000
3rd				
## 833	Goodwin, Mrs. Frederick (August	no	female	43.0000
3rd				

## 834	Green, Mr. George Henry	no	male	51.0000
3rd				
## 835	Gronnestad, Mr. Daniel Danielse	no	male	32.0000
3rd				
## 836	Guest, Mr. Robert	no	male	NA
3rd				
## 837	Gustafsson, Mr. Alfred Ossian	no	male	20.0000
3rd				
## 838	Gustafsson, Mr. Anders Vilhelm	no	male	37.0000
3rd				
## 839	Gustafsson, Mr. Johan Birger	no	male	28.0000
3rd				
## 840	Gustafsson, Mr. Karl Gideon	no	male	19.0000
3rd				
## 841	Haas, Miss. Aloisia	no	female	24.0000
3rd				
## 842	Hagardon, Miss. Kate	no	female	17.0000
3rd				
## 843	Hagland, Mr. Ingvald Olai Olsen	no	male	NA
3rd				
## 844	Hagland, Mr. Konrad Mathias Rei	no	male	NA
3rd				
## 845	Hakkarainen, Mr. Pekka Pietari	no	male	28.0000
3rd				
## 846	Hakkarainen, Mrs. Pekka Pietari	yes	female	24.0000
3rd				
## 847	Hampe, Mr. Leon	no	male	20.0000
3rd				
## 848	Hanna, Mr. Mansour	no	male	23.5000
3rd				
## 849	Hansen, Mr. Claus Peter	no	male	41.0000
3rd				
## 850	Hansen, Mr. Henrik Juul	no	male	26.0000
3rd				
## 851	Hansen, Mr. Henry Damsgaard	no	male	21.0000
3rd				
## 852	Hansen, Mrs. Claus Peter (Jenni	yes	female	45.0000
3rd				
## 853	Harknett, Miss. Alice Phoebe	no	female	NA
3rd				
## 854	Harmer, Mr. Abraham (David Lish	no	male	25.0000
3rd				
## 855	Hart, Mr. Henry	no	male	NA
3rd				
## 856	Hassan, Mr. Houssein G N	no	male	11.0000
3rd				
## 857	Healy, Miss. Hanora Nora	yes	female	NA
3rd				
## 858	Hedman, Mr. Oskar Arvid	yes	male	27.0000
3rd				

## 859	Hee, Mr. Ling	yes	male	NA
3rd				
## 860	Hegarty, Miss. Hanora Nora	no	female	18.0000
3rd				
## 861	Heikkinen, Miss. Laina	yes	female	26.0000
3rd				
## 862	Heininen, Miss. Wendla Maria	no	female	23.0000
3rd				
## 863	Hellstrom, Miss. Hilda Maria	yes	female	22.0000
3rd				
## 864	Hendekovic, Mr. Ignjac	no	male	28.0000
3rd				
## 865	Henriksson, Miss. Jenny Lovisa	no	female	28.0000
3rd				
## 866	Henry, Miss. Delia	no	female	NA
3rd				
## 867	Hirvonen, Miss. Hildur E	yes	female	2.0000
3rd				
## 868	Hirvonen, Mrs. Alexander (Helga	yes	female	22.0000
3rd				
## 869	Holm, Mr. John Fredrik Alexande	no	male	43.0000
3rd				
## 870	Holthen, Mr. Johan Martin	no	male	28.0000
3rd				
## 871	Honkanen, Miss. Eliina	yes	female	27.0000
3rd				
## 872	Horgan, Mr. John	no	male	NA
3rd				
## 873	Howard, Miss. May Elizabeth	yes	female	NA
3rd				
## 874	Humblen, Mr. Adolf Mathias Nico	no	male	42.0000
3rd				
## 875	Hyman, Mr. Abraham	yes	male	NA
3rd				
## 876	Ibrahim Shawah, Mr. Yousseff	no	male	30.0000
3rd				
## 877	Ilieff, Mr. Ylio	no	male	NA
3rd				
## 878	Ilmakangas, Miss. Ida Livija	no	female	27.0000
3rd				
## 879	Ilmakangas, Miss. Pieta Sofia	no	female	25.0000
3rd				
## 880	Ivanoff, Mr. Kanio	no	male	NA
3rd				
## 881	Jalsevac, Mr. Ivan	yes	male	29.0000
3rd				
## 882	Jansson, Mr. Carl Olof	yes	male	21.0000
3rd				
## 883	Jardin, Mr. Jose Neto	no	male	NA
3rd				

## 884	Jensen, Mr. Hans Peder	no	male	20.0000
3rd				
## 885	Jensen, Mr. Niels Peder	no	male	48.0000
3rd				
## 886	Jensen, Mr. Svend Lauritz	no	male	17.0000
3rd				
## 887	Jermyn, Miss. Annie	yes	female	NA
3rd				
## 888	Johannesen-Bratthammer, Mr. Ber	yes	male	NA
3rd				
## 889	Johanson, Mr. Jakob Alfred	no	male	34.0000
3rd				
## 890	Johansson Palmquist, Mr. Oskar	yes	male	26.0000
3rd				
## 891	Johansson, Mr. Erik	no	male	22.0000
3rd				
## 892	Johansson, Mr. Gustaf Joel	no	male	33.0000
3rd				
## 893	Johansson, Mr. Karl Johan	no	male	31.0000
3rd				
## 894	Johansson, Mr. Nils	no	male	29.0000
3rd				
## 895	Johnson, Master. Harold Theodor	yes	male	4.0000
3rd				
## 896	Johnson, Miss. Eleanor Ileen	yes	female	1.0000
3rd				
## 897	Johnson, Mr. Alfred	no	male	49.0000
3rd				
## 898	Johnson, Mr. Malkolm Joackim	no	male	33.0000
3rd				
## 899	Johnson, Mr. William Cahoone Jr	no	male	19.0000
3rd				
## 900	Johnson, Mrs. Oscar W (Elisabet	yes	female	27.0000
3rd				
## 901	Johnston, Master. William Arthu	no	male	NA
3rd				
## 902	Johnston, Miss. Catherine Helen	no	female	NA
3rd				
## 903	Johnston, Mr. Andrew G	no	male	NA
3rd				
## 904	Johnston, Mrs. Andrew G (Elizab	no	female	NA
3rd				
## 905	Jonkoff, Mr. Lallio	no	male	23.0000
3rd				
## 906	Jonsson, Mr. Carl	yes	male	32.0000
3rd				
## 907	Jonsson, Mr. Nils Hilding	no	male	27.0000
3rd				
## 908	Jussila, Miss. Katriina	no	female	20.0000
3rd				

## 909	Jussila, Miss. Mari Aina	no female	21.0000
3rd			
## 910	Jussila, Mr. Eiriik	yes male	32.0000
3rd			
## 911	Kallio, Mr. Nikolai Erland	no male	17.0000
3rd			
## 912	Kalvik, Mr. Johannes Halvorsen	no male	21.0000
3rd			
## 913	Karaic, Mr. Milan	no male	30.0000
3rd			
## 914	Karlsson, Mr. Einar Gervasius	yes male	21.0000
3rd			
## 915	Karlsson, Mr. Julius Konrad Eug	no male	33.0000
3rd			
## 916	Karlsson, Mr. Nils August	no male	22.0000
3rd			
## 917	Karun, Miss. Manca	yes female	4.0000
3rd			
## 918	Karun, Mr. Franz	yes male	39.0000
3rd			
## 919	Kassem, Mr. Fared	no male	NA
3rd			
## 920	Katavelas, Mr. Vassilios (Cata	no male	18.5000
3rd			
## 921	Keane, Mr. Andrew Andy	no male	NA
3rd			
## 922	Keefe, Mr. Arthur	no male	NA
3rd			
## 923	Kelly, Miss. Anna Katherine An	yes female	NA
3rd			
## 924	Kelly, Miss. Mary	yes female	NA
3rd			
## 925	Kelly, Mr. James (A)	no male	34.5000
3rd			
## 926	Kelly, Mr. James (B)	no male	44.0000
3rd			
## 927	Kennedy, Mr. John	yes male	NA
3rd			
## 928	Khalil, Mr. Betros	no male	NA
3rd			
## 929	Khalil, Mrs. Betros (Zahie Mar	no female	NA
3rd			
## 930	Kiernan, Mr. John	no male	NA
3rd			
## 931	Kiernan, Mr. Philip	no male	NA
3rd			
## 932	Kilgannon, Mr. Thomas J	no male	NA
3rd			
## 933	Kink, Miss. Maria	no female	22.0000
3rd			

## 934	Kink, Mr. Vincenz	no	male	26.0000
3rd				
## 935	Kink-Heilmann, Miss. Luise Gret	yes	female	4.0000
3rd				
## 936	Kink-Heilmann, Mr. Anton	yes	male	29.0000
3rd				
## 937	Kink-Heilmann, Mrs. Anton (Luis	yes	female	26.0000
3rd				
## 938	Klasen, Miss. Gertrud Emilia	no	female	1.0000
3rd				
## 939	Klasen, Mr. Klas Albin	no	male	18.0000
3rd				
## 940	Klasen, Mrs. (Hulda Kristina Eu	no	female	36.0000
3rd				
## 941	Kraeff, Mr. Theodor	no	male	NA
3rd				
## 942	Krekorian, Mr. Neshan	yes	male	25.0000
3rd				
## 943	Lahoud, Mr. Sarkis	no	male	NA
3rd				
## 944	Laitinen, Miss. Kristina Sofia	no	female	37.0000
3rd				
## 945	Laleff, Mr. Kristo	no	male	NA
3rd				
## 946	Lam, Mr. Ali	yes	male	NA
3rd				
## 947	Lam, Mr. Len	no	male	NA
3rd				
## 948	Landergren, Miss. Aurora Adelia	yes	female	22.0000
3rd				
## 949	Lane, Mr. Patrick	no	male	NA
3rd				
## 950	Lang, Mr. Fang	yes	male	26.0000
3rd				
## 951	Larsson, Mr. August Viktor	no	male	29.0000
3rd				
## 952	Larsson, Mr. Bengt Edvin	no	male	29.0000
3rd				
## 953	Larsson-Rondberg, Mr. Edvard A	no	male	22.0000
3rd				
## 954	Leeni, Mr. Fahim (Philip Zenni	yes	male	22.0000
3rd				
## 955	Lefebre, Master. Henry Forbes	no	male	NA
3rd				
## 956	Lefebre, Miss. Ida	no	female	NA
3rd				
## 957	Lefebre, Miss. Jeannie	no	female	NA
3rd				
## 958	Lefebre, Miss. Mathilde	no	female	NA
3rd				

## 959	Lefebvre, Mrs. Frank (Frances)	no female	NA
3rd			
## 960	Leinonen, Mr. Antti Gustaf	no male	32.0000
3rd			
## 961	Lemberopolous, Mr. Peter L	no male	34.5000
3rd			
## 962	Lennon, Miss. Mary	no female	NA
3rd			
## 963	Lennon, Mr. Denis	no male	NA
3rd			
## 964	Leonard, Mr. Lionel	no male	36.0000
3rd			
## 965	Lester, Mr. James	no male	39.0000
3rd			
## 966	Lievens, Mr. Rene Aime	no male	24.0000
3rd			
## 967	Lindahl, Miss. Agda Thorilda Vi	no female	25.0000
3rd			
## 968	Lindblom, Miss. Augusta Charlot	no female	45.0000
3rd			
## 969	Lindell, Mr. Edvard Bengtsson	no male	36.0000
3rd			
## 970	Lindell, Mrs. Edvard Bengtsson	no female	30.0000
3rd			
## 971	Lindqvist, Mr. Eino William	yes male	20.0000
3rd			
## 972	Linehan, Mr. Michael	no male	NA
3rd			
## 973	Ling, Mr. Lee	no male	28.0000
3rd			
## 974	Lithman, Mr. Simon	no male	NA
3rd			
## 975	Lobb, Mr. William Arthur	no male	30.0000
3rd			
## 976	Lobb, Mrs. William Arthur (Cord	no female	26.0000
3rd			
## 977	Lockyer, Mr. Edward	no male	NA
3rd			
## 978	Lovell, Mr. John Hall (Henry)	no male	20.5000
3rd			
## 979	Lulic, Mr. Nikola	yes male	27.0000
3rd			
## 980	Lundahl, Mr. Johan Svensson	no male	51.0000
3rd			
## 981	Lundin, Miss. Olga Elida	yes female	23.0000
3rd			
## 982	Lundstrom, Mr. Thure Edvin	yes male	32.0000
3rd			
## 983	Lyntakoff, Mr. Stanko	no male	NA
3rd			

## 984	MacKay, Mr. George William	no	male	NA
3rd				
## 985	Madigan, Miss. Margaret Maggie	yes	female	NA
3rd				
## 986	Madsen, Mr. Fridtjof Arne	yes	male	24.0000
3rd				
## 987	Maenpaa, Mr. Matti Alexanteri	no	male	22.0000
3rd				
## 988	Mahon, Miss. Bridget Delia	no	female	NA
3rd				
## 989	Mahon, Mr. John	no	male	NA
3rd				
## 990	Maisner, Mr. Simon	no	male	NA
3rd				
## 991	Makinen, Mr. Kalle Edvard	no	male	29.0000
3rd				
## 992	Mamee, Mr. Hanna	yes	male	NA
3rd				
## 993	Mangan, Miss. Mary	no	female	30.5000
3rd				
## 994	Mannion, Miss. Margareth	yes	female	NA
3rd				
## 995	Mardirosian, Mr. Sarkis	no	male	NA
3rd				
## 996	Markoff, Mr. Marin	no	male	35.0000
3rd				
## 997	Markun, Mr. Johann	no	male	33.0000
3rd				
## 998	Masselmani, Mrs. Fatima	yes	female	NA
3rd				
## 999	Matinoff, Mr. Nicola	no	male	NA
3rd				
## 1000	McCarthy, Miss. Catherine Kati	yes	female	NA
3rd				
## 1001	McCormack, Mr. Thomas Joseph	yes	male	NA
3rd				
## 1002	McCoy, Miss. Agnes	yes	female	NA
3rd				
## 1003	McCoy, Miss. Alicia	yes	female	NA
3rd				
## 1004	McCoy, Mr. Bernard	yes	male	NA
3rd				
## 1005	McDermott, Miss. Brigdet Delia	yes	female	NA
3rd				
## 1006	McEvoy, Mr. Michael	no	male	NA
3rd				
## 1007	McGovern, Miss. Mary	yes	female	NA
3rd				
## 1008	McGowan, Miss. Anna Annie	yes	female	15.0000
3rd				

## 1009	McGowan, Miss. Katherine	no female	35.0000
3rd			
## 1010	McMahon, Mr. Martin	no male	NA
3rd			
## 1011	McNamee, Mr. Neal	no male	24.0000
3rd			
## 1012	McNamee, Mrs. Neal (Eileen O'Le	no female	19.0000
3rd			
## 1013	McNeill, Miss. Bridget	no female	NA
3rd			
## 1014	Meanwell, Miss. (Marion Ogden)	no female	NA
3rd			
## 1015	Meek, Mrs. Thomas (Annie Louise	no female	NA
3rd			
## 1016	Meo, Mr. Alfonzo	no male	55.5000
3rd			
## 1017	Mernagh, Mr. Robert	no male	NA
3rd			
## 1018	Midtsjo, Mr. Karl Albert	yes male	21.0000
3rd			
## 1019	Miles, Mr. Frank	no male	NA
3rd			
## 1020	Mineff, Mr. Ivan	no male	24.0000
3rd			
## 1021	Minkoff, Mr. Lazar	no male	21.0000
3rd			
## 1022	Mionoff, Mr. Stoytcho	no male	28.0000
3rd			
## 1023	Mitkoff, Mr. Mito	no male	NA
3rd			
## 1024	Mockler, Miss. Helen Mary Elli	yes female	NA
3rd			
## 1025	Moen, Mr. Sigurd Hansen	no male	25.0000
3rd			
## 1026	Moor, Master. Meier	yes male	6.0000
3rd			
## 1027	Moor, Mrs. (Beila)	yes female	27.0000
3rd			
## 1028	Moore, Mr. Leonard Charles	no male	NA
3rd			
## 1029	Moran, Miss. Bertha	yes female	NA
3rd			
## 1030	Moran, Mr. Daniel J	no male	NA
3rd			
## 1031	Moran, Mr. James	no male	NA
3rd			
## 1032	Morley, Mr. William	no male	34.0000
3rd			
## 1033	Morrow, Mr. Thomas Rowan	no male	NA
3rd			

## 1034	Moss, Mr. Albert Johan	yes	male	NA
3rd				
## 1035	Moubarek, Master. Gerios	yes	male	NA
3rd				
## 1036	Moubarek, Master. Halim Gonios	yes	male	NA
3rd				
## 1037	Moubarek, Mrs. George (Omine A	yes	female	NA
3rd				
## 1038	Moussa, Mrs. (Mantoura Boulos)	yes	female	NA
3rd				
## 1039	Moutal, Mr. Rahamin Haim	no	male	NA
3rd				
## 1040	Mullens, Miss. Katherine Katie	yes	female	NA
3rd				
## 1041	Mulvihill, Miss. Bertha E	yes	female	24.0000
3rd				
## 1042	Murdlin, Mr. Joseph	no	male	NA
3rd				
## 1043	Murphy, Miss. Katherine Kate	yes	female	NA
3rd				
## 1044	Murphy, Miss. Margaret Jane	yes	female	NA
3rd				
## 1045	Murphy, Miss. Nora	yes	female	NA
3rd				
## 1046	Myhrman, Mr. Pehr Fabian Oliver	no	male	18.0000
3rd				
## 1047	Naidenoff, Mr. Penko	no	male	22.0000
3rd				
## 1048	Najib, Miss. Adele Kiamie Jane	yes	female	15.0000
3rd				
## 1049	Nakid, Miss. Maria (Mary)	yes	female	1.0000
3rd				
## 1050	Nakid, Mr. Sahid	yes	male	20.0000
3rd				
## 1051	Nakid, Mrs. Said (Waika Mary	yes	female	19.0000
3rd				
## 1052	Nancarrow, Mr. William Henry	no	male	33.0000
3rd				
## 1053	Nankoff, Mr. Minko	no	male	NA
3rd				
## 1054	Nasr, Mr. Mustafa	no	male	NA
3rd				
## 1055	Naughton, Miss. Hannah	no	female	NA
3rd				
## 1056	Nenkoff, Mr. Christo	no	male	NA
3rd				
## 1057	Nicola-Yarred, Master. Elias	yes	male	12.0000
3rd				
## 1058	Nicola-Yarred, Miss. Jamila	yes	female	14.0000
3rd				

## 1059	Nieminen, Miss. Manta Josefina	no female	29.0000
3rd			
## 1060	Niklasson, Mr. Samuel	no male	28.0000
3rd			
## 1061	Nilsson, Miss. Berta Olivia	yes female	18.0000
3rd			
## 1062	Nilsson, Miss. Helmina Josefina	yes female	26.0000
3rd			
## 1063	Nilsson, Mr. August Ferdinand	no male	21.0000
3rd			
## 1064	Nirva, Mr. Iisakki Antino Aijo	no male	41.0000
3rd			
## 1065	Niskanen, Mr. Juha	yes male	39.0000
3rd			
## 1066	Nosworthy, Mr. Richard Cater	no male	21.0000
3rd			
## 1067	Novel, Mr. Mansouer	no male	28.5000
3rd			
## 1068	Nysten, Miss. Anna Sofia	yes female	22.0000
3rd			
## 1069	Nysveen, Mr. Johan Hansen	no male	61.0000
3rd			
## 1070	O'Brien, Mr. Thomas	no male	NA
3rd			
## 1071	O'Brien, Mr. Timothy	no male	NA
3rd			
## 1072	O'Brien, Mrs. Thomas (Johanna	yes female	NA
3rd			
## 1073	O'Connell, Mr. Patrick D	no male	NA
3rd			
## 1074	O'Connor, Mr. Maurice	no male	NA
3rd			
## 1075	O'Connor, Mr. Patrick	no male	NA
3rd			
## 1076	Odahl, Mr. Nils Martin	no male	23.0000
3rd			
## 1077	O'Donoghue, Ms. Bridget	no female	NA
3rd			
## 1078	O'Driscoll, Miss. Bridget	yes female	NA
3rd			
## 1079	O'Dwyer, Miss. Ellen Nellie	yes female	NA
3rd			
## 1080	Ohman, Miss. Velin	yes female	22.0000
3rd			
## 1081	O'Keefe, Mr. Patrick	yes male	NA
3rd			
## 1082	O'Leary, Miss. Hanora Norah	yes female	NA
3rd			
## 1083	Olsen, Master. Artur Karl	yes male	9.0000
3rd			

## 1084	Olsen, Mr. Henry Margido	no	male	28.0000
3rd				
## 1085	Olsen, Mr. Karl Siegwart Andrea	no	male	42.0000
3rd				
## 1086	Olsen, Mr. Ole Martin	no	male	NA
3rd				
## 1087	Olsson, Miss. Elina	no	female	31.0000
3rd				
## 1088	Olsson, Mr. Nils Johan Goransso	no	male	28.0000
3rd				
## 1089	Olsson, Mr. Oscar Wilhelm	yes	male	32.0000
3rd				
## 1090	Olsvigen, Mr. Thor Anderson	no	male	20.0000
3rd				
## 1091	Oreskovic, Miss. Jelka	no	female	23.0000
3rd				
## 1092	Oreskovic, Miss. Marija	no	female	20.0000
3rd				
## 1093	Oreskovic, Mr. Luka	no	male	20.0000
3rd				
## 1094	Osen, Mr. Olaf Elon	no	male	16.0000
3rd				
## 1095	Osman, Mrs. Mara	yes	female	31.0000
3rd				
## 1096	O'Sullivan, Miss. Bridget Mary	no	female	NA
3rd				
## 1097	Palsson, Master. Gosta Leonard	no	male	2.0000
3rd				
## 1098	Palsson, Master. Paul Folke	no	male	6.0000
3rd				
## 1099	Palsson, Miss. Stina Viola	no	female	3.0000
3rd				
## 1100	Palsson, Miss. Torborg Danira	no	female	8.0000
3rd				
## 1101	Palsson, Mrs. Nils (Alma Cornel	no	female	29.0000
3rd				
## 1102	Panula, Master. Eino Viljami	no	male	1.0000
3rd				
## 1103	Panula, Master. Juha Niilo	no	male	7.0000
3rd				
## 1104	Panula, Master. Urho Abraham	no	male	2.0000
3rd				
## 1105	Panula, Mr. Ernesti Arvid	no	male	16.0000
3rd				
## 1106	Panula, Mr. Jaako Arnold	no	male	14.0000
3rd				
## 1107	Panula, Mrs. Juha (Maria Emilia	no	female	41.0000
3rd				
## 1108	Pasic, Mr. Jakob	no	male	21.0000
3rd				

## 1109	Patchett, Mr. George	no	male	19.0000
3rd				
## 1110	Paulner, Mr. Uscher	no	male	NA
3rd				
## 1111	Pavlovic, Mr. Stefo	no	male	32.0000
3rd				
## 1112	Peacock, Master. Alfred Edward	no	male	0.7500
3rd				
## 1113	Peacock, Miss. Treasteall	no	female	3.0000
3rd				
## 1114	Peacock, Mrs. Benjamin (Edith N	no	female	26.0000
3rd				
## 1115	Pearce, Mr. Ernest	no	male	NA
3rd				
## 1116	Pedersen, Mr. Olaf	no	male	NA
3rd				
## 1117	Peduzzi, Mr. Joseph	no	male	NA
3rd				
## 1118	Pekoniemi, Mr. Edvard	no	male	21.0000
3rd				
## 1119	Peltomaki, Mr. Nikolai Johannes	no	male	25.0000
3rd				
## 1120	Perkin, Mr. John Henry	no	male	22.0000
3rd				
## 1121	Persson, Mr. Ernst Ulrik	yes	male	25.0000
3rd				
## 1122	Peter, Master. Michael J	yes	male	NA
3rd				
## 1123	Peter, Miss. Anna	yes	female	NA
3rd				
## 1124	Peter, Mrs. Catherine (Catherin	yes	female	NA
3rd				
## 1125	Peters, Miss. Katie	no	female	NA
3rd				
## 1126	Petersen, Mr. Marius	no	male	24.0000
3rd				
## 1127	Petranec, Miss. Matilda	no	female	28.0000
3rd				
## 1128	Petroff, Mr. Nedelio	no	male	19.0000
3rd				
## 1129	Petroff, Mr. Pastcho (Pentcho	no	male	NA
3rd				
## 1130	Petterson, Mr. Johan Emil	no	male	25.0000
3rd				
## 1131	Pettersson, Miss. Ellen Natalia	no	female	18.0000
3rd				
## 1132	Pickard, Mr. Berk (Berk Trembis	yes	male	32.0000
3rd				
## 1133	Plotcharsky, Mr. Vasil	no	male	NA
3rd				

## 1134	Pokrnic, Mr. Mate	no	male	17.0000
3rd				
## 1135	Pokrnic, Mr. Tome	no	male	24.0000
3rd				
## 1136	Radeff, Mr. Alexander	no	male	NA
3rd				
## 1137	Rasmussen, Mrs. (Lena Jacobsen	no	female	NA
3rd				
## 1138	Razi, Mr. Raihed	no	male	NA
3rd				
## 1139	Reed, Mr. James George	no	male	NA
3rd				
## 1140	Rekic, Mr. Tido	no	male	38.0000
3rd				
## 1141	Reynolds, Mr. Harold J	no	male	21.0000
3rd				
## 1142	Rice, Master. Albert	no	male	10.0000
3rd				
## 1143	Rice, Master. Arthur	no	male	4.0000
3rd				
## 1144	Rice, Master. Eric	no	male	7.0000
3rd				
## 1145	Rice, Master. Eugene	no	male	2.0000
3rd				
## 1146	Rice, Master. George Hugh	no	male	8.0000
3rd				
## 1147	Rice, Mrs. William (Margaret No	no	female	39.0000
3rd				
## 1148	Riihivouri, Miss. Susanna Juhan	no	female	22.0000
3rd				
## 1149	Rintamaki, Mr. Matti	no	male	35.0000
3rd				
## 1150	Riordan, Miss. Johanna Hannah	yes	female	NA
3rd				
## 1151	Risien, Mr. Samuel Beard	no	male	NA
3rd				
## 1152	Risien, Mrs. Samuel (Emma)	no	female	NA
3rd				
## 1153	Robins, Mr. Alexander A	no	male	50.0000
3rd				
## 1154	Robins, Mrs. Alexander A (Grace	no	female	47.0000
3rd				
## 1155	Rogers, Mr. William John	no	male	NA
3rd				
## 1156	Rommetvedt, Mr. Knud Paust	no	male	NA
3rd				
## 1157	Rosblom, Miss. Salli Helena	no	female	2.0000
3rd				
## 1158	Rosblom, Mr. Viktor Richard	no	male	18.0000
3rd				

## 1159	Rosblom, Mrs. Viktor (Helena Wi 3rd	no	female	41.0000
## 1160	Roth, Miss. Sarah A 3rd	yes	female	NA
## 1161	Rouse, Mr. Richard Henry 3rd	no	male	50.0000
## 1162	Rush, Mr. Alfred George John 3rd	no	male	16.0000
## 1163	Ryan, Mr. Edward 3rd	yes	male	NA
## 1164	Ryan, Mr. Patrick 3rd	no	male	NA
## 1165	Saad, Mr. Amin 3rd	no	male	NA
## 1166	Saad, Mr. Khalil 3rd	no	male	25.0000
## 1167	Saade, Mr. Jean Nassr 3rd	no	male	NA
## 1168	Sadlier, Mr. Matthew 3rd	no	male	NA
## 1169	Sadowitz, Mr. Harry 3rd	no	male	NA
## 1170	Saether, Mr. Simon Sivertsen 3rd	no	male	38.5000
## 1171	Sage, Master. Thomas Henry 3rd	no	male	NA
## 1172	Sage, Master. William Henry 3rd	no	male	14.5000
## 1173	Sage, Miss. Ada 3rd	no	female	NA
## 1174	Sage, Miss. Constance Gladys 3rd	no	female	NA
## 1175	Sage, Miss. Dorothy Edith Doll 3rd	no	female	NA
## 1176	Sage, Miss. Stella Anna 3rd	no	female	NA
## 1177	Sage, Mr. Douglas Bullen 3rd	no	male	NA
## 1178	Sage, Mr. Frederick 3rd	no	male	NA
## 1179	Sage, Mr. George John Jr 3rd	no	male	NA
## 1180	Sage, Mr. John George 3rd	no	male	NA
## 1181	Sage, Mrs. John (Annie Bullen) 3rd	no	female	NA
## 1182	Salander, Mr. Karl Johan 3rd	no	male	24.0000
## 1183	Salkjelsvik, Miss. Anna Kristin 3rd	yes	female	21.0000

## 1184	Salonen, Mr. Johan Werner	no	male	39.0000
3rd				
## 1185	Samaan, Mr. Elias	no	male	NA
3rd				
## 1186	Samaan, Mr. Hanna	no	male	NA
3rd				
## 1187	Samaan, Mr. Youssef	no	male	NA
3rd				
## 1188	Sandstrom, Miss. Beatrice Irene	yes	female	1.0000
3rd				
## 1189	Sandstrom, Mrs. Hjalmar (Agnes	yes	female	24.0000
3rd				
## 1190	Sandstrom, Miss. Marguerite Rut	yes	female	4.0000
3rd				
## 1191	Sap, Mr. Julius	yes	male	25.0000
3rd				
## 1192	Saundercock, Mr. William Henry	no	male	20.0000
3rd				
## 1193	Sawyer, Mr. Frederick Charles	no	male	24.5000
3rd				
## 1194	Scanlan, Mr. James	no	male	NA
3rd				
## 1195	Sdycoff, Mr. Todor	no	male	NA
3rd				
## 1196	Shaughnessy, Mr. Patrick	no	male	NA
3rd				
## 1197	Sheerlinck, Mr. Jan Baptist	yes	male	29.0000
3rd				
## 1198	Shellard, Mr. Frederick William	no	male	NA
3rd				
## 1199	Shine, Miss. Ellen Natalia	yes	female	NA
3rd				
## 1200	Shorney, Mr. Charles Joseph	no	male	NA
3rd				
## 1201	Simmons, Mr. John	no	male	NA
3rd				
## 1202	Sirayanian, Mr. Orsen	no	male	22.0000
3rd				
## 1203	Sirota, Mr. Maurice	no	male	NA
3rd				
## 1204	Sivic, Mr. Husein	no	male	40.0000
3rd				
## 1205	Sivola, Mr. Antti Wilhelm	no	male	21.0000
3rd				
## 1206	Sjoblom, Miss. Anna Sofia	yes	female	18.0000
3rd				
## 1207	Skoog, Master. Harald	no	male	4.0000
3rd				
## 1208	Skoog, Master. Karl Thorsten	no	male	10.0000
3rd				

## 1209	Skoog, Miss. Mabel	no female	9.0000
3rd			
## 1210	Skoog, Miss. Margit Elizabeth	no female	2.0000
3rd			
## 1211	Skoog, Mr. Wilhelm	no male	40.0000
3rd			
## 1212	Skoog, Mrs. William (Anna Bernh	no female	45.0000
3rd			
## 1213	Slabenoff, Mr. Petco	no male	NA
3rd			
## 1214	Slocovski, Mr. Selman Francis	no male	NA
3rd			
## 1215	Smiljanic, Mr. Mile	no male	NA
3rd			
## 1216	Smith, Mr. Thomas	no male	NA
3rd			
## 1217	Smyth, Miss. Julia	yes female	NA
3rd			
## 1218	Soholt, Mr. Peter Andreas Lauri	no male	19.0000
3rd			
## 1219	Somerton, Mr. Francis William	no male	30.0000
3rd			
## 1220	Spector, Mr. Woolf	no male	NA
3rd			
## 1221	Spinner, Mr. Henry John	no male	32.0000
3rd			
## 1222	Staneff, Mr. Ivan	no male	NA
3rd			
## 1223	Stankovic, Mr. Ivan	no male	33.0000
3rd			
## 1224	Stanley, Miss. Amy Zillah Elsie	yes female	23.0000
3rd			
## 1225	Stanley, Mr. Edward Roland	no male	21.0000
3rd			
## 1226	Storey, Mr. Thomas	no male	60.5000
3rd			
## 1227	Stoytcheff, Mr. Ilia	no male	19.0000
3rd			
## 1228	Strandberg, Miss. Ida Sofia	no female	22.0000
3rd			
## 1229	Stranden, Mr. Juho	yes male	31.0000
3rd			
## 1230	Strilic, Mr. Ivan	no male	27.0000
3rd			
## 1231	Strom, Miss. Telma Matilda	no female	2.0000
3rd			
## 1232	Strom, Mrs. Wilhelm (Elna Matil	no female	29.0000
3rd			
## 1233	Sunderland, Mr. Victor Francis	yes male	16.0000
3rd			

## 1234	Sundman, Mr. Johan Julian	yes	male	44.0000
3rd				
## 1235	Sutehall, Mr. Henry Jr	no	male	25.0000
3rd				
## 1236	Svensson, Mr. Johan	no	male	74.0000
3rd				
## 1237	Svensson, Mr. Johan Cervin	yes	male	14.0000
3rd				
## 1238	Svensson, Mr. Olof	no	male	24.0000
3rd				
## 1239	Tenglin, Mr. Gunnar Isidor	yes	male	25.0000
3rd				
## 1240	Theobald, Mr. Thomas Leonard	no	male	34.0000
3rd				
## 1241	Thomas, Master. Assad Alexander	yes	male	0.4167
3rd				
## 1242	Thomas, Mr. Charles P	no	male	NA
3rd				
## 1243	Thomas, Mr. John	no	male	NA
3rd				
## 1244	Thomas, Mr. Tannous	no	male	NA
3rd				
## 1245	Thomas, Mrs. Alexander (Thamine	yes	female	16.0000
3rd				
## 1246	Thomson, Mr. Alexander Morrison	no	male	NA
3rd				
## 1247	Thorneycroft, Mr. Percival	no	male	NA
3rd				
## 1248	Thorneycroft, Mrs. Percival (Fl	yes	female	NA
3rd				
## 1249	Tikkanen, Mr. Juho	no	male	32.0000
3rd				
## 1250	Tobin, Mr. Roger	no	male	NA
3rd				
## 1251	Todoroff, Mr. Lallo	no	male	NA
3rd				
## 1252	Tomlin, Mr. Ernest Portage	no	male	30.5000
3rd				
## 1253	Torber, Mr. Ernst William	no	male	44.0000
3rd				
## 1254	Torfa, Mr. Assad	no	male	NA
3rd				
## 1255	Tornquist, Mr. William Henry	yes	male	25.0000
3rd				
## 1256	Toufik, Mr. Nakli	no	male	NA
3rd				
## 1257	Touma, Master. Georges Youssef	yes	male	7.0000
3rd				
## 1258	Touma, Miss. Maria Youssef	yes	female	9.0000
3rd				

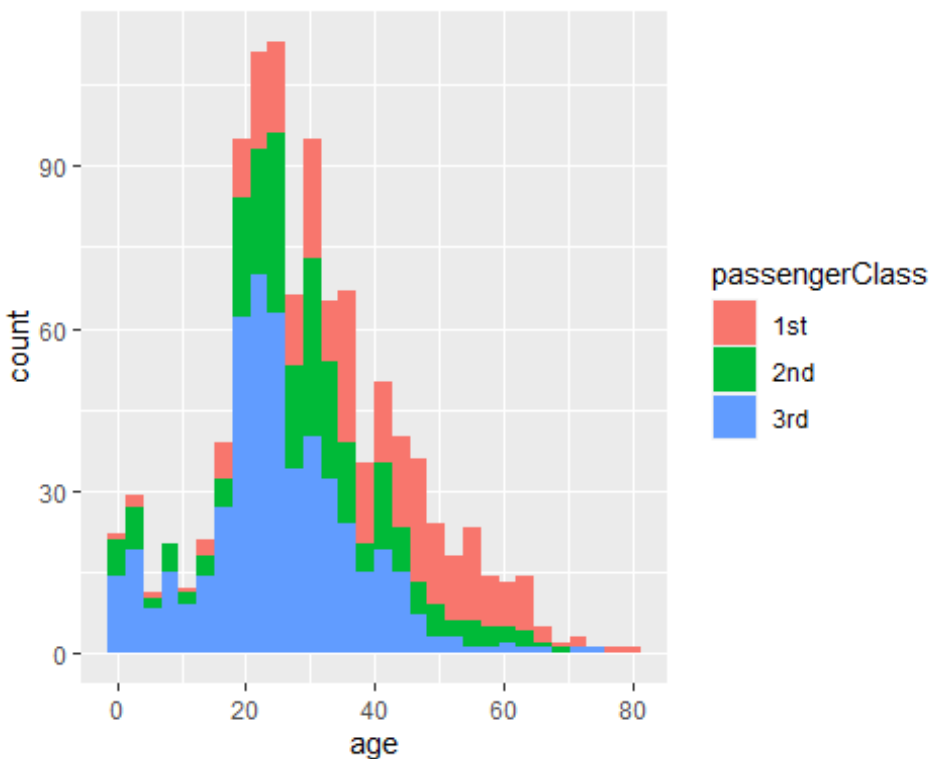
## 1259	Touma, Mrs. Darwis (Hanne Youss 3rd	yes	female	29.0000
## 1260	Turcin, Mr. Stjepan 3rd	no	male	36.0000
## 1261	Turja, Miss. Anna Sofia 3rd	yes	female	18.0000
## 1262	Turkula, Mrs. (Hedwig) 3rd	yes	female	63.0000
## 1263	van Billiard, Master. James Wil 3rd	no	male	NA
## 1264	van Billiard, Master. Walter Jo 3rd	no	male	11.5000
## 1265	van Billiard, Mr. Austin Blyler 3rd	no	male	40.5000
## 1266	Van Impe, Miss. Catharina 3rd	no	female	10.0000
## 1267	Van Impe, Mr. Jean Baptiste 3rd	no	male	36.0000
## 1268	Van Impe, Mrs. Jean Baptiste (R 3rd	no	female	30.0000
## 1269	van Melkebeke, Mr. Philemon 3rd	no	male	NA
## 1270	Vande Velde, Mr. Johannes Josep 3rd	no	male	33.0000
## 1271	Vande Walle, Mr. Nestor Cyriel 3rd	no	male	28.0000
## 1272	Vanden Steen, Mr. Leo Peter 3rd	no	male	28.0000
## 1273	Vander Cruyssen, Mr. Victor 3rd	no	male	47.0000
## 1274	Vander Planke, Miss. Augusta Ma 3rd	no	female	18.0000
## 1275	Vander Planke, Mr. Julius 3rd	no	male	31.0000
## 1276	Vander Planke, Mr. Leo Edmondus 3rd	no	male	16.0000
## 1277	Vander Planke, Mrs. Julius (Eme 3rd	no	female	31.0000
## 1278	Vartanian, Mr. David 3rd	yes	male	22.0000
## 1279	Vendel, Mr. Olof Edvin 3rd	no	male	20.0000
## 1280	Vestrom, Miss. Hulda Amanda Ado 3rd	no	female	14.0000
## 1281	Vovk, Mr. Janko 3rd	no	male	22.0000
## 1282	Waelens, Mr. Achille 3rd	no	male	22.0000
## 1283	Ware, Mr. Frederick 3rd	no	male	NA

## 1284	Warren, Mr. Charles William	no	male	NA
3rd				
## 1285	Webber, Mr. James	no	male	NA
3rd				
## 1286	Wenzel, Mr. Linhart	no	male	32.5000
3rd				
## 1287	Whabee, Mrs. George Joseph (Sha	yes	female	38.0000
3rd				
## 1288	Widegren, Mr. Carl/Charles Pete	no	male	51.0000
3rd				
## 1289	Wiklund, Mr. Jakob Alfred	no	male	18.0000
3rd				
## 1290	Wiklund, Mr. Karl Johan	no	male	21.0000
3rd				
## 1291	Wilkes, Mrs. James (Ellen Needs	yes	female	47.0000
3rd				
## 1292	Willer, Mr. Aaron (Abi Weller	no	male	NA
3rd				
## 1293	Willey, Mr. Edward	no	male	NA
3rd				
## 1294	Williams, Mr. Howard Hugh Harr	no	male	NA
3rd				
## 1295	Williams, Mr. Leslie	no	male	28.5000
3rd				
## 1296	Windelov, Mr. Einar	no	male	21.0000
3rd				
## 1297	Wirz, Mr. Albert	no	male	27.0000
3rd				
## 1298	Wiseman, Mr. Phillippe	no	male	NA
3rd				
## 1299	Wittevrongel, Mr. Camille	no	male	36.0000
3rd				
## 1300	Yasbeck, Mr. Antoni	no	male	27.0000
3rd				
## 1301	Yasbeck, Mrs. Antoni (Selini Al	yes	female	15.0000
3rd				
## 1302	Youseff, Mr. Gerious	no	male	45.5000
3rd				
## 1303	Yousif, Mr. Wazli	no	male	NA
3rd				
## 1304	Yousseff, Mr. Gerious	no	male	NA
3rd				
## 1305	Zabour, Miss. Hileni	no	female	14.5000
3rd				
## 1306	Zabour, Miss. Thamine	no	female	NA
3rd				
## 1307	Zakarian, Mr. Mapriededer	no	male	26.5000
3rd				
## 1308	Zakarian, Mr. Ortin	no	male	27.0000
3rd				

```
## 1309          Zimmerman, Mr. Leo      no   male 29.0000  
3rd
```

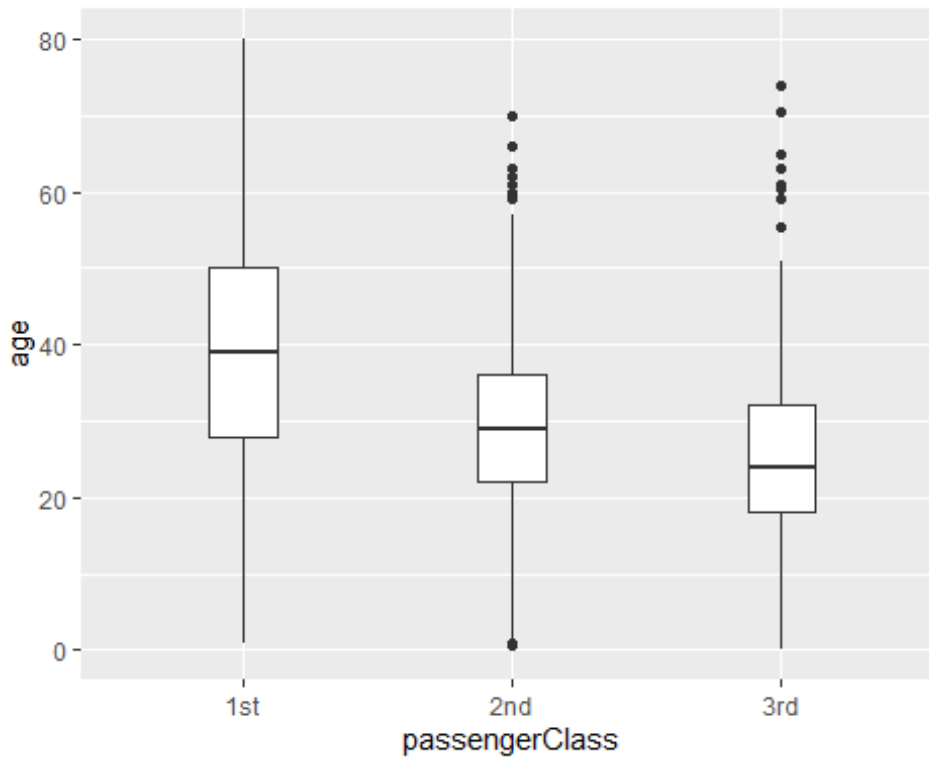
#Histogram

```
ggplot(ship_df,  
mapping = aes(x = age, fill = passengerClass)  
) + geom_histogram()  
  
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.  
## Warning: Removed 263 rows containing non-finite values (`stat_bin()`).
```



#Boxplot

```
ggplot(ship_df,  
mapping = aes(x = passengerClass, y = age)  
) + geom_boxplot(width = 0.25)  
  
## Warning: Removed 263 rows containing non-finite values (`stat_boxplot()`).
```

#Measure of central tendency

```
summary(ship_df)
```

```
##          X              survived              sex              age
## Length:1309      Length:1309      Length:1309      Min.   : 0.1667
## Class :character  Class :character  Class :character  1st Qu.:21.0000
## Mode  :character  Mode  :character  Mode  :character  Median :28.0000
##                                     Mean   :29.8811
##                                     3rd Qu.:39.0000
##                                     Max.   :80.0000
##                                     NA's   :263
## passengerClass
## Length:1309
## Class :character
## Mode  :character
##
##
##
##
```

#The mean, median, and mode can all be seen given below.

```
age_df <- data.frame(age =ship_df$age)
age_df
```

```
##          age
## 1      29.0000
```

## 2	0.9167
## 3	2.0000
## 4	30.0000
## 5	25.0000
## 6	48.0000
## 7	63.0000
## 8	39.0000
## 9	53.0000
## 10	71.0000
## 11	47.0000
## 12	18.0000
## 13	24.0000
## 14	26.0000
## 15	80.0000
## 16	NA
## 17	24.0000
## 18	50.0000
## 19	32.0000
## 20	36.0000
## 21	37.0000
## 22	47.0000
## 23	26.0000
## 24	42.0000
## 25	29.0000
## 26	25.0000
## 27	25.0000
## 28	19.0000
## 29	35.0000
## 30	28.0000
## 31	45.0000
## 32	40.0000
## 33	30.0000
## 34	58.0000
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##	963	NA
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##	965	39.0000
##	966	24.0000
##	967	25.0000
##	968	45.0000
##	969	36.0000
##	970	30.0000
##	971	20.0000
##	972	NA
##	973	28.0000
##	974	NA
##	975	30.0000
##	976	26.0000
##	977	NA
##	978	20.5000
##	979	27.0000
##	980	51.0000
##	981	23.0000
##	982	32.0000
##	983	NA
##	984	NA
##	985	NA
##	986	24.0000
##	987	22.0000
##	988	NA
##	989	NA
##	990	NA
##	991	29.0000
##	992	NA
##	993	30.5000
##	994	NA
##	995	NA
##	996	35.0000
##	997	33.0000
##	998	NA
##	999	NA
##	1000	NA
##	1001	NA

##	1002	NA
##	1003	NA
##	1004	NA
##	1005	NA
##	1006	NA
##	1007	NA
##	1008	15.0000
##	1009	35.0000
##	1010	NA
##	1011	24.0000
##	1012	19.0000
##	1013	NA
##	1014	NA
##	1015	NA
##	1016	55.5000
##	1017	NA
##	1018	21.0000
##	1019	NA
##	1020	24.0000
##	1021	21.0000
##	1022	28.0000
##	1023	NA
##	1024	NA
##	1025	25.0000
##	1026	6.0000
##	1027	27.0000
##	1028	NA
##	1029	NA
##	1030	NA
##	1031	NA
##	1032	34.0000
##	1033	NA
##	1034	NA
##	1035	NA
##	1036	NA
##	1037	NA
##	1038	NA
##	1039	NA
##	1040	NA
##	1041	24.0000
##	1042	NA
##	1043	NA
##	1044	NA
##	1045	NA
##	1046	18.0000
##	1047	22.0000
##	1048	15.0000
##	1049	1.0000
##	1050	20.0000
##	1051	19.0000

1052 33.0000
1053 NA
1054 NA
1055 NA
1056 NA
1057 12.0000
1058 14.0000
1059 29.0000
1060 28.0000
1061 18.0000
1062 26.0000
1063 21.0000
1064 41.0000
1065 39.0000
1066 21.0000
1067 28.5000
1068 22.0000
1069 61.0000
1070 NA
1071 NA
1072 NA
1073 NA
1074 NA
1075 NA
1076 23.0000
1077 NA
1078 NA
1079 NA
1080 22.0000
1081 NA
1082 NA
1083 9.0000
1084 28.0000
1085 42.0000
1086 NA
1087 31.0000
1088 28.0000
1089 32.0000
1090 20.0000
1091 23.0000
1092 20.0000
1093 20.0000
1094 16.0000
1095 31.0000
1096 NA
1097 2.0000
1098 6.0000
1099 3.0000
1100 8.0000
1101 29.0000

```
## 1102 1.0000
## 1103 7.0000
## 1104 2.0000
## 1105 16.0000
## 1106 14.0000
## 1107 41.0000
## 1108 21.0000
## 1109 19.0000
## 1110 NA
## 1111 32.0000
## 1112 0.7500
## 1113 3.0000
## 1114 26.0000
## 1115 NA
## 1116 NA
## 1117 NA
## 1118 21.0000
## 1119 25.0000
## 1120 22.0000
## 1121 25.0000
## 1122 NA
## 1123 NA
## 1124 NA
## 1125 NA
## 1126 24.0000
## 1127 28.0000
## 1128 19.0000
## 1129 NA
## 1130 25.0000
## 1131 18.0000
## 1132 32.0000
## 1133 NA
## 1134 17.0000
## 1135 24.0000
## 1136 NA
## 1137 NA
## 1138 NA
## 1139 NA
## 1140 38.0000
## 1141 21.0000
## 1142 10.0000
## 1143 4.0000
## 1144 7.0000
## 1145 2.0000
## 1146 8.0000
## 1147 39.0000
## 1148 22.0000
## 1149 35.0000
## 1150 NA
## 1151 NA
```


##	1152	NA
##	1153	50.0000
##	1154	47.0000
##	1155	NA
##	1156	NA
##	1157	2.0000
##	1158	18.0000
##	1159	41.0000
##	1160	NA
##	1161	50.0000
##	1162	16.0000
##	1163	NA
##	1164	NA
##	1165	NA
##	1166	25.0000
##	1167	NA
##	1168	NA
##	1169	NA
##	1170	38.5000
##	1171	NA
##	1172	14.5000
##	1173	NA
##	1174	NA
##	1175	NA
##	1176	NA
##	1177	NA
##	1178	NA
##	1179	NA
##	1180	NA
##	1181	NA
##	1182	24.0000
##	1183	21.0000
##	1184	39.0000
##	1185	NA
##	1186	NA
##	1187	NA
##	1188	1.0000
##	1189	24.0000
##	1190	4.0000
##	1191	25.0000
##	1192	20.0000
##	1193	24.5000
##	1194	NA
##	1195	NA
##	1196	NA
##	1197	29.0000
##	1198	NA
##	1199	NA
##	1200	NA
##	1201	NA

```
## 1202 22.0000
## 1203      NA
## 1204 40.0000
## 1205 21.0000
## 1206 18.0000
## 1207  4.0000
## 1208 10.0000
## 1209  9.0000
## 1210  2.0000
## 1211 40.0000
## 1212 45.0000
## 1213      NA
## 1214      NA
## 1215      NA
## 1216      NA
## 1217      NA
## 1218 19.0000
## 1219 30.0000
## 1220      NA
## 1221 32.0000
## 1222      NA
## 1223 33.0000
## 1224 23.0000
## 1225 21.0000
## 1226 60.5000
## 1227 19.0000
## 1228 22.0000
## 1229 31.0000
## 1230 27.0000
## 1231  2.0000
## 1232 29.0000
## 1233 16.0000
## 1234 44.0000
## 1235 25.0000
## 1236 74.0000
## 1237 14.0000
## 1238 24.0000
## 1239 25.0000
## 1240 34.0000
## 1241  0.4167
## 1242      NA
## 1243      NA
## 1244      NA
## 1245 16.0000
## 1246      NA
## 1247      NA
## 1248      NA
## 1249 32.0000
## 1250      NA
## 1251      NA
```

```
## 1252 30.5000
## 1253 44.0000
## 1254      NA
## 1255 25.0000
## 1256      NA
## 1257  7.0000
## 1258  9.0000
## 1259 29.0000
## 1260 36.0000
## 1261 18.0000
## 1262 63.0000
## 1263      NA
## 1264 11.5000
## 1265 40.5000
## 1266 10.0000
## 1267 36.0000
## 1268 30.0000
## 1269      NA
## 1270 33.0000
## 1271 28.0000
## 1272 28.0000
## 1273 47.0000
## 1274 18.0000
## 1275 31.0000
## 1276 16.0000
## 1277 31.0000
## 1278 22.0000
## 1279 20.0000
## 1280 14.0000
## 1281 22.0000
## 1282 22.0000
## 1283      NA
## 1284      NA
## 1285      NA
## 1286 32.5000
## 1287 38.0000
## 1288 51.0000
## 1289 18.0000
## 1290 21.0000
## 1291 47.0000
## 1292      NA
## 1293      NA
## 1294      NA
## 1295 28.5000
## 1296 21.0000
## 1297 27.0000
## 1298      NA
## 1299 36.0000
## 1300 27.0000
## 1301 15.0000
```

```
## 1302 45.5000
## 1303      NA
## 1304      NA
## 1305 14.5000
## 1306      NA
## 1307 26.5000
## 1308 27.0000
## 1309 29.0000
```

```
#var(age_df$age)
#mad(age_df$age)
#skewness(age_df)
```

Due to some errors on R studio, most of the codes would not register with the data set. For all the codes it kept outputting NA, so i have removed them and added the simple ones as a placeholder.

#I will however try to analyse using the summary stats as well as the histogram and the bar plots.

#using the plots a lot of visual aspects can help understand the details of the data set and some require summary stats for that.

#Central tendency, things such as the mean, median and mode can be seen observed using a histogram and a box plot, the mean, mode and the median can be seen on a histogram. In a histogram, the central tendency can be seen as the peak or the most frequencies peak in the distribution, and in a box plot, the median is represented by a line in the box which refers to the center of the distribution.

#The measure of dispersion, is the variance of the data points. The standard deviation and the range of the data points are analysed in the summary stats. The exact amount for these are not apparent from the histograms and the box plots but they do give the general idea. For a narrow histogram, the variance is low, and there is less dispersion, and when its wide there is a greater dispersion. For a box plot the dispersion is represented as the length of the box, bigger box = more variance.

#The measure of skewness is the measure of asymmetry of the distribution. Left skewed histograms are called negative and right skewed are called positive. This tells us the spread of data points on the histogram. On the histogram the skewness can be clearly seen and the box plot it is shown by its position on the y axis, understood by the asymmetry of its whiskers.

#Kurtosis is the measure of tailedness or the shape of the distribution when compared to a normal distribution. In a histogram it is the height and shape of the bins. A taller, more peaked distribution indicates higher kurtosis, while a flatter distribution suggests lower kurtosis. In a box plot it is the the legth and thickness of the whiskers. If the whiskers are really long and

thick, the outliers are persistently present so it has a higher Kurtosis and a shorter suggest less outliers.